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MILANO 1863**



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1. Introduction

1.1 Purpose

Triplo is a cross-platform mobile application for smartphones and smartwatches, conceived as a digital logbook for trekking activities.

The application aims to enhance the experience of hiking in mountain environments by combining route information, environmental data, and recording of personal activity.

Users can explore predefined trekking routes and access useful information about the paths, including navigation support and contextual data such as weather conditions.

At the same time, the application allows hikers to document their experiences by creating logbook entries enriched with photos and personal diary notes.

The platform also enables users to discover and follow other hikers, and to browse selected entries from their trekking diaries, enhancing the sharing of experiences and inspiration for future hikes.

During a hike, users can optionally complete challenges, such as time-based goals designed to enrich the trekking experience.

The following sections describe the system architecture, the main functionalities of the application, and the technological choices adopted in its development.

1.2 Scope

Triplo supports users during trekking activities by providing guidance, contextual information, and tools to document their experiences in mountain environments.

The application focuses on the Madonna di Campiglio area and offers a set of predefined trekking routes with different levels of difficulty.

During a hike, users can access information related to the selected route, including navigation support and environmental conditions that may influence the trekking experience.

The application also encourages engagement through optional challenges that users may complete along the route. Each trekking activity can be recorded in a personal digital logbook, where users can document their experience by adding photos and diary notes. Users can also browse selected entries from the logbooks of other hikers they follow, allowing them to discover new routes and gain inspiration from shared experiences.

The application aims to support outdoor exploration while enabling users to record, revisit, and share meaningful moments from their trekking activities.

1.3 Definitions, Acronyms, Abbreviations

- **GPS:** Generic term used in this document to indicate satellite-based positioning systems available on the device (such as GPS, Galileo, and others)
- **LTE:** Long Term Evolution is a mobile communication standard. LTE is also referred to as “4G”, the fourth generation of mobile communication technology. LTE consists of a number of enhancements to the “Universal Mobile Telecommunications System” (UMTS) – the third generation of mobile communications.
- **URI:** A Uniform Resource Identifier is a string of characters used to identify or name a resource

on the internet. The pairing mechanism between smartphone and smartwatch is based on a custom URI scheme. The general format of the URI in this application is:

```
triplo://watch-pair/{watchId}?t={token}
```

This structure encodes:

- the unique identifier of the watch (`watchId`)
- a temporary token (`token`) used to validate the pairing request

The URI is embedded into a QR code displayed on the smartwatch.

1.4 References

1.4.1 Main References

- Design and Implementation of Mobile Applications Lectures and Slides
 - Introduction <https://webeep.polimi.it/mod/resource/view.php?id=399280>
 - Flutter <https://webeep.polimi.it/mod/resource/view.php?id=399287>
 - Android <https://webeep.polimi.it/mod/resource/view.php?id=411641>
 - iOS <https://webeep.polimi.it/mod/resource/view.php?id=417791>
 - Lectures <https://webeep.polimi.it/mod/url/view.php?id=376880>

1.4.2 Other References

- Flutter Documentation <https://docs.flutter.dev/>
- Firebase Documentation <https://firebase.google.com/docs/>
- OpenWeather API Documentation
https://openweathermap.org/api/one-call-3?collection=one_call_api_3.0
- Severe Weather Alerts API
<https://www.weatherbit.io/api/alerts>
- Oracle Documentation: What Is Model-View-Controller (MVC)?
<https://www.oracle.com/technical-resources/articles/javase/application-design-with-mvc.html>
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<https://www.telekom.com/en/company/details/the-nine-most-important-facts-about-lte-606>

2. Architectural Design

2.1 Overview

2.1.1 High-level components

The Triplo system is designed according to a modular architecture in which the main application logic is implemented on the client side, while backend functionalities are delegated to managed services and external APIs, as illustrated in Figures 2.1 and 2.2.

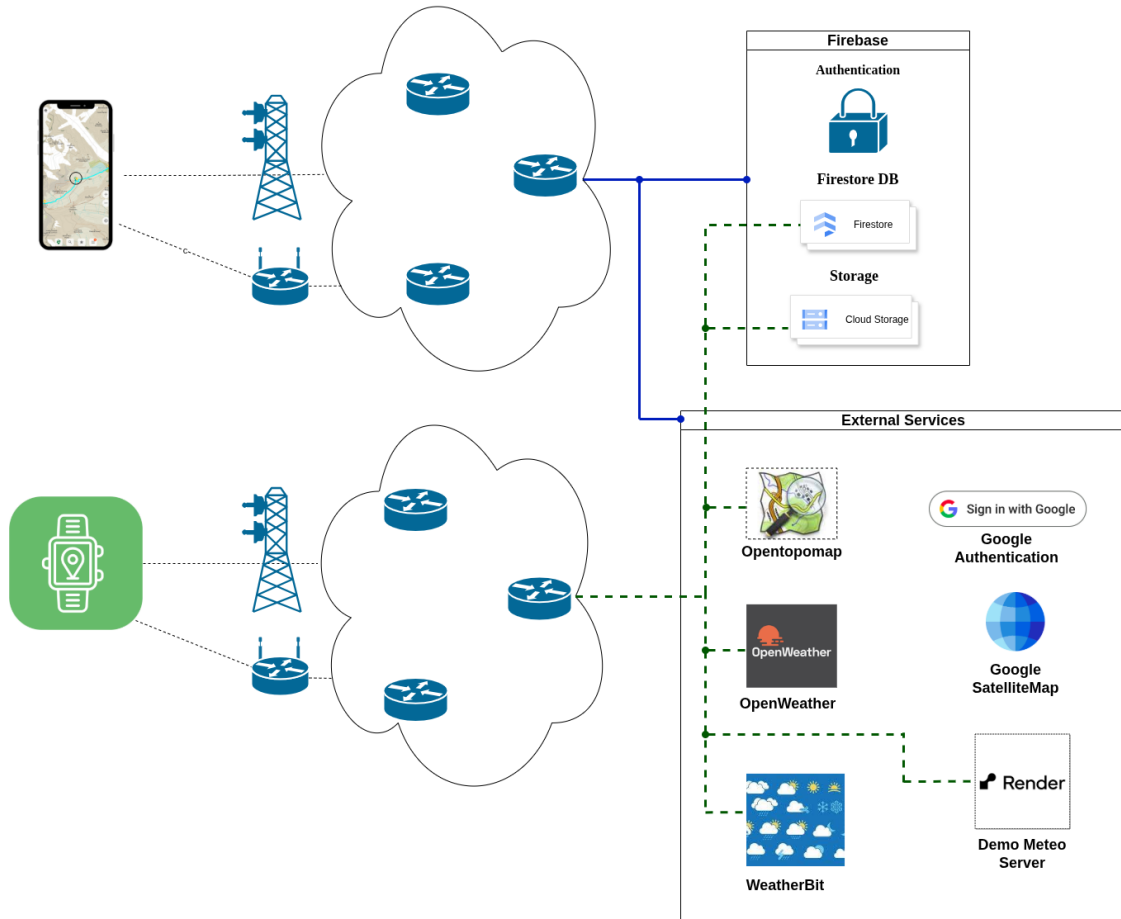


Figure 2.1: Overview Diagram: Blue lines denote the smartphone application's connectivity with Firebase and external services, providing full access to all Firebase modules and external APIs. Dashed green lines denote the smartwatch application's connectivity, which is limited to selected Firebase services, Firestore and Storage and specific external services, for example OpenTopoMap. The smartphone and smartwatch applications access the network independently, reflecting their ability to operate as standalone clients, while still relying on a shared set of backend services for data and functionality.

2.1.2 High level components interaction

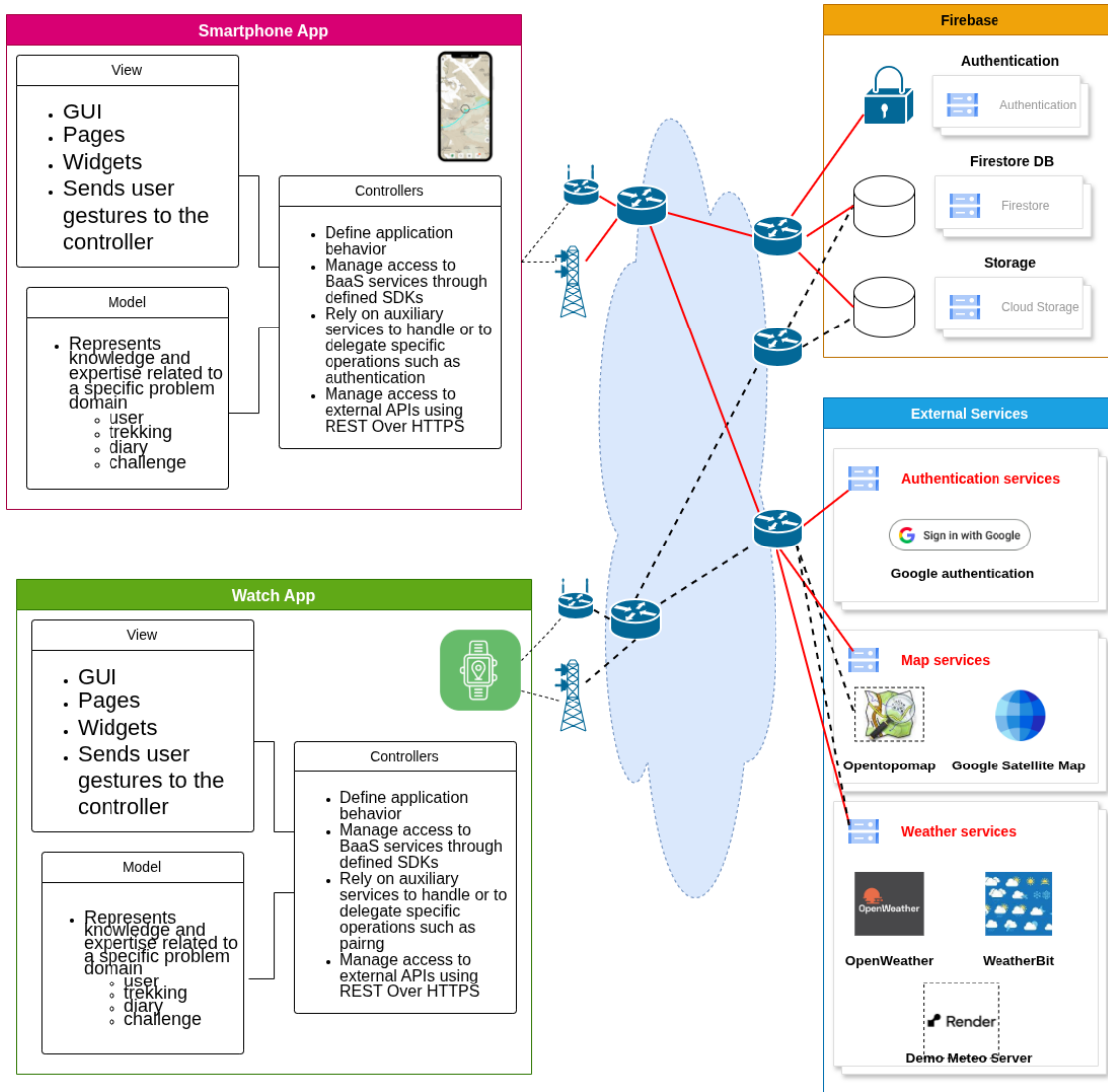


Figure 2.2: Detailed Overview Diagram

The system separates concerns between presentation, application logic, and data management by adopting a Model–View–Controller (MVC) architectural pattern. Both the presentation layer and the core application logic are implemented within the Flutter-based applications, while remaining logically decoupled through the use of Controllers that mediate between Views and Models.

In particular, the mobile application represents the main client, while the smartwatch application provides a lightweight extension offering a subset of functionalities adapted to the constraints and interaction model of wearable devices, relying on the same backend services and data sources.

User interactions are handled by the View layer, while business logic and coordination between components are delegated to Controllers, which act as the central mediation layer between the user interface and the underlying data models.

Data management and authentication are handled through Firebase services, specifically Firebase Authentication, Cloud Firestore, and Firebase Storage. The system leverages Firebase as a Backend-as-a-Service (BaaS) platform, providing managed services for authentication, data storage and user management, relying on platform-specific SDKs for access to these resources. The adoption of Backend-as-a-Service (BaaS) solutions enables the system to replace a traditional custom server-side layer with a more lightweight and scalable architecture. By delegating data management, authentication, and storage to managed services, architectural complexity is reduced by eliminating the need for maintaining a dedicated custom backend infrastructure. In addition to Firebase, the system integrates external third-party services (e.g., weather, map, and satellite APIs), which are accessed via RESTful APIs over

HTTPS.

This architectural approach emphasizes a client-driven design, reduces backend complexity, and enables scalability by leveraging managed cloud services.

This approach removes the need for a dedicated backend gateway or custom server-side components, allowing the application to directly orchestrate interactions with backend services and external resources.

2.2 Component view

2.2.1 High level component view

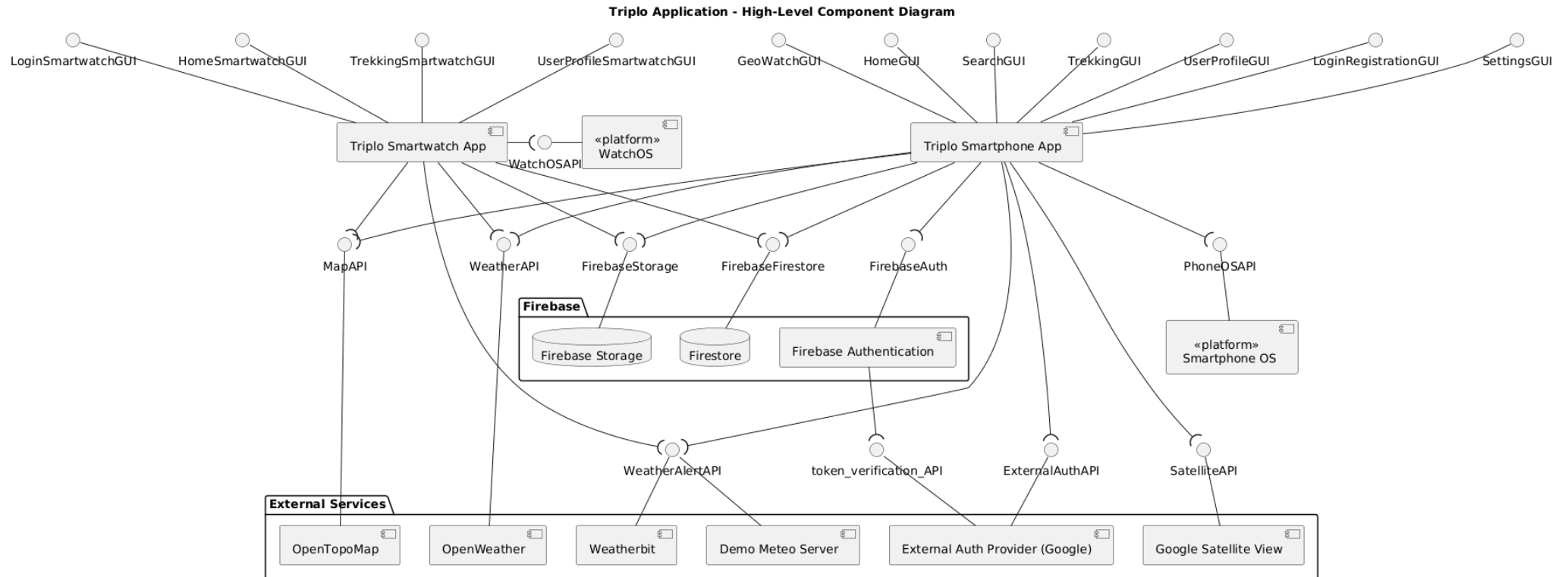


Figure 2.3: High level component view

The high-level component diagram provides an overview of the main architectural elements of the system and their interactions. The system is composed of two client applications, namely the smartphone application and the smartwatch application, both relying on a shared set of backend services and external resources.

The graphical user interfaces (GUIs) are represented following the convention adopted in both the *Software Engineering* and *UML Distilled* books quoted in the references. In this context, each GUI represents a group of related pages or a major functional area of the application, rather than a single page.

The diagram highlights the presence of two versions of the application. The smartphone application represents the main client and provides the full set of functionalities. It is implemented using Flutter, which enables cross-platform development and allows the application to run on both Android and iOS devices, as explained in section 2.8.

The smartwatch application represents a lightweight extension of the system and is deployed on Wear OS, as discussed in section 2.8. It provides a subset of functionalities adapted to the constraints of wearable devices.

Both applications interact with a common set of backend services and selected external resources. In particular, both the smartphone and smartwatch applications access Firestore for data storage and Firebase Storage for media management, as well as shared external services such as OpenTopoMap for map visualization. However, the smartwatch application is intentionally restricted to a subset of services, reflecting its role as a lightweight client and the constraints of wearable devices.

An architectural difference regards authentication. While the smartphone application directly relies on Firebase Authentication, the smartwatch application does not perform authentication independently. Instead, it adopts a pairing mechanism based on QR code scanning. Through this process, the smartphone securely associates an authenticated user with the smartwatch by storing the corresponding identifier in Firestore.

This design choice provides several advantages, detailed in section 2.8.3

Access to backend resources such as Firestore, Firebase Storage, and selected external services is otherwise shared between the smartphone and smartwatch applications. The diagram explicitly shows which services are accessible from each client, clarifying the distribution of responsibilities.

The system also integrates multiple external services, for example map providers such as OpenTopoMap and Google Satellite View, as well as external authentication providers. Weather alert functionalities rely on both Weatherbit and a Demo Meteo Server. The latter is used exclusively for demonstration purposes, allowing the simulation of weather alerts without depending on real external conditions.

The operating system interfaces are modeled as platform components, Smartphone OS and Watch OS. These platforms are not accessed directly by the application; instead, access is mediated through Flutter packages that provide abstractions over native operating system services. The specific packages used to interact with such services are detailed in the diagrams of the smartphone application components, section 2.2.2 and of the smartwatch application components, section 2.2.3.

The interfaces associated with Firebase services, Firebase Authentication, Firestore, and Firebase Storage are based on the SDKs provided by the Backend-as-a-Service platform. These SDKs abstract the underlying communication mechanisms and offer a structured way to access backend functionalities, as documented in the Firebase official documentation.

In contrast, the remaining external interfaces, such as Weather API, Weather Alert API, Satellite API, are accessed through RESTful communication. In these cases, the application uses HTTP-based requests via Flutter libraries, handling data exchange in JSON format and performing client-side decoding.

The token verification API is not directly managed within the application logic but is represented the complete authentication flow. In particular, when authentication is performed through an external provider, Google, the application obtains an authentication token which is then passed to Firebase Authentication. Firebase subsequently validates this token with the external provider as part of the OAuth2-based authentication process. This additional interaction is abstracted by the SDK but is included in the diagram to provide a more accurate representation of the overall system behavior.

2.2.2 Mobile Smartphone App component view

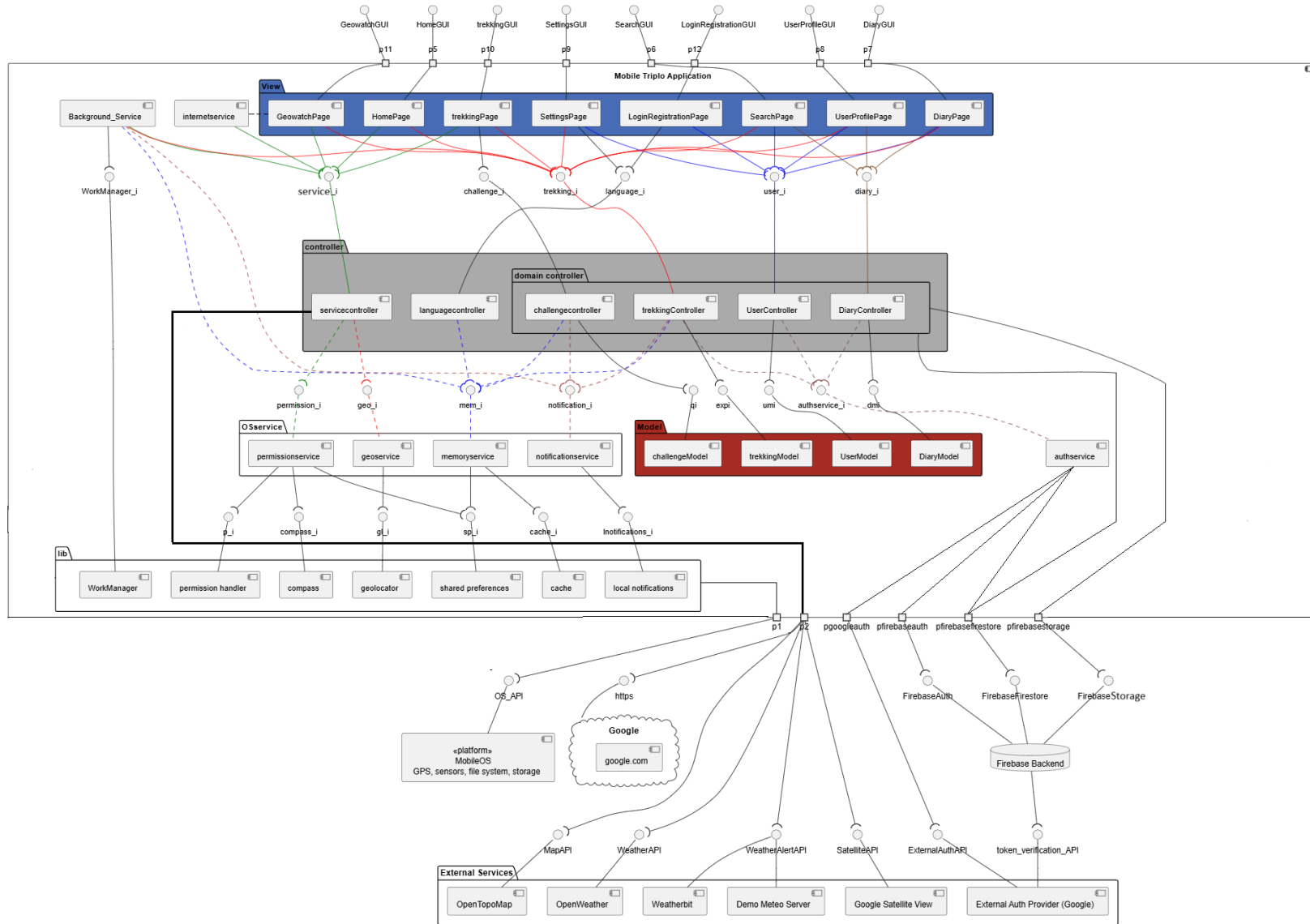


Figure 2.4: Smartphone Application component view

View Components

The UI of the application is organized into components that group together multiple views related to the same functionality, rather than representing isolated pages. This choice reflects how users interact with the system, as each View component encapsulates a complete feature.

The **GeoWatch Page component** includes all functionalities related to navigation and environmental awareness during a trekking activity. It is responsible for presenting weather information along the route, weather alerts, and a satellite map with meteorological layers. It also provides navigation support through compass heading, altitude, and user position.

The **Home Page component** represents the main map-based point of the application. It displays trekking routes on an OpenTopoMap layer and allows users to explore available paths geographically.

The **Trekking Page component** provides detailed information about a selected route, including a description and photos. It also allows users to access the component that handles the creation of a diary associated with the route and to save the route to their profile, the **Diary Page component**.

The **Diary Page component** handles the creation and modification of diary entries related to trekking experiences, including notes, images, and activity-related data.

The **User Profile Page component** is responsible for displaying user information, including saved trekkings, diary related, follower and following users' data.

The **Settings Page component** allows users to manage application and account settings, including profile information, security options such as password updates, smartwatch pairing, and weather notification preferences.

The **LoginRegistration Page component** manages login and registration workflows. It supports authentication via email and password, performs simple format validation operations, supports Google authentication, and password recovery.

The **Search Page component** enables users to search for other users and access their public profiles or to search for a trekking by its name.

Controller components

The controller package coordinates the application logic and mediates interactions between UI components, models, and backend services.

Domain controllers, such as **UserController**, **DiaryController**, **TrekkingController**, and **ChallengeController**, are responsible for handling use cases specific to their domain. They orchestrate data flow between the UI and Firebase services.

The **ServiceController** acts as a unified facade for functionalities that do not belong to an application domain defined in the model, such as access to external APIs and operating system services, reducing coupling between UI components and low-level implementations.

The **LanguageController** is responsible for managing localization and language-related logic.

OSService Package components

The OSService package contains services that directly interact with Flutter libraries responsible to communicate with system APIs and services managed by the operating system.

These include components such as **GeoService**, **PermissionService**, **MemoryService**, and **NotificationService**, each encapsulating a specific interaction with the operating system, such as location access, permission handling, local storage, and notifications.

This package isolates dependencies on external libraries and platform APIs. If a library needs to be replaced or updated, the changes are confined to the corresponding service, without affecting the rest of the application.

Internet Service

The **InternetService** is responsible for monitoring network connectivity. It periodically relies on the **ServiceController** to attempt connections to external endpoints (e.g., Google services). Based on the outcome, the application switches between offline and online modes, ensuring consistent behavior in varying network conditions.

Background Service

The `BackgroundService` enables the execution of tasks even when the application is not actively open, depending on operating system constraints.

It is used to perform periodic operations such as checking weather conditions and generating notifications. These tasks are scheduled through the mechanisms `WorkManager` mechanism, a Flutter wrapper around Android's `Workmanager` and iOS Background tasks. Background tasks are executed based on system availability.

AuthService

The `AuthService` is responsible not only for authentication but also for managing the smartwatch pairing process.

It handles user authentication through `Firebase Auth` and `Google`, password reset, session management, coordinates the pairing workflow between the smartphone application and the smartwatch, centralizing all identity-related logic.

Backend and External Services

The application relies on `Firebase` as a `Backend-as-a-Service`, including `Authentication`, `Firestore`, and `Storage`.

External services are used for specific functionalities, such as `OpenTopoMap` for map visualization, `OpenWeather` and `Weatherbit` for weather data and alerts, and `Google` services for authentication and connectivity verification.

Architectural Summary

The architecture is organized in packages that separate concerns clearly. Domain controllers manage application logic and backend interaction, the `ServiceController` provides a unified access point to external and system services, and `OSService` components encapsulate platform dependencies. This structure improves modularity, maintainability, and adaptability of the system.

Domain controllers, such as `UserController`, `DiaryController`, `TrekkingController`, and `ChallengeCon` are responsible for coordinating the application logic related to their respective functional domains. These controllers manage interactions between the View layer and the Model layer, and they also mediate access to backend resources such as `Firebase` services.

Therefore, in the proposed architecture:

- domain controllers manage the application use cases and the interaction with `Firebase`-related resources;
- The `ServiceController` acts as a unified facade aggregating multiple underlying services, including external APIs (weather, maps) and device-level services (location, permissions, navigation)
- `Firebase` is modeled as a set of backend services used by the application, rather than as a custom backend.

Note: the `GeoWatch` component lightly depends on the `InternetService` to determine the current connectivity state, only the *Navigation Page* with the compass and altimeter depends on it just for hiding the drawer menu in offline mode. This interaction is not based on direct method invocation, but the `InternetService` exposes a boolean connectivity state, which is observed by the UI through a provider-based mechanism. When the connectivity status changes, the service notifies listeners, and the UI updates accordingly. Since it is not a method invocation, this has not been modelled as an interface but just as a light dependency with a dashed line in the diagram.

Package	Purpose
flutter	Core UI toolkit for building cross-platform applications
provider	State management and implementation of the Observer pattern
firebase_core	Initialization of Firebase services
firebase_auth	User authentication management
cloud_firestore	Firebase cloud database for application data
firebase_storage	Cloud storage for images and media files
google_sign_in	Google authentication integration
flutter_map	Map rendering using OpenStreetMap tiles
latlong2	Geographic coordinate utilities
geolocator	Device location services
geocoding	Address and location conversion
flutter_compass	Compass and orientation data
image_picker	Image selection from device gallery or camera
mobile_scanner	QR code scanning functionality
shared_preferences	Local persistent key-value storage
flutter_local_notifications	Local notification management
workmanager	Background tasks
permission_handler	Runtime permission handling
flutter_dotenv	Environment variable management
uuid	Unique identifiers
intl	Internationalization and localization utilities
share_plus	Content sharing across apps
flutter_cache_manager	Image and file caching
cupertino_icons	iOS-style icon set

Table 2.1: Main dependencies used in the smartphone application

2.2.3 Mobile Smartwatch App component view

Navigation Page Disambiguation

The term *Navigation Page* is used in both the smartphone and smartwatch applications, but it refers to different UIs with distinct responsibilities. In the smartphone application, the Navigation Page corresponds to a view within the `GeoWatchPage` component. It provides sensor-based navigation functionalities, such as compass orientation and altitude information, supporting the user during outdoor activities. In the smartwatch application, the Navigation Page is a standalone component (`NavigationPage`) that acts as a central access point for the main features of the wearable app. It allows the user to navigate to the Home Page, User Page, and language-related functionalities, serving primarily as a presentation hub. This distinction is important to avoid ambiguity when referring to navigation-related components across the two platforms.

2.3 Firebase BaaS Management

Imagine Database

The database adopted for the Triplo application is based on a NoSQL approach using Firebase Firestore. Unlike traditional relational databases, Firestore stores data as collections of JSON-like documents, where each document can contain nested structures and arrays.

For documentation purposes, the database structure has been represented using a diagram similar to a relational schema. This representation has been introduced for clarity and readability, allowing an easier understanding of entities and their relationships, even though the underlying data model is document-oriented rather than relational.

The main collections include `user`, `diary`, `trekking`, `challenge`, and auxiliary collections such as `user_index` and `trekking_index`, which are used to support efficient search operations. Additional collections such as `watch`, `watch_pair`, and `weather_notification` handle device pairing and notification functionalities. From an infrastructure perspective, the Firestore database is deployed in the *europa-west 8* region, while Firebase Storage is currently hosted in the United States. This configuration may introduce additional latency when retrieving large media assets such as images. A potential improvement would be to migrate storage resources to a European region, which is supported by Firebase through paid plans, in order to reduce latency and improve performance.

To mitigate latency issues, especially for images retrieved from Firebase Storage, a caching strategy has been adopted. Some images are locally cached on the device, reducing repeated network requests and improving user experience. This design decision is further discussed in [Section 2.8.8](#).

2.4 Deployment view

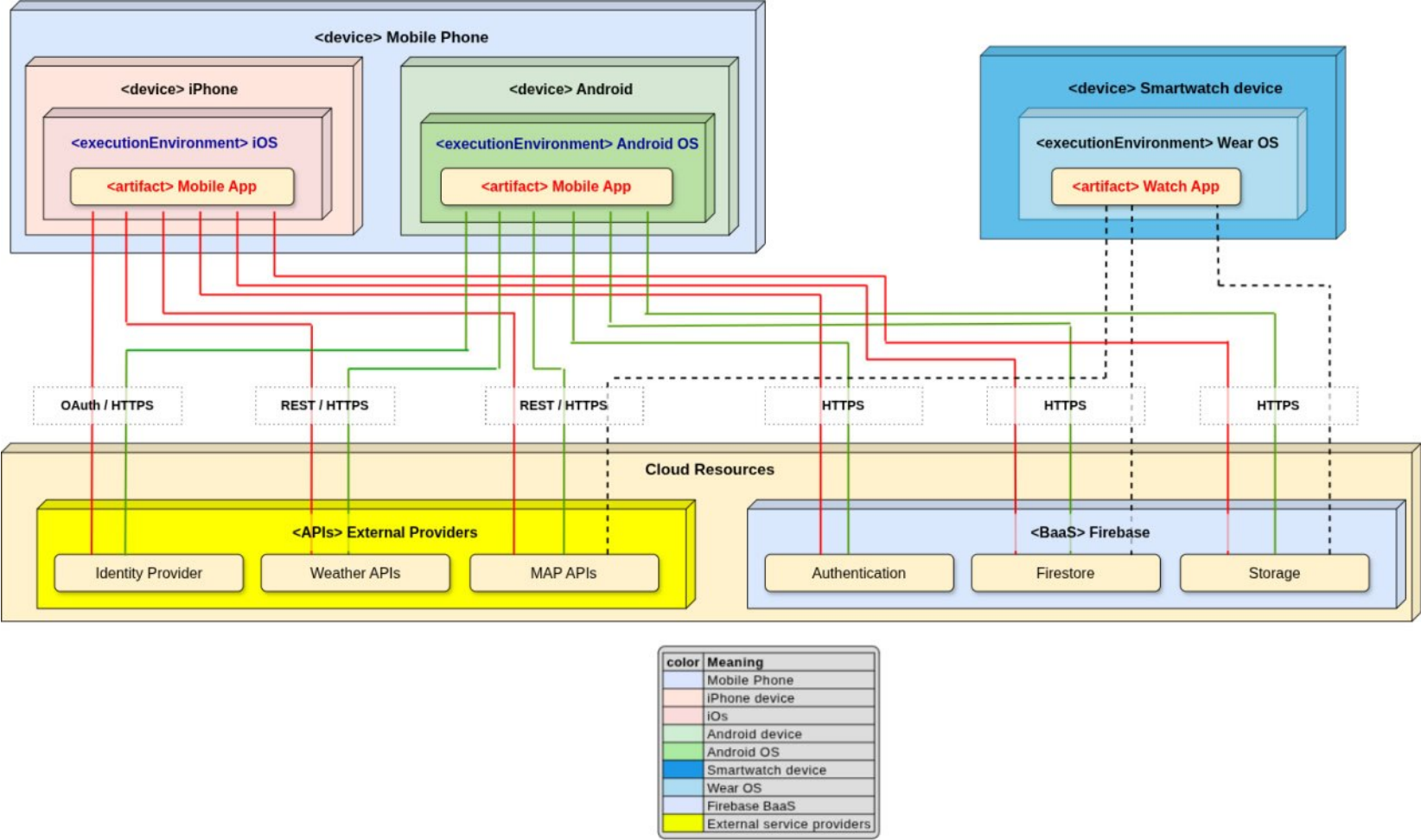


Figure 2.5: Deplyment Diagram

The deployment diagram is consistent with the UML deployment notation described in chapter 8 of the UML Distilled book and on page 52 of the Software Engineering book listed in the references. Physical hardware units are modeled through the `«device»` stereotype, while software platforms and runtime contexts are represented through the `«executionEnvironment»` stereotype.

The deployed software elements are shown as `«artifact»`. Communication paths are represented by links annotated with the corresponding protocol or interaction type, such as HTTPS, REST/HTTPS, and OAuth/HTTPS.

In particular, the diagram distinguishes the mobile deployment on Android and iOS devices, while the smartwatch deployment is limited to Wear OS. The cloud side is modeled as a device containing different execution environments, namely Firebase and the set of external providers, which host the artifacts used by the applications.

The deployment diagram represents the distribution of the application artifacts across different execution environments. In particular, the smartphone application is deployed as an artifact on both iOS and Android execution environments, while the smartwatch application is deployed separately on a Wear OS execution environment.

Although the smartphone application is shown as deployed on two distinct platforms, it is important to note that it is based on a single shared codebase. The application is developed using Flutter, a cross-platform framework that allows the same source code to be compiled and executed on both iOS and Android devices. As a result, the two deployments correspond to different execution environments of the same application, rather than distinct implementations. In contrast, the smartwatch application is implemented as a separate artifact. While it shares the same backend services and part of the architectural structure, it is specifically designed for the Wear OS environment and provides a reduced set of functionalities for wearable devices. Therefore, it is represented as a distinct deployment unit in the diagram.

2.5 Runtime view

The identified use cases provide an abstract representation of the interactions between the user and the system. They have been selected to capture the essential functionalities of the application and to model its observable behavior from the user's perspective.

This set of use cases is representative of the main system capabilities and supports a clear understanding of the services offered to the user.

The reported use cases are twelve in total. Some of them include sub-diagrams to represent important phases of the respective use case or to provide a more detailed view of key processes. Although these sub-diagrams are not always strictly use cases, they are intended as supporting diagrams to improve the clarity of the system's runtime behavior within the documentation

1. Add Diary Page - Section [2.5.1](#)
2. Explore trekkings on Map - Section [2.5.2](#)
3. View trekking information -Section [2.5.3](#)
 - Image caching for trekking images
4. View weather information - Section [2.5.4](#)
 - View Satellite Weather Information
5. Search Users - Section [2.5.5](#)
6. Search trekking - Section [2.5.6](#)
7. View compass and altitude information - Section [2.5.7](#)
8. Login - Section [2.5.8](#)
9. View User Information - Section [2.5.9](#)
10. Register - Section [2.5.10](#)
11. Update User Information - Section [2.5.11](#)
 - change language
12. Phone Watch Pair - Section [2.5.12](#)
 - Phone-side Smartwatch Pairing
 - Watch bootstrap
 - Watch-side Smartwatch Pairing

Use Case	Device	Notes
Login / Register	Smartphone	Full authentication flow managed on smartphone
Explore Trekkings on Map	Smartphone Smartwatch	Smartphone used as reference for sequence diagram
View Trekking Information	Smartphone Smartwatch	Smartphone used as reference for sequence diagram
View Weather Information	Smartphone Smartwatch	Smartphone used as reference for sequence diagram
Add Diary Page	Smartphone	
Search Users / Trekking	Smartphone	Search functionalities handled on smartphone
View Compass and Altitude Information	Smartphone	
Phone Watch Pair	Smartphone Smartwatch	Process modelled with phone-side and watch-side sequence diagrams
View User Information	Smartphone Smartwatch	Smartphone used as reference for sequence diagram
Update User Information	Smartphone	Profile management handled on smartphone

Table 2.2: Comparison of use cases between smartphone and smartwatch

2.5.1 Add Diary Page use case sequence diagram

The most important part of the *Add Diary Page* sequence diagram is the save phase, where the data entered by the user is stored in the system. When the user presses the save button, the `DiaryPage` starts the process by showing a loading page and then it formats the date, computing the duration from hours and minutes. If the user has selected photos, the page calls the `DiaryController` to upload them to `FirestoreStorage`. The controller uploads each image and returns a list of paths, which will be stored in the diary instance.

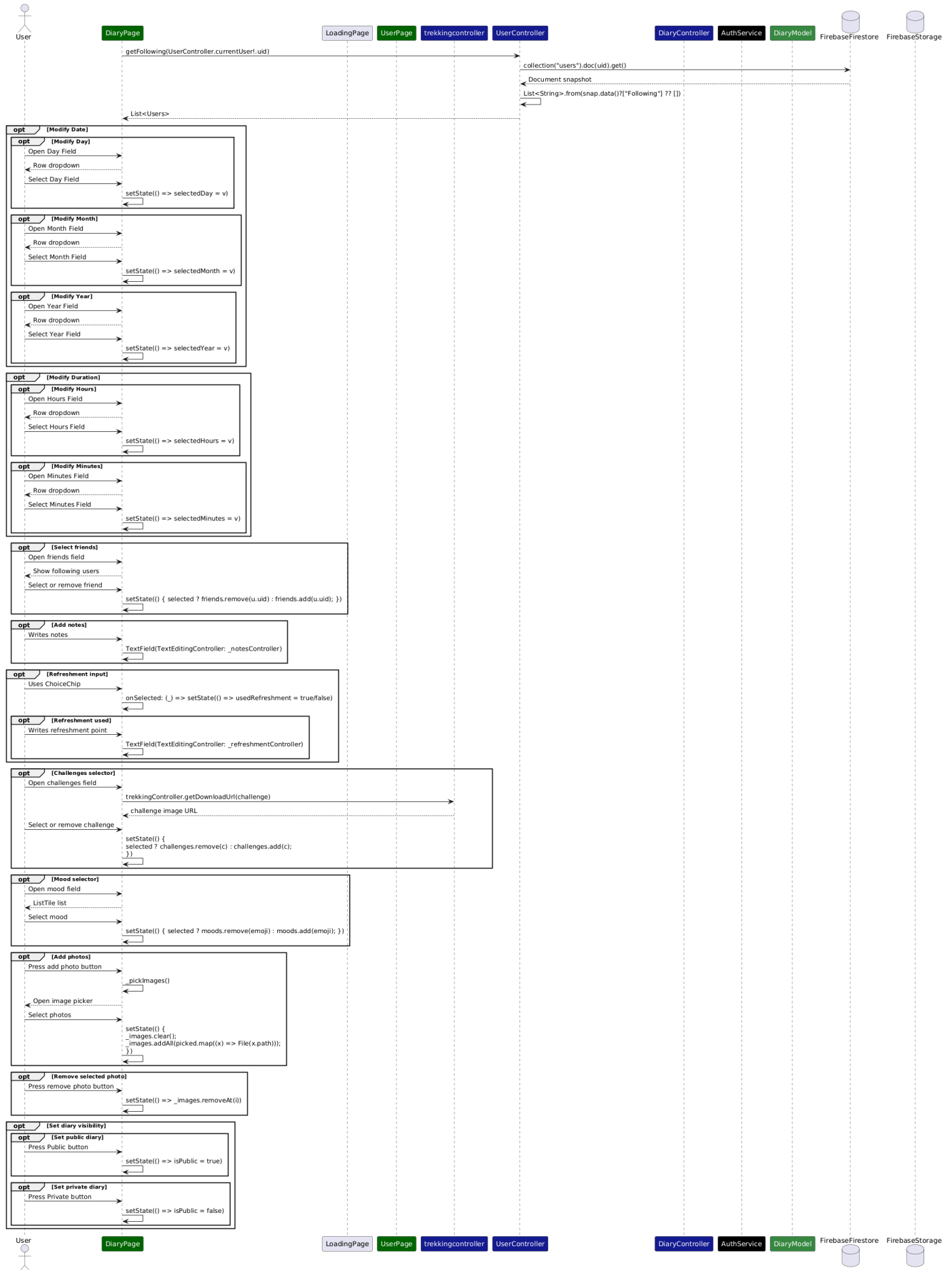
After this step, the `DiaryPage` calls the `addDiary()` method of the `DiaryController`, passing all the collected data. The controller retrieves the current user identifier from `FirebaseAuth`, generates a new unique identifier for the diary, and creates a `Diary` object.

Before saving the diary in `Firestore`, the object is converted into a `Map<String, dynamic>`. This is necessary because Firestore stores data in a JSON-like format, which is based on key-value pairs. The map represents the diary as a set of fields (such as date, duration, notes, etc.), where each field is associated with a value. The type `dynamic` is used because these values can have different types, such as strings, numbers, booleans, or lists.

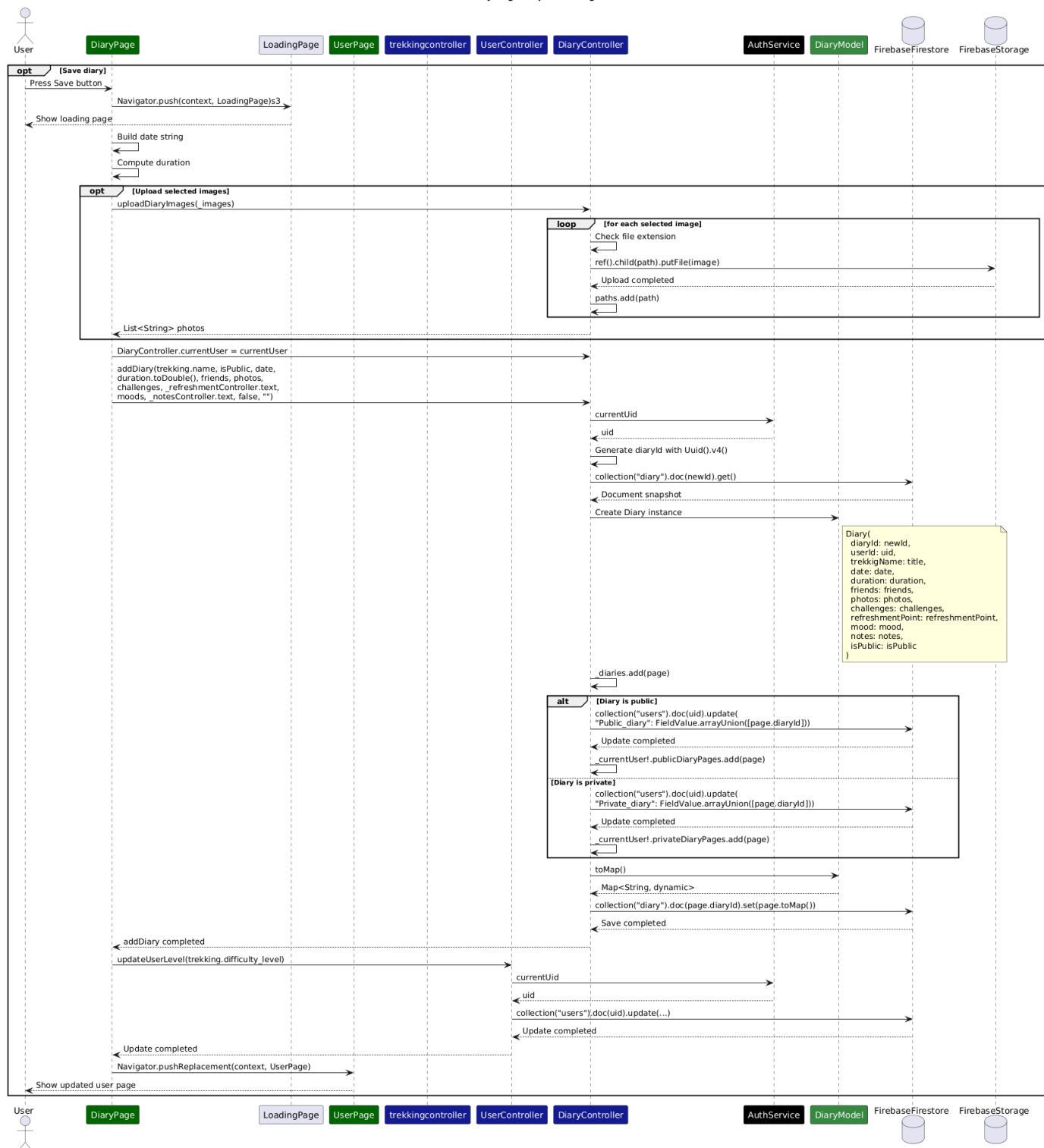
Once converted, the map is stored in the `diary` collection in Firestore. At the same time, the controller updates the user document, adding the diary identifier to either the public or private diary list, depending on the selected visibility.

Finally, the control returns to the `DiaryPage`, which updates the user level through the `UserController` and navigates to the `UserPage`, completing the operation.

Add Diary Page Sequence Diagram

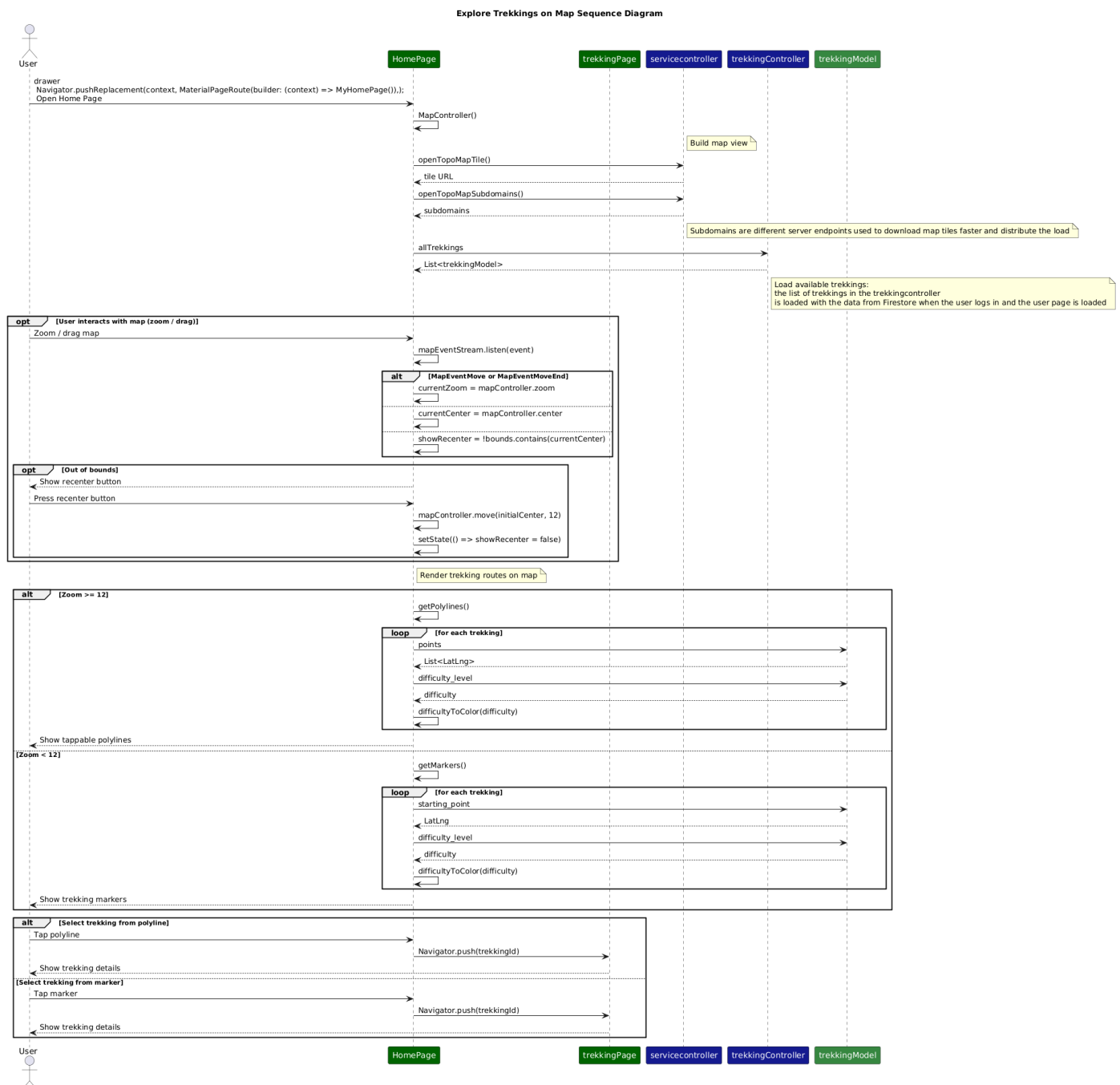


Add Diary Page Sequence Diagram



2.5.2 Explore Trekkings on Map Sequence Diagram

The sequence diagram for the *Explore Trekkings on Map* use case describes how the user interacts with the main map interface in order to visualize the available trekking routes. The interaction starts when the user opens the `HomePage`, which initializes the map component and requests the information needed to render the map background through the `servicecontroller`. In particular, the page retrieves the tile URL template and the corresponding subdomains used by the map provider. Once the map has been initialized, the `HomePage` accesses the trekking data exposed by the `trekkingController`. The available trekkings are then rendered differently depending on the zoom level and the rendering logic uses the trekking information, such as route points, starting point, and difficulty level, in order to determine the visual representation. The diagram also highlights the dynamic behavior of the map during user interaction. When the user drags or zooms the map, the page updates the current zoom and center, and checks whether the visible area is still inside the bounds. If the user moves outside the expected area, a recenter button is shown. Finally, when the user selects a trekking, the system navigates from the `HomePage` to the `trekkingPage`, where the detailed trekking information is displayed.



2.5.3 View Trekking Information Sequence Diagram

The sequence diagram for the *View Trekking Information* use case describes how the system presents the details of a selected trekking. The interaction starts when the user opens the `trekkingPage`. The page initializes its internal state and verifies which user is authenticated through the `UserController`. The trekking data is then retrieved through the `trekkingController`. In particular, the controller attempts to search in its local collection.

After the trekking has been obtained, the page checks whether the route is already saved by the current user.

The page also requests weather information near the trail through the `servicecontroller`.

A further part of the interaction regards the trekking images, such as the map photo, the ending point photo, and the challenge images. In the main sequence diagram, this step is represented at a high level through calls from the `trekkingPage` to the `trekkingController`. The detailed behavior of the cache mechanism used for image retrieval is described separately in a separate diagram, in order to keep the main diagram focused on the overall interaction flow.

Finally, the page allows the user to perform additional actions related to the trekking. The route can be saved or removed from the saved list, a diary entry can be created through the `DiaryPage`, and the user can navigate to other trekking-related views such as the route details page or the geowatch and map page. In this way, the `trekkingPage` acts as the central entry point for the main information and actions associated with a trekking.

Image Caching for Trekking Images Sequence Diagram

This sequence diagram illustrates, through a representative example, how the image caching mechanism is implemented within the application. Its purpose is not to introduce a new use case, but rather to clarify the internal behavior of image retrieval, which is abstracted in the main sequence diagram describing the trekking visualization flow.

The process starts when the `trekkingPage` requests an image through the `trekkingController`. The controller delegates the retrieval to the `memoryservice`, which implements a multi-level caching strategy.

First, the system attempts to retrieve the image from the in-memory RAM cache, using a `Map<String, File>` structure. If the image is found, it is immediately returned to the page, ensuring minimal latency. If the image is not available in memory, the system checks the disk cache through the `CacheManager`. This involves interacting with the underlying operating system to load a previously stored file. If the image is found on disk, it is reinserted into the in-memory cache and returned to the UI, improving performance for subsequent accesses.

If the image is not available in either RAM or disk cache, the system performs a download from `FirestoreStorage`. In the case of paths expressed using the `gs://` protocol, the controller first resolves them into a downloadable URL. The image is then downloaded, stored on disk, and finally cached in memory before being returned to the page.

The same logic is applied in a batched manner when multiple images, for example trekking challenges, are requested. In this case, the controller iterates over all image paths and applies the same caching strategy to each element.

View Trekking Information Sequence Diagram

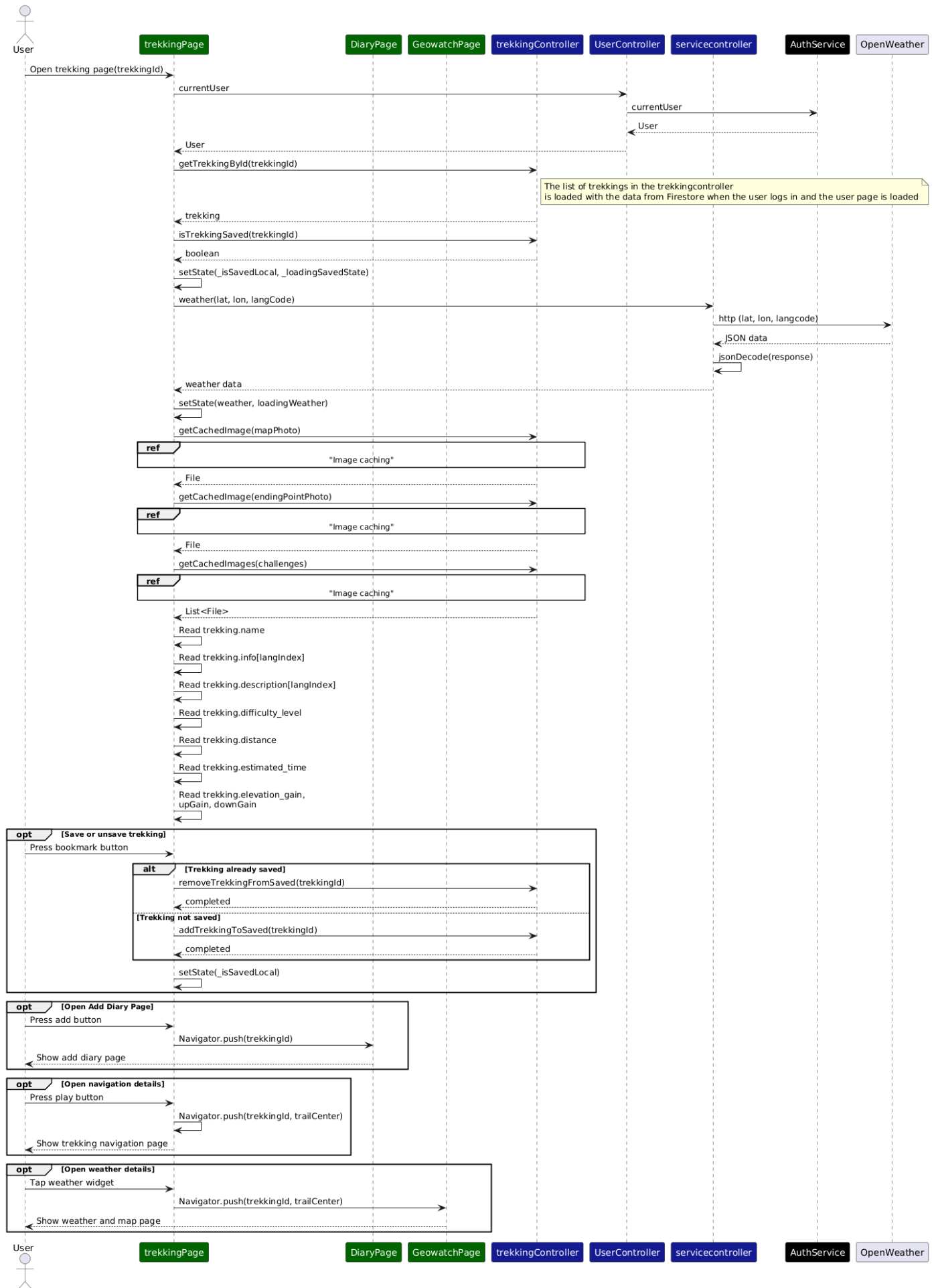
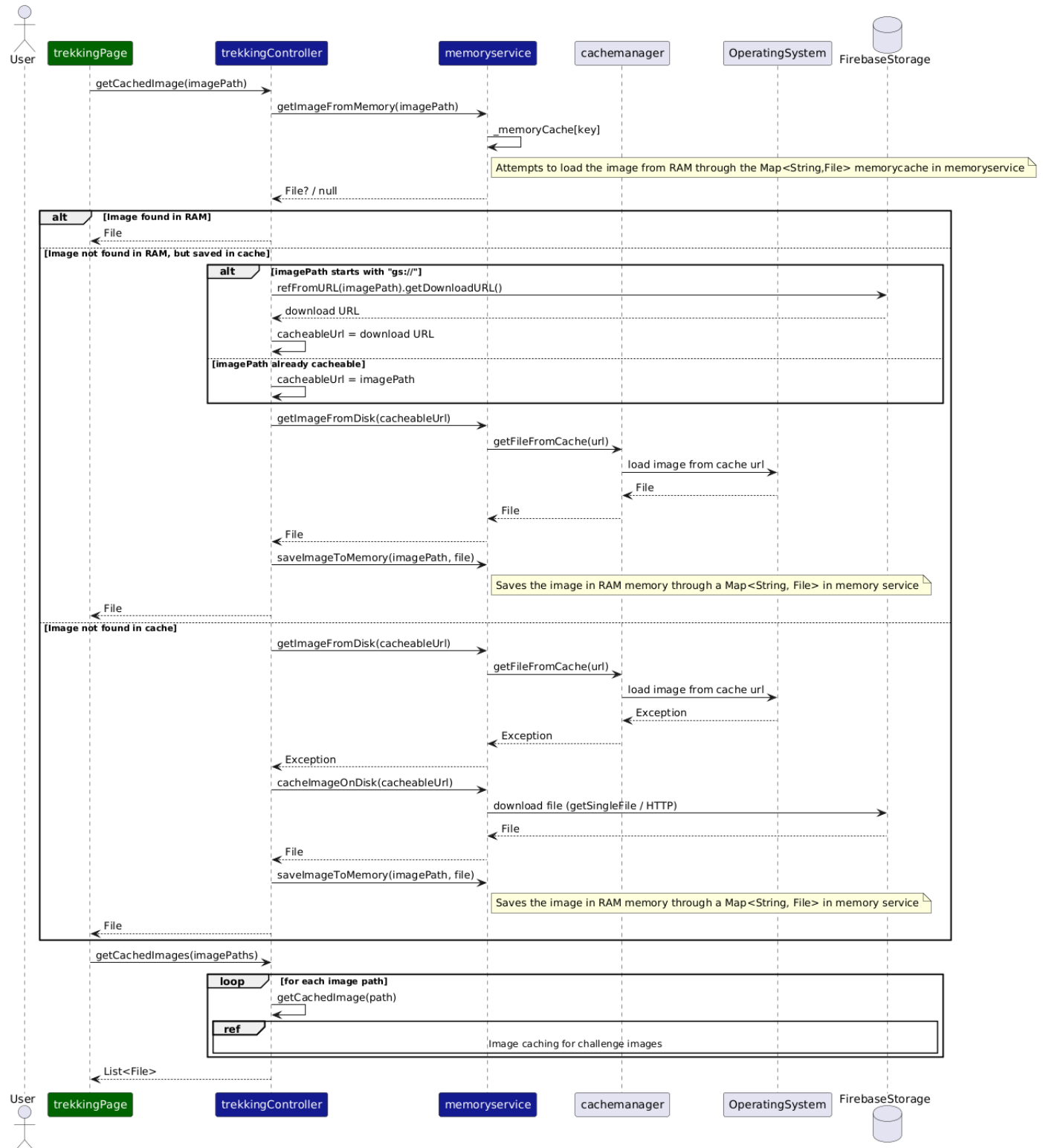


Image caching for trekking images sequence diagram



2.5.4 View Weather Information Sequence Diagram

The **Geowatch** component is responsible for retrieving, processing, and displaying weather information related to a trekking route or the user's current position. This functionality is implemented through the **ServiceController**, which acts as a facade for external APIs. The main logic is handled by the method `_loadAll()`, which is responsible for orchestrating the full weather retrieval process. This method:

- determines the target location (trekking route or user GPS position),
- retrieves current weather data,
- retrieves forecast data,
- parses the forecast into a simplified structure,
- retrieves weather alerts.

The method is invoked:

- once during initialization,
- periodically through a timer (every 5 minutes),
- when the user switches between trail-based and GPS-based weather.

The method `_loadAlertState()` is responsible for retrieving the current notification status for the selected trekking route. It queries the **TrekkingController**, which:

- retrieves the current user identifier through **AuthService**,
- queries Firestore to check whether the trekking is present in the user's alert list.

The result determines the state of the notification icon in the UI. The method `weather(lat, lon, lang)` performs an HTTP request to the OpenWeather API. The JSON response is decoded into a `Map<String, dynamic>` and sent over to the UI. The method `forecast(lat, lon, lang)` retrieves raw forecast data from OpenWeather. Since the response contains multiple time-based entries, a parsing phase is required. This parsing is handled by the following methods in the **ServiceController**:

- `parseForecastItem(raw)`: converts a single raw JSON entry into a simplified structure (date, temperature, icon, description),
- `parseForecast(rawList)`: applies the transformation to all entries.

Weather alerts are obtained from two sources:

- `weatherbitAlerts(lat, lon)` for real-time alerts,
- `mockAlerts()` for testing purposes.

The results are merged within the UI layer to provide a unified alert visualization. The **GoogleSatellitePage** part of the **Geowatch** component, provides an alternative map visualization using satellite imagery. The page retrieves tile URLs through the **ServiceController** and optionally overlays weather layers, such as pressure, rain and others using OpenWeather tile services. Weather data is handled using `Map<String, dynamic>` structures. This allows flexible interaction with JSON responses, avoiding the need for rigid model definitions.

Handling Asynchronous Updates in Dynamic Pages The `mounted` property indicates whether the current page is still visible and active. When a page is removed (for example, after navigating to another screen), it is no longer safe to update its UI.

Since weather data is loaded asynchronously, it may happen that:

- a request is started,
- the user leaves the page,
- the request completes afterward.

If `setState` is called after the page has been removed, the application would crash.

For this reason, the code checks `if (!mounted) return;` before updating the UI, ensuring that updates are performed only when the page is still active.

Google Satellite integration The Google Satellite page does not directly perform HTTP requests to OpenWeather. Instead, it delegates the construction of the tile URL to the `ServiceController`.

The base map is not a single image but is composed of multiple small image tiles. The `ServiceController` provides a URL template for the map provider, for example Google Satellite for Geowatch or OpenTopoMap for the `HomePage`.

This URL contains placeholders such as `{z}`, `{x}`, and `{y}`, which represent:

- the zoom level (`z`)
- the horizontal tile index (`x`)
- the vertical tile index (`y`)

The map component replaces these placeholders and loads only the tiles needed for the current view. When the user moves or zooms the map, new tiles are requested dynamically.

Architectural Separation: UI and API Responsibilities The UI does not directly perform HTTP requests for weather data. Instead, it delegates all external API interactions to the `ServiceController`.

An exception is the map rendering system, for both this map and the map in the home page, sequence diagram 2.5.2: once a tile URL template is provided, the map component internally handles tile requests to external providers. This behavior is part of the rendering library `FlutterMap` and does not violate the architectural separation, as the application logic does not manage these requests.

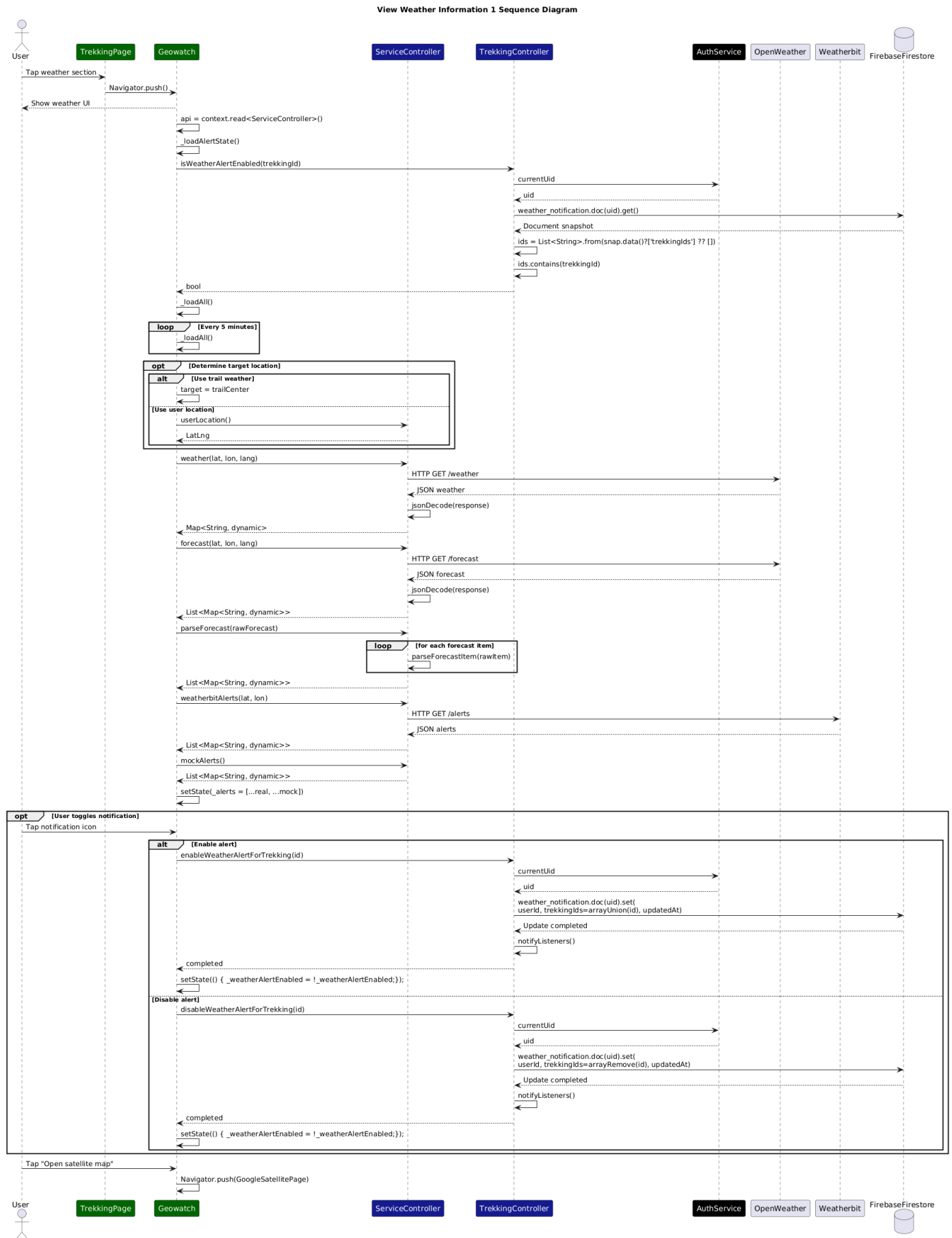


Figure 2.6: View Weather Information sequence diagram

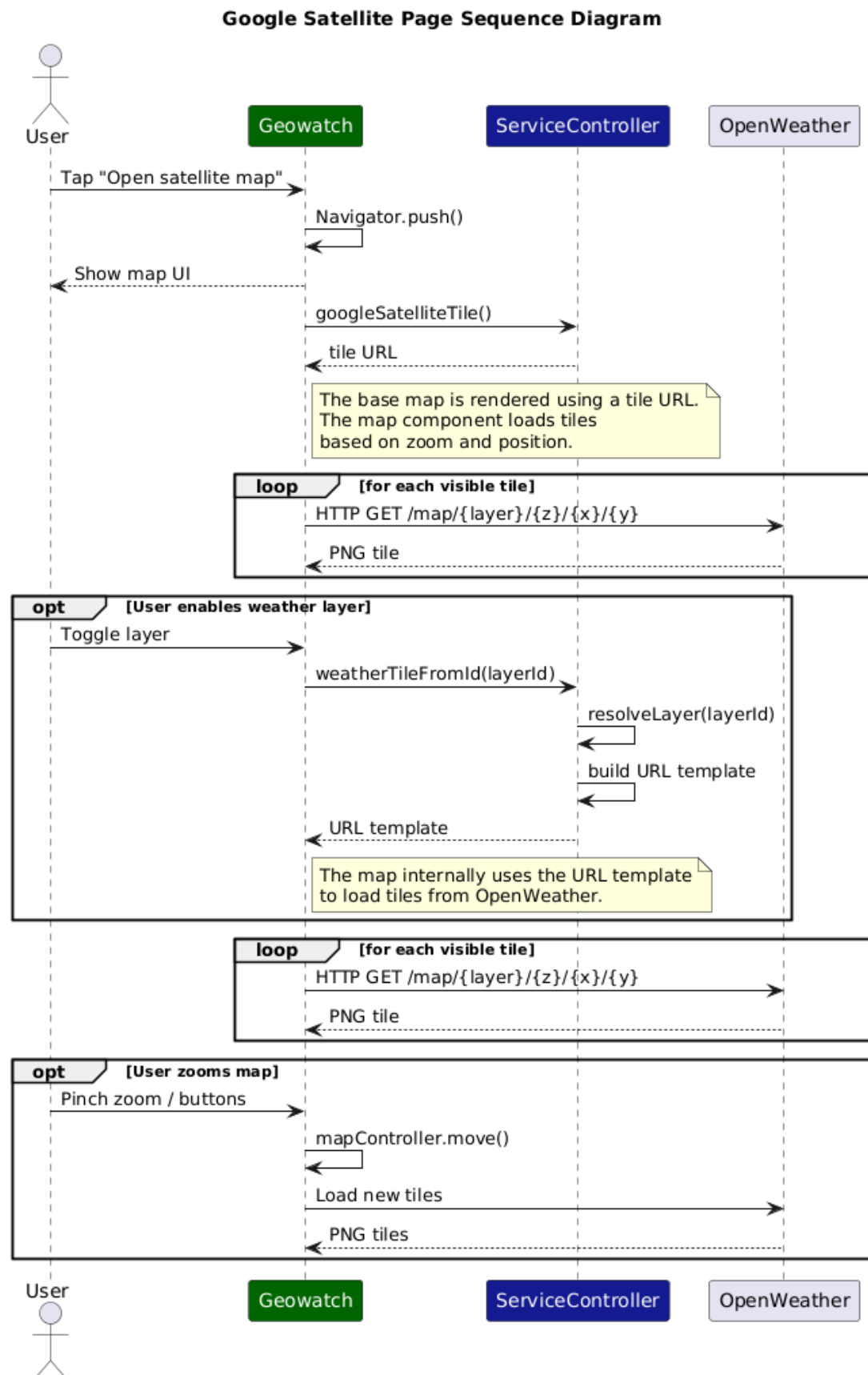


Figure 2.7: View Satellite Weather Information sequence diagram

2.5.5 Search Users Sequence Diagram

The SearchPage component is responsible for handling user search interactions. The UI elements belong to the SearchPage component and communicate with the UserController.

When the user types a query, the method `_runSearch(query)` is run by the User controller. Before performing the search, the UI state is updated using `setState` to indicate that a search operation is in progress.

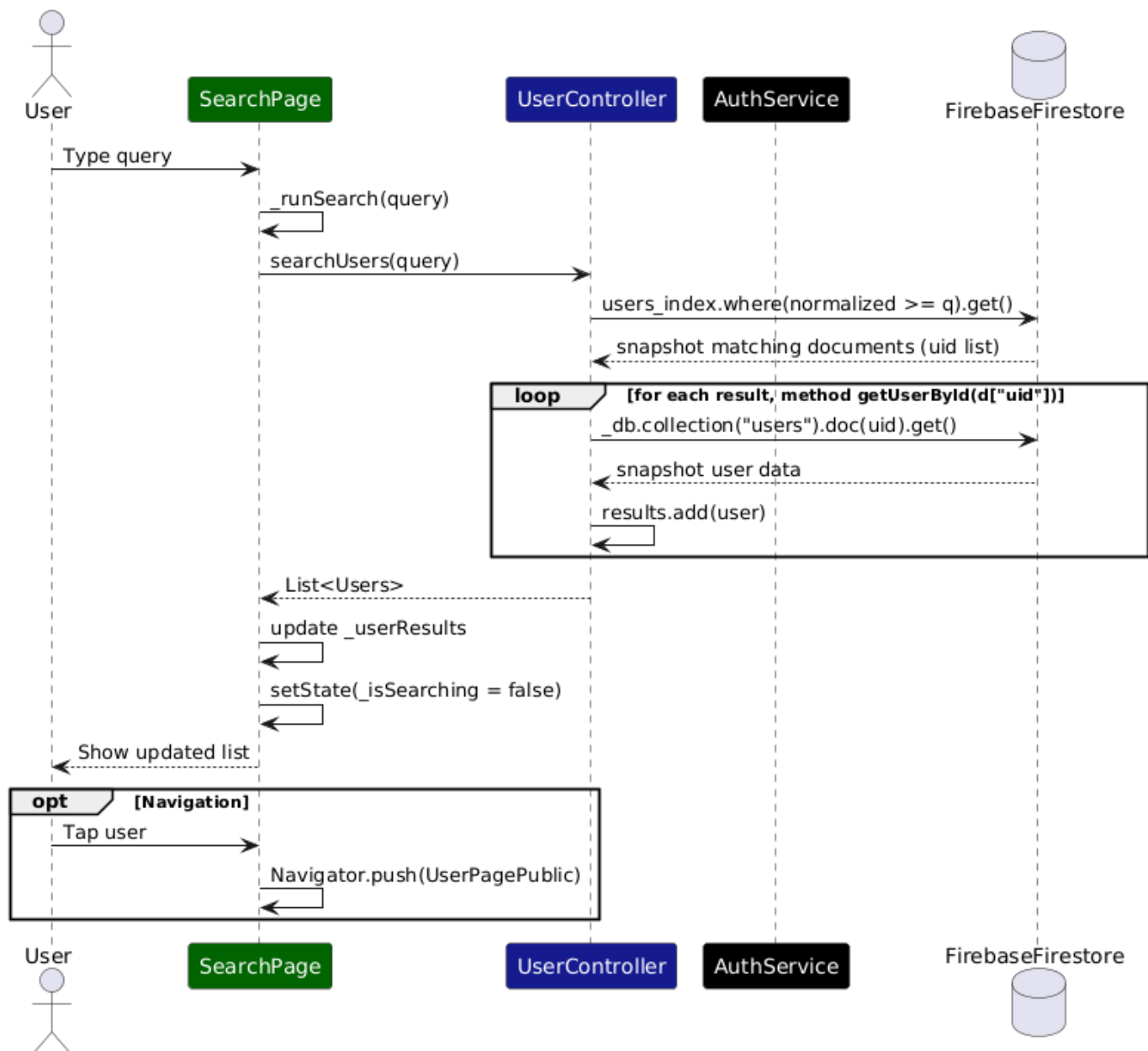
The method `searchUsers(query)` is then invoked in the UserController.

The search process is performed in two steps:

- a query is executed on the `users_index` collection using a normalized version of the input string,
- for each result, the corresponding user document is retrieved from the `users` collection.

The results are returned as a list of `Users` objects. The SearchPage updates its internal state by assigning the results to `_userResults` and calling `setState`.

Search Users Sequence Diagram



2.5.6 Search Trekking Sequence Diagram

The trekking search functionality follows a similar structure but introduces an additional optimization layer through local caching.

When the user enters a query, the method `_runSearch(query)` is executed, and the UI is updated using `setState` to reflect the loading state.

The request is delegated to the `searchTrekking(query)` method of the `TrekkingController`.

The search is performed in two phases:

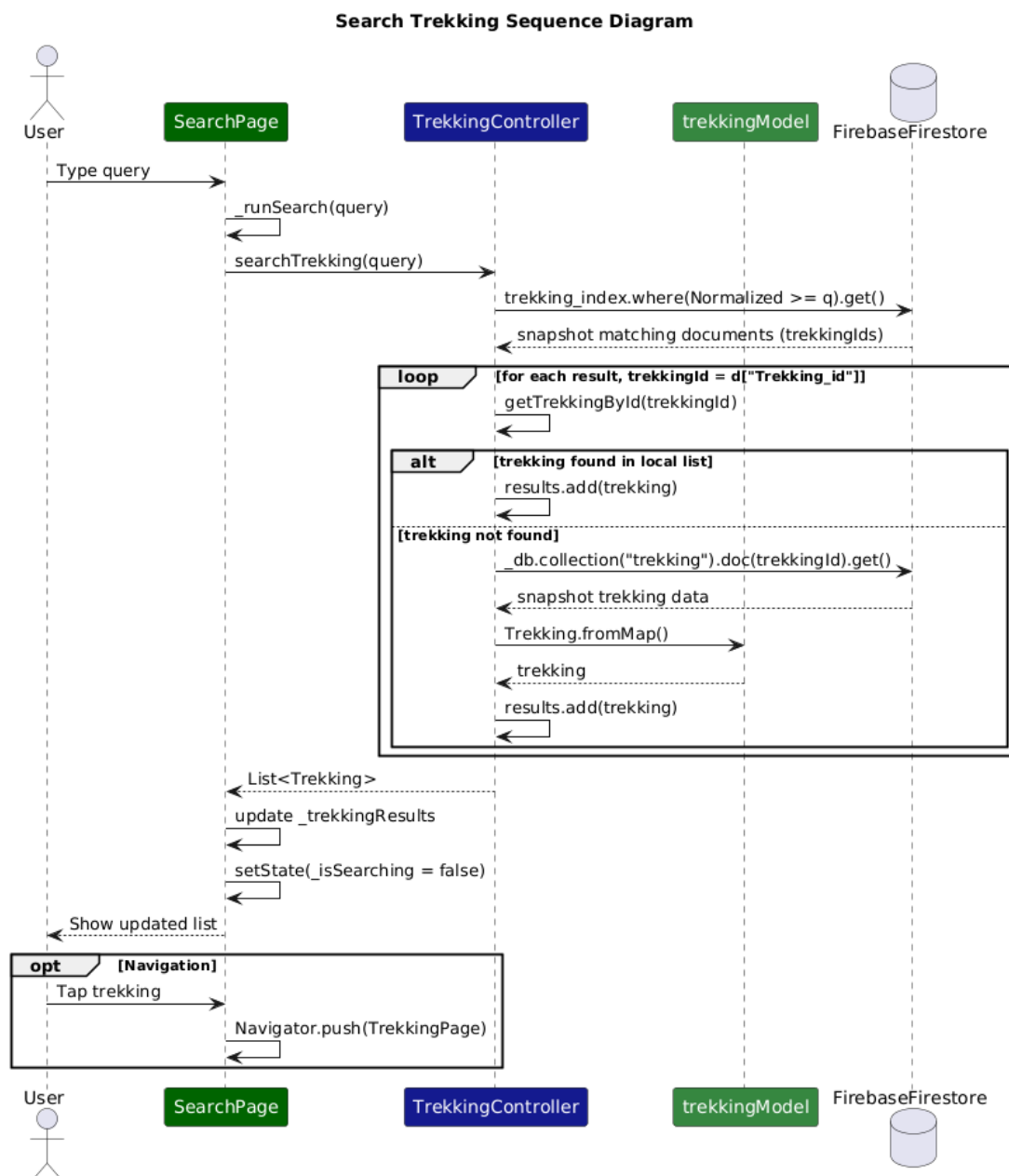
- the `trekking_index` collection is queried using a normalized string,
- each resulting trekking identifier is resolved into a full trekking object.

Before querying Firestore, the controller checks whether the trekking is already available in the local list. If the object is found locally, it is reused; otherwise, it is fetched from Firestore.

This approach reduces network overhead and improves performance.

Once the results are collected, they are returned to the `SearchPage`, which updates its internal state and rebuilds the UI using `setState`.

When the user selects a trekking, the application navigates to the corresponding `TrekkingPage`.



2.5.7 View Compass and Altitude Information Sequence Diagram

The **Geowatch** component is responsible for displaying compass, altitude, and current position information. This functionality is implemented through the interaction between the page, the **servicecontroller**, the **geoservice**, and the **permissionservice**. When the navigation page is opened, the initialization starts in `initState()`, where the page registers itself as an app lifecycle observer and schedules the execution of `_initLocation()`. The observer is used to detect when the application returns to the foreground, for example after the user has changed GPS or permission settings. The method `_initLocation()` delegates the preliminary checks to the **servicecontroller** through `loadNavigationLocation()`. This method verifies whether the location permission has been granted and whether the GPS service is enabled. If one of these conditions is not satisfied, the page enters an error state and shows a dedicated message to the user. If permission is denied, the returned state contains the error `PERMISSION_DENIED`. If the GPS service is disabled, the returned state contains the error `GPS_DISABLED`. In both cases, the page updates its local state through `setState()` and displays an error instead of the navigation information. Only when both checks are successful does the page subscribe to the position stream. The stream is obtained through the **servicecontroller**, which delegates the request to the **geoservice**. Every new **Position** object received from the stream is used to update altitude, latitude, longitude. The compass information is handled through a separated stream. In the architecture, the UI does not access the **FlutterCompass** library. Instead, it requests the compass stream through the **servicecontroller**, which delegates the operation to the **geoservice**. The UI then receives heading values and converts them into cardinal directions through the helper method `_direction()`. When an error state is shown, the page also provides a button that helps the user solve the problem. If the GPS is disabled, the button invokes `openGpsSettings()`, which is delegated through the **servicecontroller** to the **geoservice**. If the problem is related to permissions, the page instead invokes `openPermissionSettings()`, which is delegated to the **permissionservice**. In this way, the UI does not directly interact with operating system APIs. The diagram also includes the handling of lifecycle changes. When the app returns to the foreground, it handles the app state change through method `didChangeAppLifecycleState()`. The page sets the checking state again and re-runs `_initLocation()`. This is useful, for example, when the user comes back from the system settings after enabling GPS or granting permissions. Finally, when the page is closed, the observer and the active position stream subscription are disposed through the `dispose()` method. This avoids unnecessary updates when the page is no longer visible.

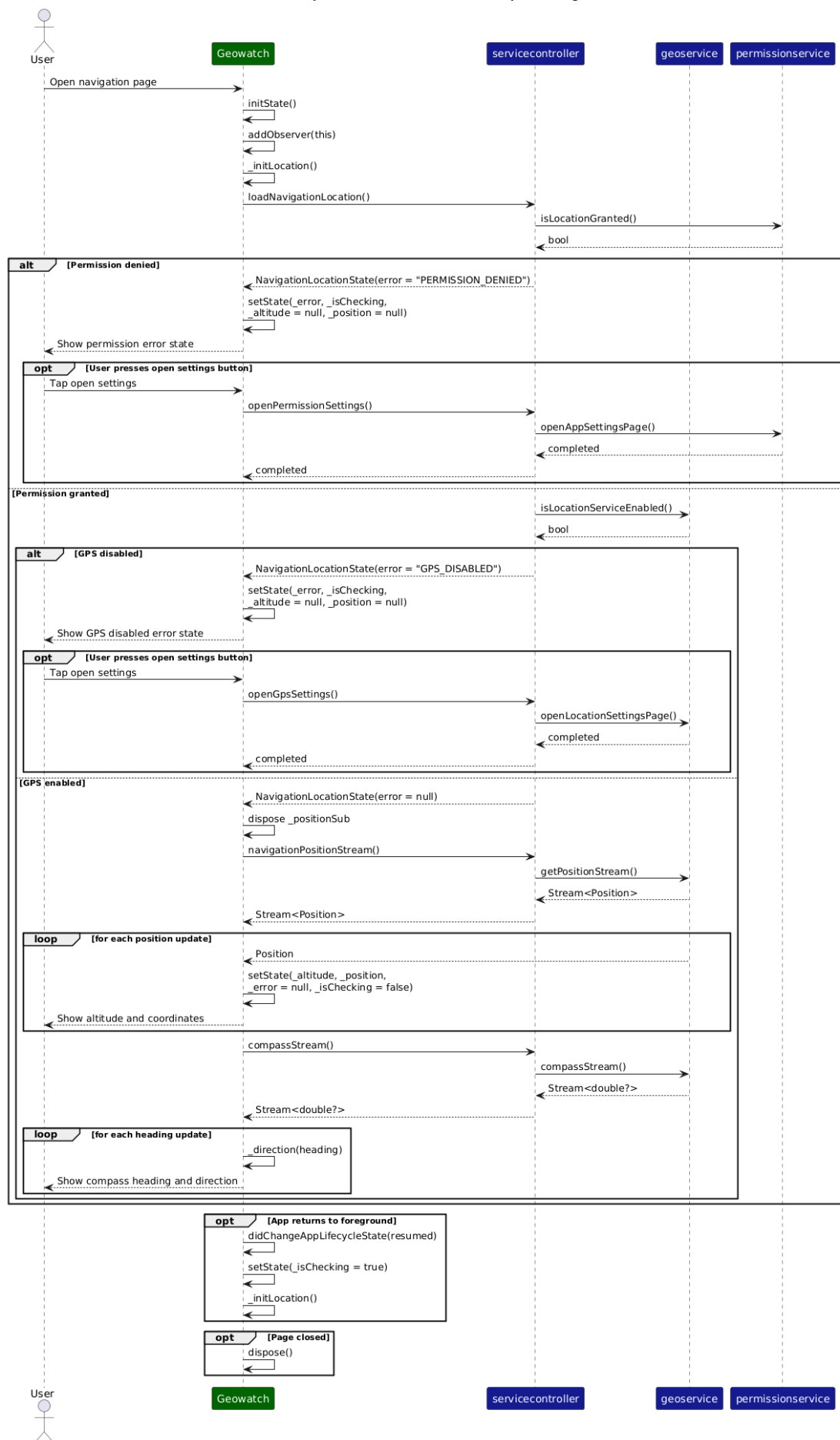
Application Lifecycle in Flutter

The **application lifecycle in Flutter** refers to the sequence of states an app goes through during its execution, such as *resumed*, *inactive*, *paused*, and *detached* and the mechanisms provided to handle transitions between these states. Flutter provides a platform-independent abstraction of the lifecycle through the `AppLifecycleState` enum. Developers can monitor these changes by implementing the `WidgetsBindingObserver`, which allows the application to listen for lifecycle updates. In particular, the method `didChangeAppLifecycleState(AppLifecycleState state)` is overridden to react to state transitions, enabling behaviors such as pausing tasks, saving data, or managing background operations when the app moves between foreground and background.

Difference between permission denied and GPS disabled

The two error conditions shown in the diagram represent different problems. The `PERMISSION_DENIED` state means that the application is not allowed to access the device location. In this case, the problem is related to the application permission, so the user must enable the permission from the app settings. The `GPS_DISABLED` state means that the application has permission to access location, but the location service of the device is currently turned off. In this case, the problem is not related to the app permission, but to the system location service, so the user must enable GPS from the device settings. For this reason, the page provides two different actions: opening the application settings in the case of denied permission, and opening the location settings in the case of disabled GPS.

View Compass and Altitude Information Sequence Diagram



2.5.8 Login Sequence Diagram

The login process supports two authentication mechanisms: email/password and Google sign-in. These are modeled as alternative flows within the same sequence diagram.

Email and password login When the user presses the login button, the page executes the method `_loginEmailPwd(context)`. The method first validates that both email and password are not empty. If one of the two fields is missing, the page shows an error message through a `SnackBar`.

If the input is valid, the page invokes `userController.login(email, password)`. The `UserController` delegates the operation to the `AuthService`, which uses Firebase Authentication to authenticate the user through `signInWithEmailAndPassword()`.

Firebase Authentication is used exclusively to verify user identity and retrieve a unique user identifier (UID). The method `signInWithEmailAndPassword()` returns a `UserCredential` object, which contains the authenticated Firebase user. After authentication, the system invokes `loadUserCore()`, which retrieves the user document from Firestore:

- Username
- Email
- Profile data
- Level and statistics

This method intentionally avoids loading relational or large data such as followers, following, or diaries. This design prevents blocking the UI during login and improves perceived performance.

Instead, additional data is loaded lazily in the `UserPage` using `FutureBuilder`, enabling asynchronous and incremental UI updates.

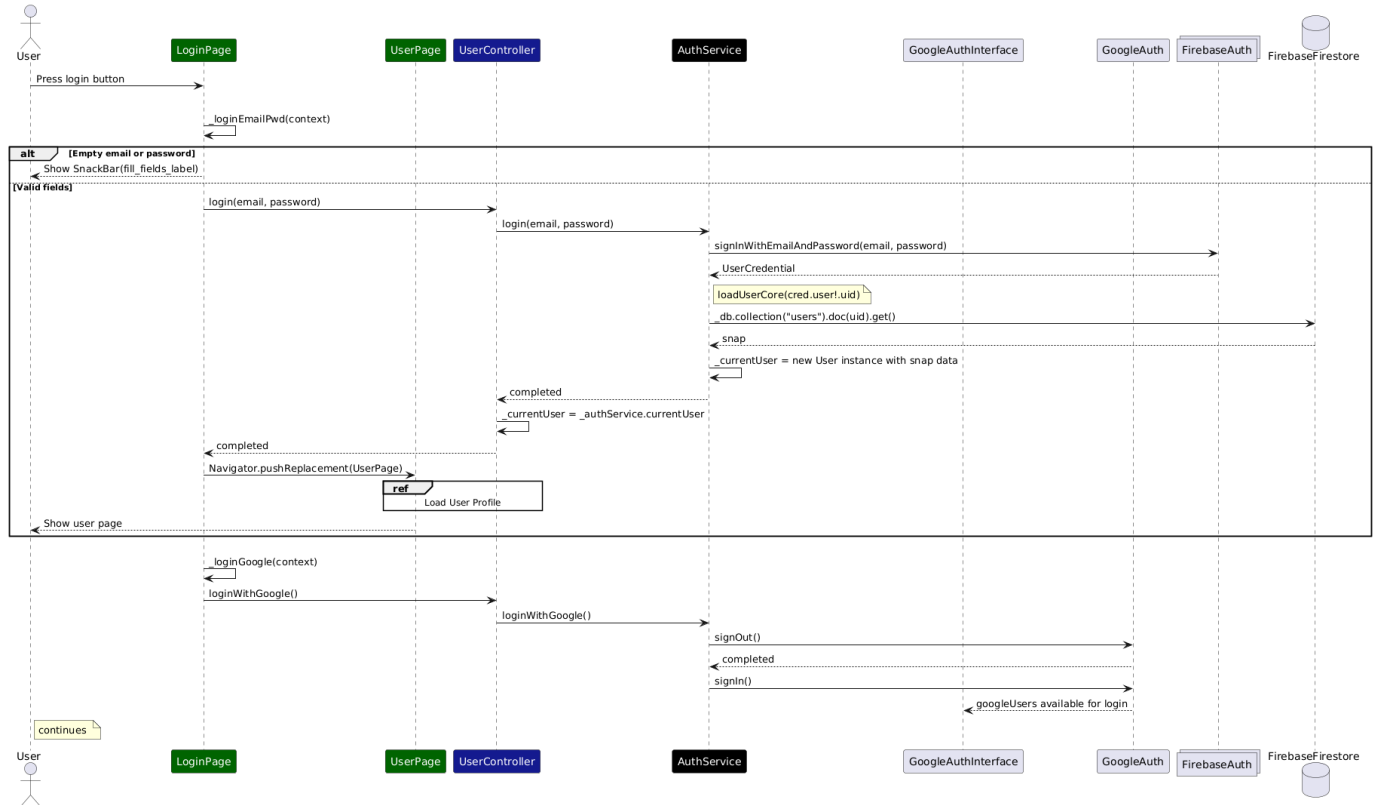
After successful login, the application navigates to the `UserPage`, whose loading behavior is described in a separate sequence diagram.

Google login When the user presses the Google Sign-In button, the page executes `_loginGoogle(context)`. The request is delegated to `userController.loginWithGoogle()`, which calls the corresponding method in the `AuthService`.

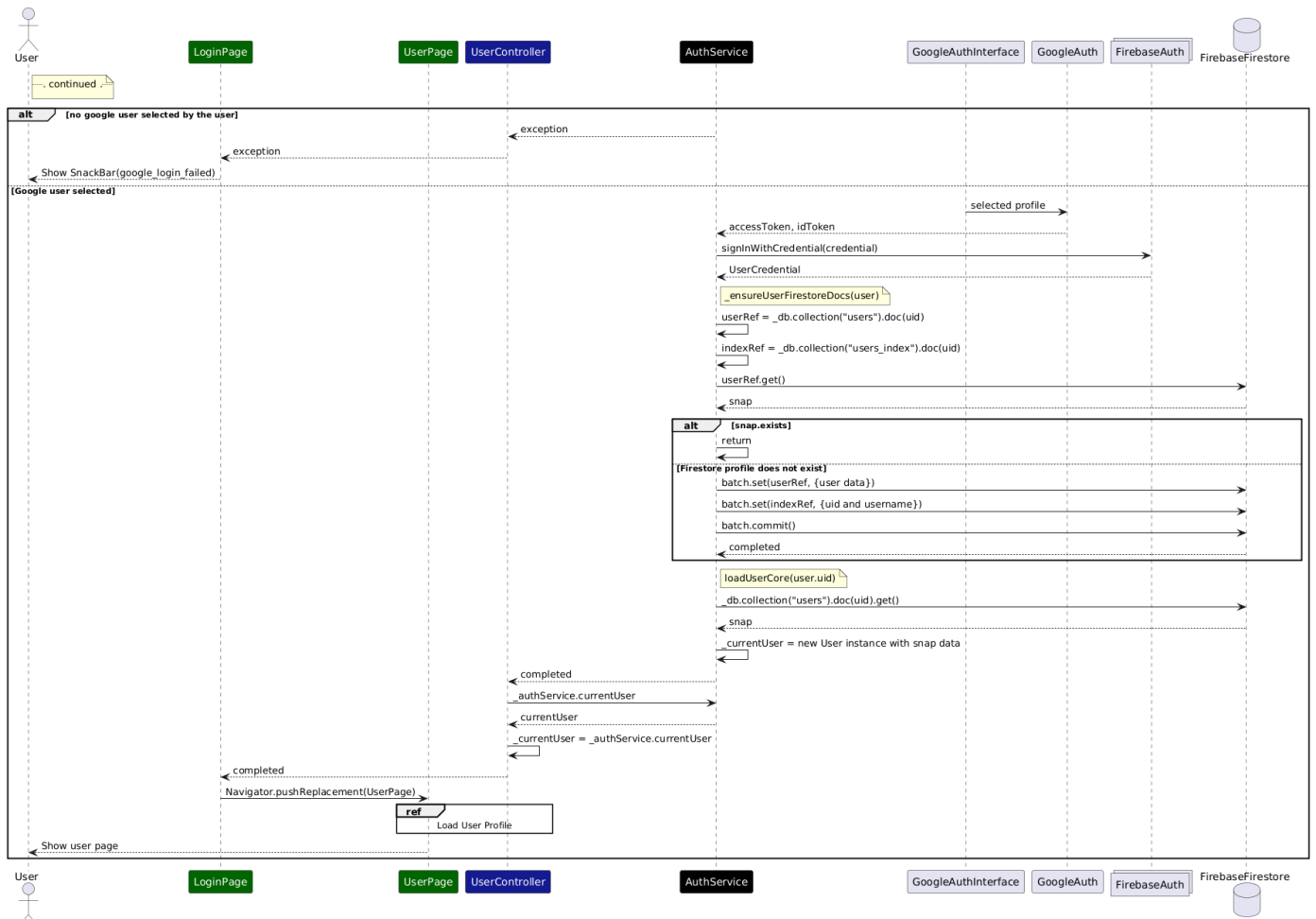
The Google login flow first forces a sign-out from the previous Google session and then opens the Google account interface. If the user does not select a profile, the flow terminates with an error message shown in the UI.

If the user selects a Google account, the `AuthService` retrieves the Google authentication tokens and exchanges them for Firebase credentials through `signInWithCredential()`. Once Firebase authentication succeeds, the service verifies whether the user already has a corresponding Firestore profile through `_ensureUserFirestoreDocs(user)`. If the profile does not exist, the service creates the required documents in both the `users` and `users_index` collections. After that, the method `loadUserCore(uid)` is executed, and the page navigates to the `UserPage`.

Login Sequence Diagram



Login Sequence Diagram



2.5.9 View User Information Sequence Diagram

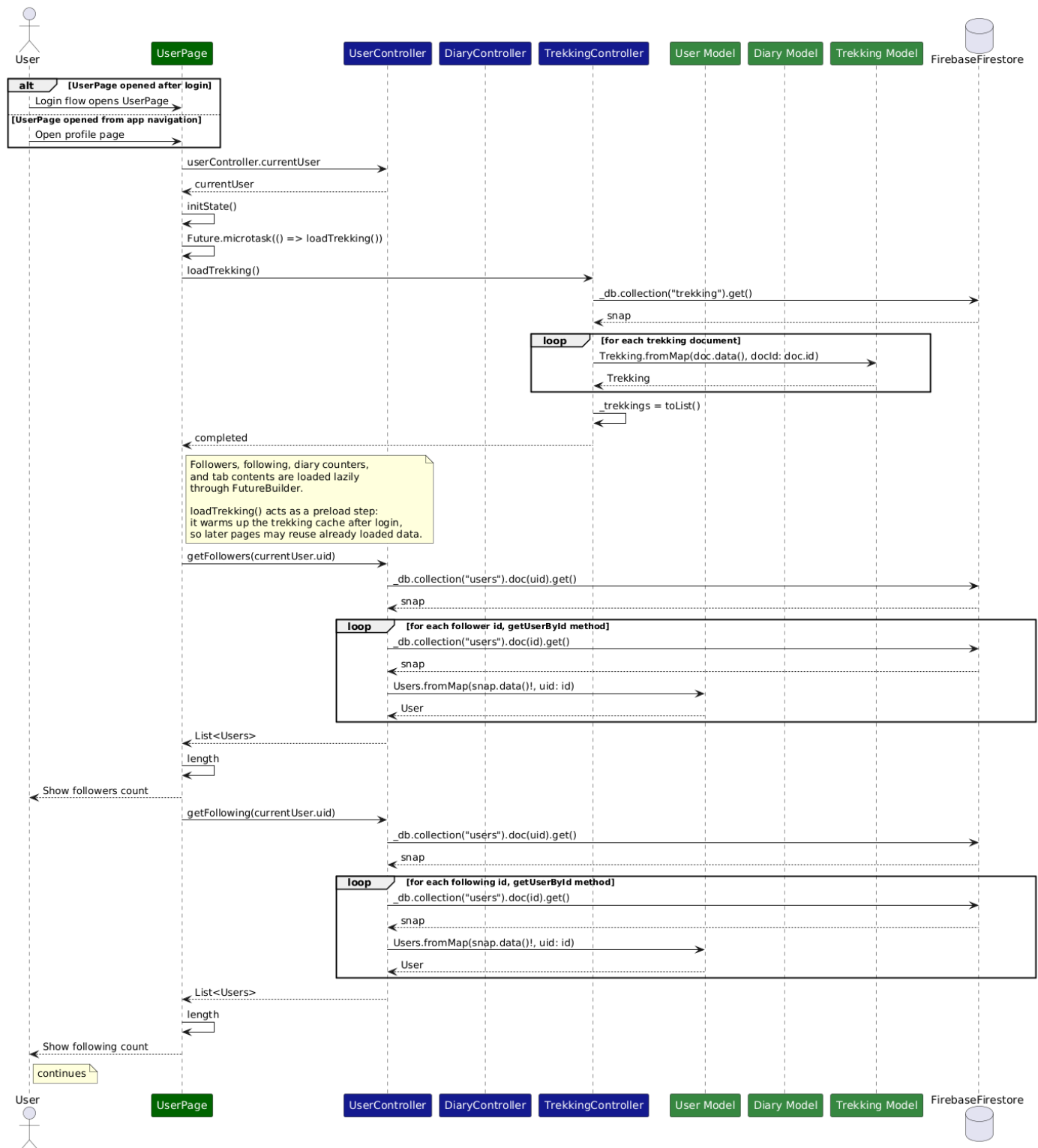
The `UserPage` can be opened either as the landing page after a successful login or through the normal application navigation. Once the page is displayed, it retrieves the current authenticated user from the `UserController`.

During initialization, the page schedules the invocation of `loadTrekking()` through `Future.microtask()`, instead of calling it directly inside `initState()`. This choice postpones the trekking load operation until the initialization phase is completed, keeping the widget initialization lighter. Although trekking loading is not strictly part of the core profile information, this preload can still be justified for the following reasons

- the `UserPage` is often the first page shown after login, so loading trekking data here can anticipate work that will likely be needed shortly afterward by pages such as the `HomePage`, `TrekkingPage`, `SearchPage`, and the saved trekking tab itself
- the `UserPage` allows navigation toward saved trekkings and trekking detail pages, so having the trekking cache already populated may reduce subsequent loading work
- `loadTrekking()` acts as a warm-up step for the controller-side cache, because it populates the local `_trekkings` list and makes data immediately available for methods such as `getTrekkingById()`, `getTrekkingId()`, using Firestore only as secondary solution

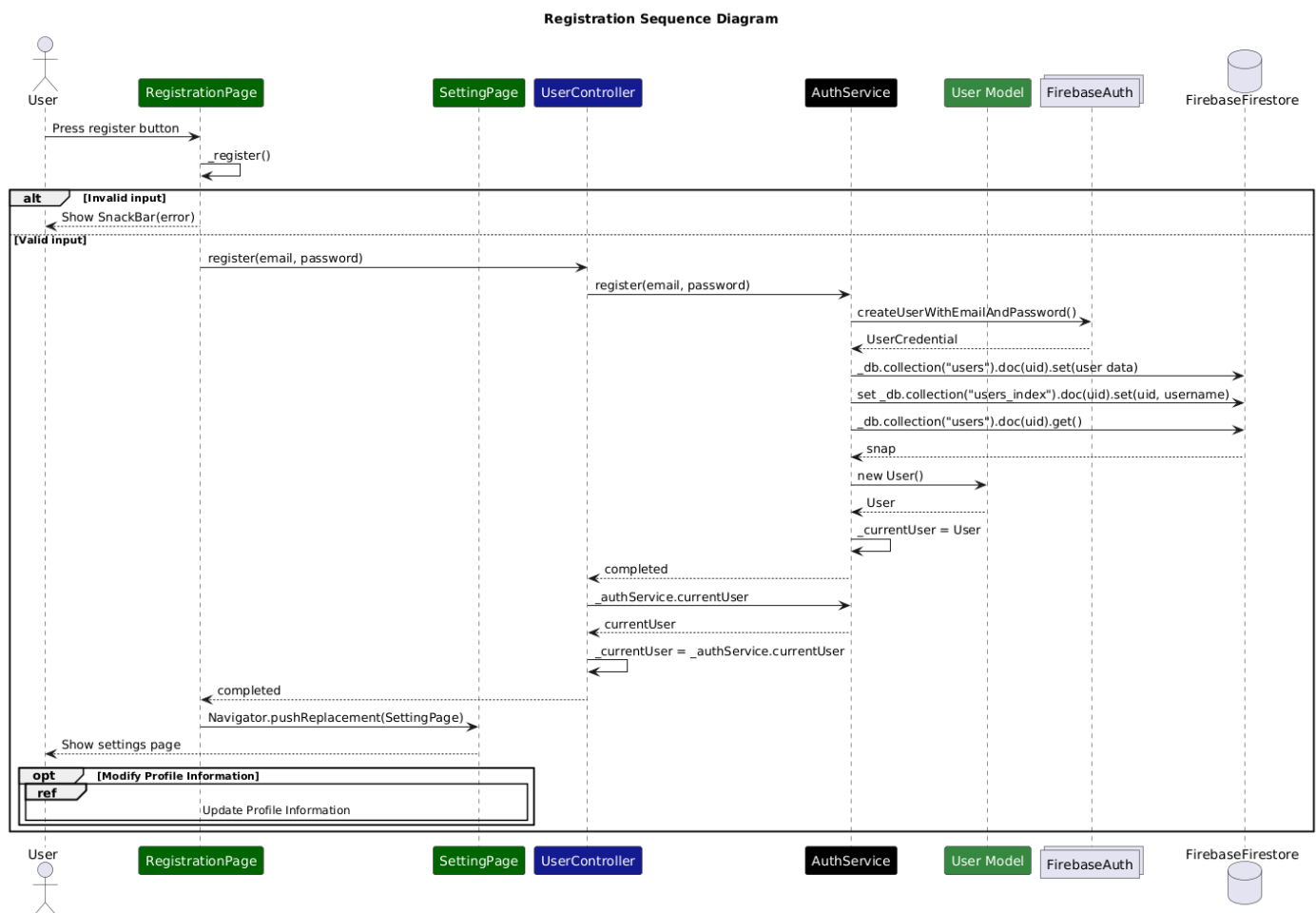
The main profile information shown in the page is obtained from the current user already available in the `UserController`. Additional information is retrieved lazily through `FutureBuilder`. This applies to followers, following users, diary counters, public diaries, private diaries, and saved trekkings. For followers and following, the `UserController` first retrieves the current user document from Firestore in order to obtain the identifiers stored in the `Followers` and `Following` fields. Then, for each identifier, the controller invokes `getUserById()`, which retrieves the corresponding Firestore document and builds a `Users` object through `Users.fromMap()`. The total diary count is computed through the helper method `_loadAllDiaries(context, uid)` defined in the `UserPage`. This method invokes both `getPublicDiaries(uid)` and `getPrivateDiaries(uid)` from the `DiaryController`, then merges the two returned lists into a single collection. For the diary tabs, the `DiaryController` queries Firestore and converts each result document into a `Diary` object through `Diary.fromMap()`. For the saved trekking tab, the `TrekkingController` first retrieves the current user document to obtain the identifiers stored in `Saved_trekkings`. Then, for each identifier, it loads the corresponding trekking document and converts it into a `Trekking` object through `Trekking.fromMap()`. This design combines a lightweight preload of trekking data with lazy loading of profile-related collections, keeping the `UserPage` responsive while still preparing application data that is likely to be useful immediately after login.

User Page Sequence Diagram



2.5.10 Registration Sequence Diagram

The registration functionality is handled by the `RegistrationPage`, which delegates the creation of the new account to the `UserController` and the `AuthService`. When the user presses the register button, the page executes the method `_register(context)`. The method first validates the input fields. If one or more fields are empty, the UI shows a `SnackBar` requesting the user to fill all fields. If the password and confirmation password do not match, another `SnackBar` is shown. If the data is valid, the page activates the loading state and calls `UserController.register(email, password)`. The `UserController` delegates the operation to the `AuthService`, which creates the account in Firebase Authentication through `createUserWithEmailAndPassword()`. After the account is successfully created, the `AuthService` derives a default username from the email prefix and creates the corresponding Firestore documents in the `users` and `users_index` collections. The first document stores the full user profile data, while the second supports search functionality through a normalized username representation. Once the Firestore documents have been created, the `AuthService` loads the newly created user profile through `loadUserCore(uid)`. The resulting user object is returned to the `UserController`, which updates its internal state. Finally, the `RegistrationPage` shows a success message and navigates to the `SettingPage`. The loading indicator is then disabled. It is important to note that, in the current implementation, the registration page only supports the traditional email/password flow. Google-based account creation is not handled by the registration page itself, but is implicitly managed during the first successful Google login through the creation of the Firestore profile documents.



2.5.11 Update User Information Sequence Diagram

The update of user information is handled by the **SettingPage**, which allows the user to modify several profile attributes such as username, name, surname, birthdate, profile image, and email. The page delegates saving user information to the **UserController**, which is responsible for uploading the local user state to Firestore.

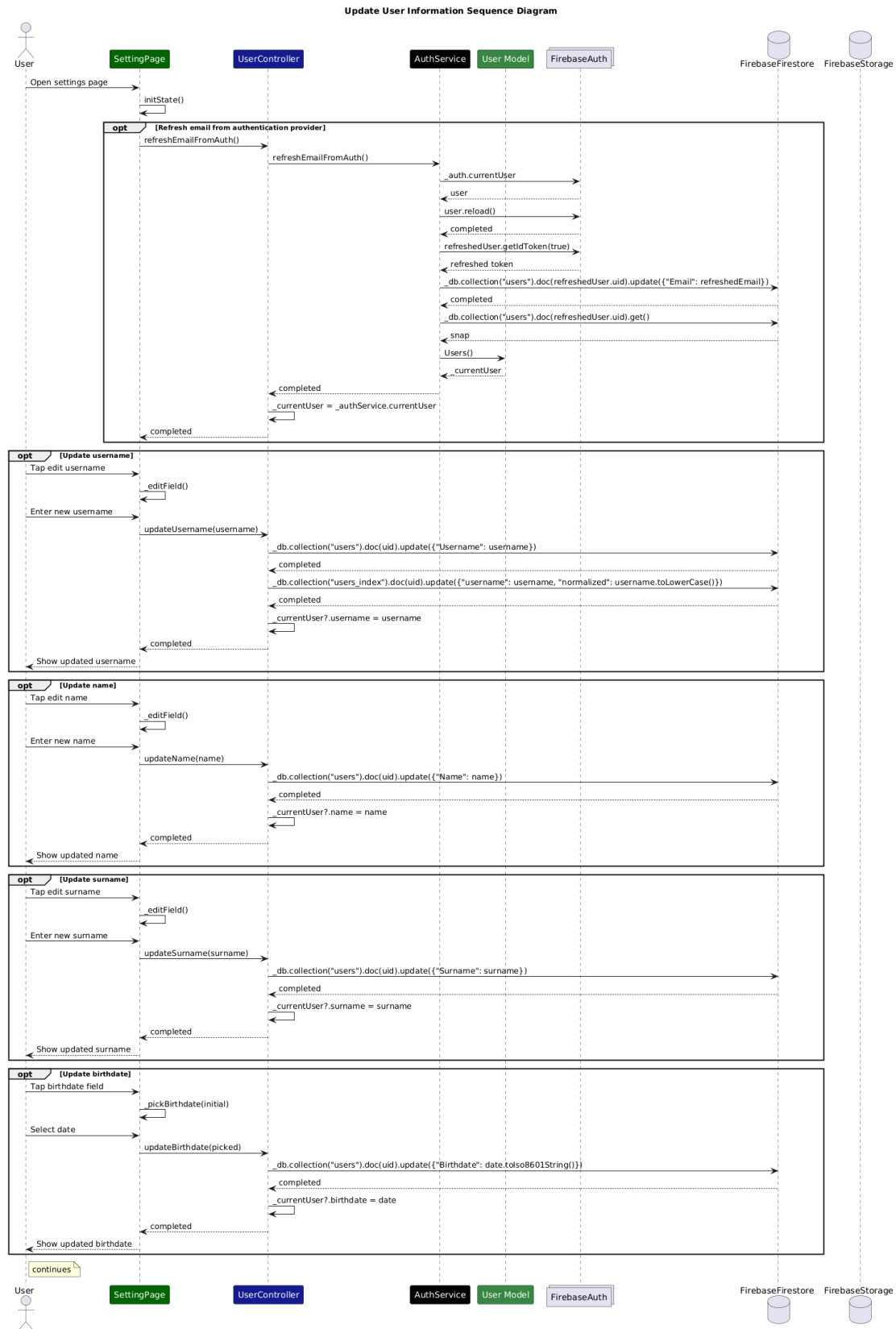
When the page is opened, it executes an initialization step by invoking `refreshEmailFromAuth()`, in order to check that the email stored in Firestore is aligned with the current value available in Firebase Authentication. Infact, if the user has requested to change its email and has verified the new one by clicking the link in the mail inbox, Firebase Auth may have registered that the user is associated to a new email, so this reference shall be updated also in the Firestore user record. The **AuthService** first reloads the authenticated Firebase user, loads the updated email, updates the Firestore user document, and finally rebuilds the local **User** through the method `loadUserCore()`.

The update of textual profile fields, such as username, name, and surname, follows a common pattern. The UI opens a dialog through `_editField()`, the user enters a new value, and the page delegates the save to Firestore operation to the **UserController**. The controller then updates the Firestore document and synchronizes the corresponding field inside the local `_currentUser` object. The username update also requires an update of the `users_index` collection, so that the user remains searchable through the normalized username.

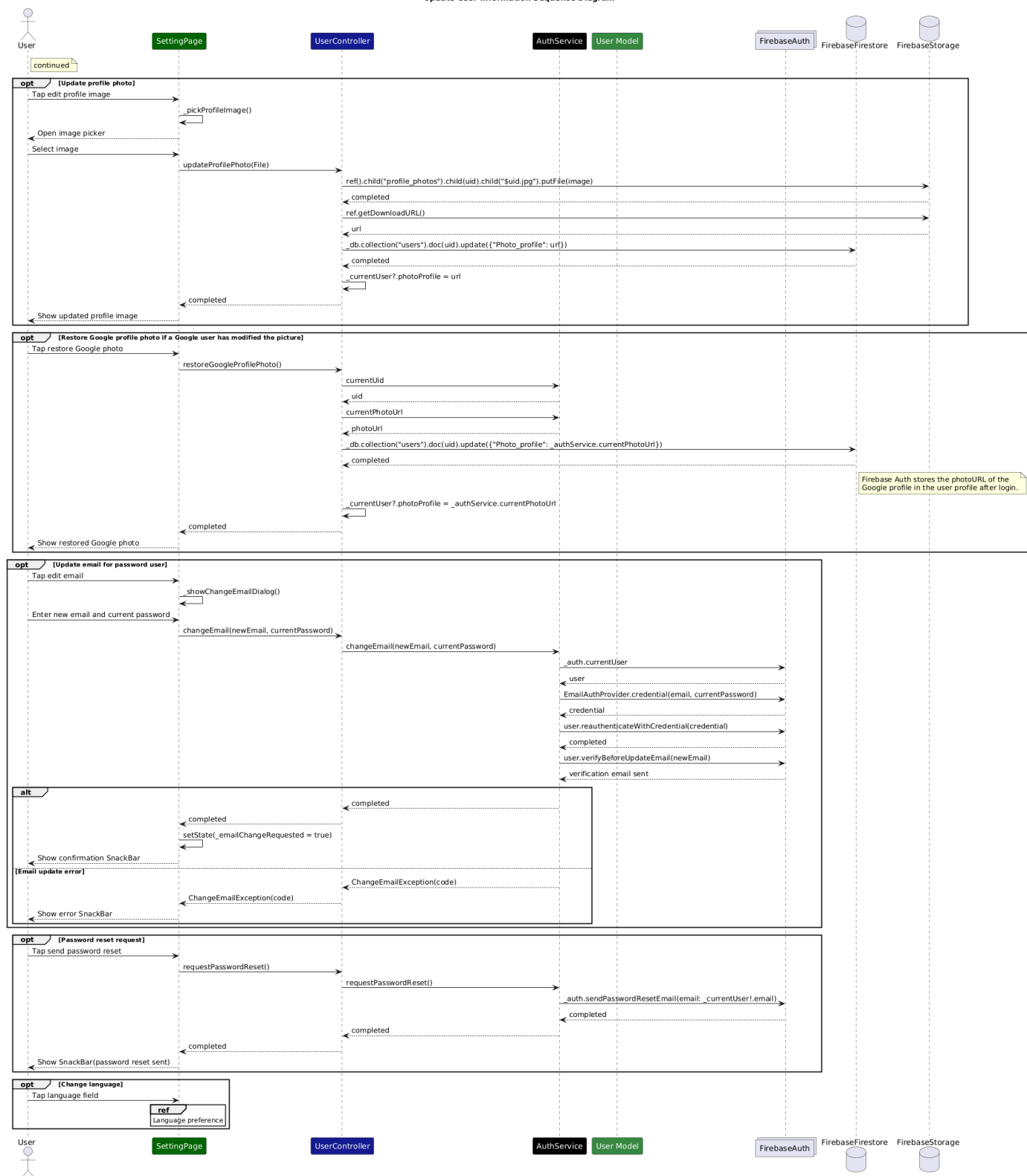
The birthdate update is handled similarly, but the input is collected through `showDatePicker()` instead of a text dialog. Once the user selects a date, the **UserController** updates Firestore and updates the local model field.

The profile image update uses the image picker to let the user select a file from the gallery. The **UserController** uploads the file to Firebase Storage, obtains the public download URL, stores that URL inside the Firestore user document, and updates the local `_currentUser.photoProfile` field. In this way, the UI can show the new profile image after the operation completes.

The email update is more complex because it involves Firebase Authentication. This operation is available only for password-based accounts, since Google profiles are managed by the external auth provider **GoogleAuth**. The page opens a dedicated dialog where the user enters the new email and the current password. The **UserController** delegates the operation to the **AuthService**, which first reauthenticates the current user through `reauthenticateWithCredential()` because, according to Firebase documentation, some security sensitive actions such as setting a new email address require that the user has recently signed in, then invokes `verifyBeforeUpdateEmail(newEmail)`. Firebase Authentication sends a verification link to the new email address. If the request succeeds, the page records that an email change has been requested and informs the user through a confirmation message. If Firebase returns an error, the exception is shown through a **SnackBar**.

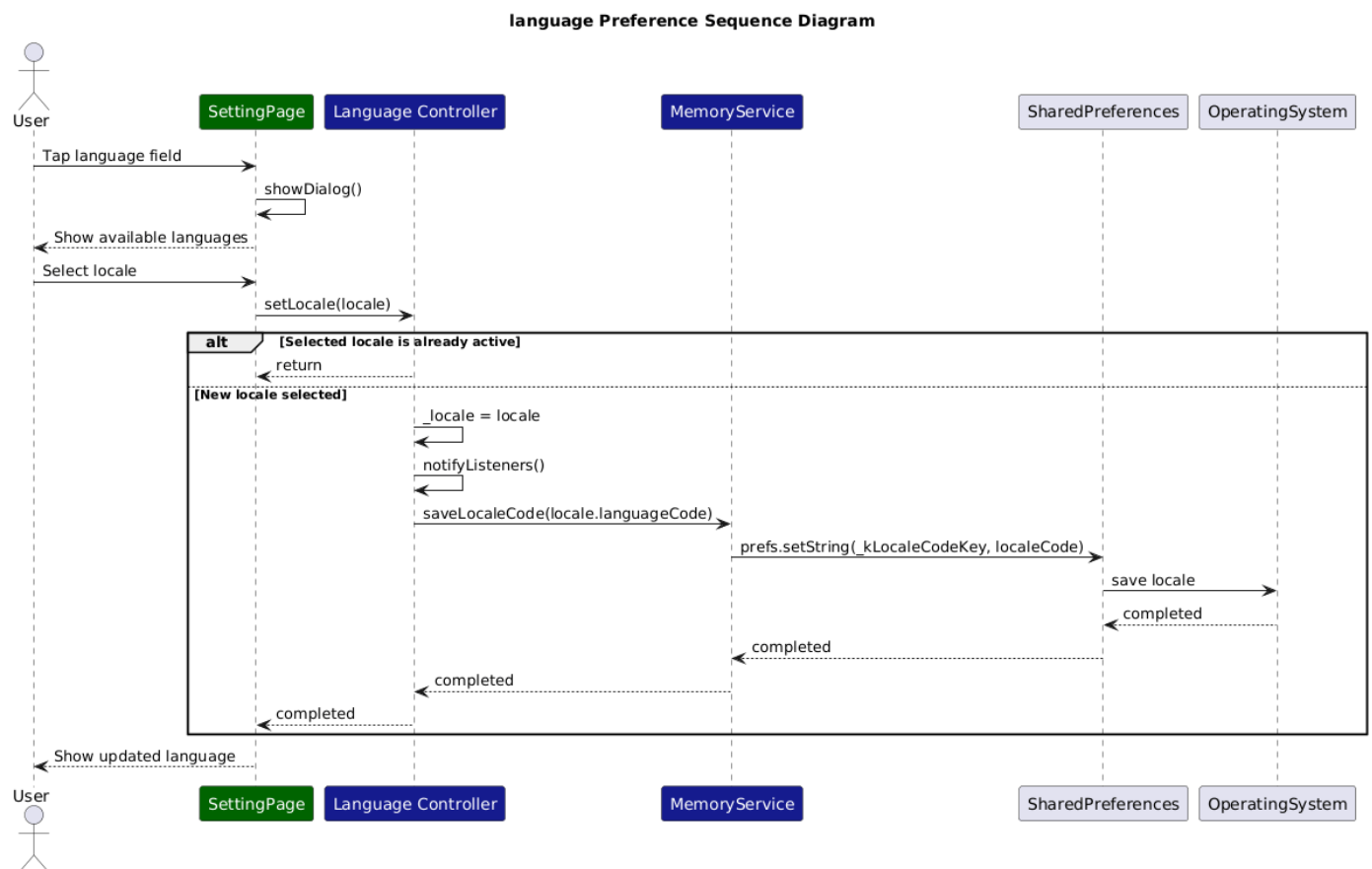


Update User Information Sequence Diagram



Change Language Sequence Diagram

The language change process is handled by the **Language** controller, and saving the preference on the phone memory is delegated to the **MemoryService**. When the user selects the language field in the **SettingPage**, the application opens a dialog that displays the list of available locales. Once the user chooses a language, the dialog invokes `setLocale(locale)` on the Language controller. The controller first checks if the selected locale is already active. If so, no further action is needed. Otherwise, it updates the internal `_locale` field, calls `notifyListeners()`, and then delegates the saving the preference to the **MemoryService** through `saveLocaleCode(locale.languageCode)`. The **MemoryService** stores the selected language code locally using shared preferences. This allows the application to restore the previously selected language when it is reopened, using the method `loadSavedLocale()`.



2.5.12 Phone Watch Pair

High level description of the pairing mechanism

The pairing mechanism associates a Wear OS device with an authenticated mobile user through a backend-mediated approval process. The watch does not authenticate directly with Firebase Auth; instead, device identity and user identity are linked via Firestore. Reasons behind this design decision are detailed in Section 2.8.3

Device Identity

Each watch either makes or restores from memory a `watchId`, stored securely on-device. A corresponding document exists in Firestore:

```
watch/{watchId}
```

The `watchId` uniquely identifies the physical device and is never deleted during logout.

Pairing Flow

1. Initialization (Watch) The watch provides a new random UUID (`qrToken`) and writes:

```
watch_pair/{watchId}
```

with:

- `status = "waiting"`
- `uid = null`
- `qrToken`
- `createdAt, expiresAt`

A QR code encoding `watchId` and `qrToken` is displayed. URI format:

```
triplo://watch-pair/<watchId>?t=<token>
```

2. Approval (Phone) After scanning the QR, the authenticated mobile app:

- Verifies token validity and expiration.
- Updates the document with:
 - `status = "approved"`
 - `uid = request.auth.uid`

3. Completion (Watch) The watch listens to the document in real time. When `status == "approved"` and `uid` is set, pairing is considered complete and the watch loads user data using that `uid`.

Session Restoration

On startup, the watch checks:

```
watch_pair/{watchId}
```

If the document is already approved, the session is restored.

Logout

Logout does not delete the device identity. Instead, the watch resets:

- `status = "waiting"`
- `uid = null`

This preserves device identity while removing user association.

Security Properties

- Only the watch may create/reset pairing in `waiting` state.
- Only authenticated users may approve pairing.
- QR tokens are random and time-limited.
- The smartphone application allows the user to remotely disable or re-enable an existing pairing through a backend flag

This design cleanly separates device identity from user authentication while minimizing credential exposure on the wearable device.

Remote Pairing Management and Anti-Theft Property

The system provides a remote management mechanism that allows users to enable or disable the usability of a paired wearable device directly from the smartphone application. Instead of removing the association between the user and the device, the system maintains the pairing information in the backend and relies on a dedicated status flag to control whether the pairing can be effectively used by the watch. If the pairing is disabled, the wearable device cannot access or display the associated user profile. However, the underlying association remains stored in the backend. If the pairing is re-enabled, the wearable device regains access to the associated user profile without requiring a new pairing procedure. This design introduces an anti-theft and safety property: in case of device loss or unauthorized access, the user can quickly disable the pairing, effectively blocking access to personal data while preserving the association for future recovery.

Phone-side smartwatch pairing approval from the `SettingPage` component

Phone-side smartwatch pairing sequence diagram illustrates the phone-side portion of the smartwatch pairing. In the phone UI implementation, the QR scanning interaction belongs to the `SettingPage` component.

The interaction starts when the user opens the pairing section and scans the QR code on the smartwatch. Once a QR code is detected, the page extracts the raw payload and checks if that code has already been processed. This behaviour is handled by the boolean flag `_handled`.

This flag is used to not allow repeated processing of the same scanner detection while the page is still active.

If a valid raw payload is available, `SettingPage` delegates the parsing step to

`UserController.extractWatchPair()`, which delegates the actual logic of the phone side pairing process to `AuthService.extractWatchPair()`. This service parses the scanned URI and verifies that it matches the following structure:

```
triplo://watch-pair/<watchId>?t=<token>
```

The service also validates the presence of both the smartwatch identifier and the temporary QR token in the URI payload. If the payload is malformed, the page shows an error message and enables a new scan by resetting `_handled`.

If the QR payload is valid, the page shows a confirmation dialog to the user. If the user does not confirm the operation, the page allows the next scan. If the user confirms, the page invokes `UserController.approveWatchPair()`, which forwards the request to `AuthService.approveWatchPair()`. The actual approval is performed inside `AuthService`, since this step depends on the currently authenticated phone user and on the pairing state stored in Firestore. The service executes a Firestore transaction on the document `watch_pair/<watchId>`. The transaction reads the document snapshot, denoted as `snap`, and validates the pairing request by checking four conditions: the document must exist, its status must be `waiting`, the token stored in Firestore must match the token contained in the scanned QR code, and the expiration timestamp must still be valid.

Only if all these conditions hold, the transaction updates the pairing document by setting `status = approved` and by storing the UID of the authenticated phone user. From an architectural perspective, the transaction is important because it makes validation and approval atomic. This reduces the risk of inconsistent state transitions. The page itself does not directly interact with Firestore or Firebase Authentication; it only reacts to success or failure by updating local UI state.

Watch bootstrap and initialization of the pairing-aware user flow

When the smartwatch application starts, it first executes a lightweight bootstrap phase whose purpose is to restore a stable device identity and initialize the user-related flow in a pairing-aware way, as illustrated in the *Watch bootstrap sequence diagram*. This bootstrap does not directly perform the pairing itself, but it prepares the conditions under which the pairing logic can later determine whether the watch should show the QR-based login screen or the actual user area. The first relevant step regards the retrieval of the watch identifier. During application startup, the watch obtains a `watchId` through `WatchIdService.getOrCreateWatchId()`. If a valid identifier is already available in secure local storage, that value is reused. Otherwise, the application makes or retrieves a new identifier and stores it locally so that the same watch can be recognized across future executions. This mechanism ensures that the pairing protocol is tied to a stable device identity. Once the `watchId` is available, the application initializes the services and controllers involved in the watch-side user flow. In particular, `PairingService` is created with the current `watchId`, while `UserController` is connected to the pairing service so that it can later load the profile corresponding to the paired user. The entry point for the user-related interaction is not the pairing screen itself, but `NavigationPage`. When the user selects the user area from the navigation menu, the application pushes `PairingGateway(child: UserPage())`. At this point, `PairingGateway` performs the bootstrap of the pairing route.

Watch-side access to the user area through the pairing flow

Watch-side smartwatch pairing sequence diagram describes how the smartwatch application grants access to the user area through the pairing mechanism. In the watch app implementation, the user does not open a pairing screen, the interaction starts from `NavigationPage`: when the user taps the `User` button, the application performs a `Navigator.push()` toward `PairingGateway(child: UserPage())`.

Pairing Gateway

From an architectural perspective, `PairingGateway` should not be interpreted as an autonomous functional component, it is considered part of the `PairingService` and its role is to decide whether the watch should render the QR-based pairing screen or the actual `UserPage`.

The pairing bootstrap starts in the state of `PairingGateway`, which is the widget pushed when the user opens the user area from `NavigationPage`. Its responsibility is not to implement the pairing itself, but to bootstrap the pairing state and decide which page should be rendered, either the `LoginPage` or `UserPage`.

Restoring the session if the user has already paired the watch

The pairing logic relies on two complementary mechanisms. First, `PairingService.restoreWatchPairing()` performs a read of the Firestore document `watch_pair/<watchId>` in order to determine the current pairing state. Second, the service installs a real-time subscription through `docRef.snapshots().listen()`, so that later updates to the same Firestore document, such as approval from the phone or a reset to the `waiting` state, are seen by the watch without requiring to implement a more complicated ping-like mechanism to listen to updates of the document on Firestore.

If the pairing document is already in `approved` state and contains a non-empty UID, `PairingService` stores that UID in its internal variable `_pairedUid` and calls `notifyListeners()`. In this context, the listeners are the Flutter widgets that observe `PairingService` through `Provider`, notably `PairingGateway`.

At this point, `PairingGateway` requests `UserController.loadCurrentPairedUser()`. This step is necessary because the pairing service only knows which user UID is associated with the watch, but it does not load the complete user profile. The actual profile loading is delegated to `UserController`, which reads the corresponding document from the `User` collection by calling `loadUserCore(uid)`. The returned Firestore snapshot, denoted as `snap`, is converted into a `User` domain object through `Users.fromMap()` and stored in `UserController` as `_currentUser`. When `loadUserCore()` completes, `UserController` also calls `notifyListeners()`.

Here again, the listeners are the widgets that observe `UserController` through `Provider` and the `PairingGateway` is a listener.

Then the UI is rebuilt because the `PairingGateway` returns the widget `child` that was set by the `Navigation Page` through `Navigator.push()` toward `PairingGateway(child: UserPage())`. The `UserPage` is then able to see that a paired UID is available.

In this sense, the pairing service is responsible for determining whether the watch is associated with a user, whereas the user controller is responsible for loading the actual user data needed by `UserPage`.

Pair the watch if the user has not paired it

If no approved pairing is found, the `PairingService` starts a new pairing cycle through `PairingService.startWatchPairing()`. In this branch, the service makes a new QR token, stores it in local state, writes the Firestore document `watch_pair/<watchId>` with status `waiting`, a null UID, an expiration timestamp, and a new token, and then installs a snapshot listener on that document. A local expiration timer is also started. The `PairingService` then renders `LoginPage`, which reads the QR payload exposed by `PairingService` and displays it to the user. The `LoginPage` is only the visual pairing screen, it does not control the pairing itself.

The most relevant transition regards the phone approving the QR code. This operation updates the Firestore document `watch_pair/<watchId>` by setting its status to `approved` and assigning the corre-

sponding user UID. The update is received by the active snapshot listener previously installed by `PairingService.startWatchPairing()` through `docRef.snapshots().listen()`.

Inside the listener, the service reads the updated document and retrieves the fields `status`, `uid`, and `qrToken`. At this point, the service compares the token stored in Firestore with the locally active token (`_qrToken`). This comparison is performed inside the listener itself through the instruction `if (tokenOnDb != _qrToken) return;` and ensures that only updates referring to the currently displayed QR code are accepted. If the tokens do not match, the update is ignored, in order not to approve stale QR codes.

If the token matches and the status is `approved`, `PairingService` stores the paired UID in `_pairedUid`, stops the expiration timer, and calls `notifyListeners()`. This notification is sent over to the `PairingGateway`, that rebuilds the UI. The `PairingGateway` reacts to the new state by invoking `UserController.loadCurrentPairedUser()`, which loads the corresponding user document from Firestore and reconstructs the `Users` model. As a result, the UI transitions from the pairing screen to `UserPage`.

The logout and re-pairing flow clarifies an additional architectural point. When the user logs out from the smartwatch user page, `PairingService.logoutWatch()` clears the active timer, stops the current Firestore pairing subscription, resets the pairing document by making a new QR token and restoring the `waiting` state, and clears the locally paired UID. However, the reset of the backend document alone is not sufficient, the observation of the pairing document must be reinstalled through a new call to `restoreWatchPairing()`. This step is crucial because otherwise the watch could display a new QR code while failing to display the `UserPage` correctly after the first approval performed by the phone.

Phone-side management of connected watches and remote logout

Phone-side view paired watches and remote logout management sequence diagram describes the phone-side flow used to inspect the currently connected smartwatch devices and manage the remote logout flag associated with them. This functionality does not replace the ordinary pairing protocol; rather, it provides an additional management layer on top of the live pairing state already stored in Firestore.

The interaction starts from `SettingPage` component, where the user opens the dedicated paired watches page. Once the route is pushed, `PairedWatchesPage`, which is part of the `SettingPage` component requests the list of paired watches through

`UserController.getConnectedWatches()`, that invokes `AuthService.getConnectedWatches()`.

`AuthService` then queries the `watch_pair` collection by selecting the documents whose `uid` matches the authenticated user and whose `status` is `approved`. The result is returned to the page and rendered as a list of paired smartwatch devices.

It is important to clarify the semantics of this page. The data shown here does not represent a historical registry of every device ever associated with the account. Instead, it reflects the current pairing state stored in Firestore, according to the active content of the `watch_pair` documents.

For each listed watch, the page also provides an operation to enable or disable remote logout. These actions are again delegated through `UserController` to `AuthService`. When the user enables remote logout, the phone updates the selected `watch_pair/<watchId>` document by writing a timestamp into the optional field `remoteLogoutAt`. When the user clears remote logout, the same field is set back to `null`. From the phone-side perspective, these operations are intentionally simple and do not alter the ordinary pairing fields such as `status`, `uid`, or `qrToken`.

This design choice is architecturally relevant because it keeps the original pairing protocol unchanged. The field `remoteLogoutAt` acts as an additional control flag, not as a replacement for the existing pairing state. The watch continues to observe the same `watch_pair` document through its existing Firestore listener and may interpret the presence or absence of `remoteLogoutAt` as a condition that temporarily blocks or restores access to the user area. In this way, remote logout can be managed from the phone while leaving the main pairing flow conceptually intact.

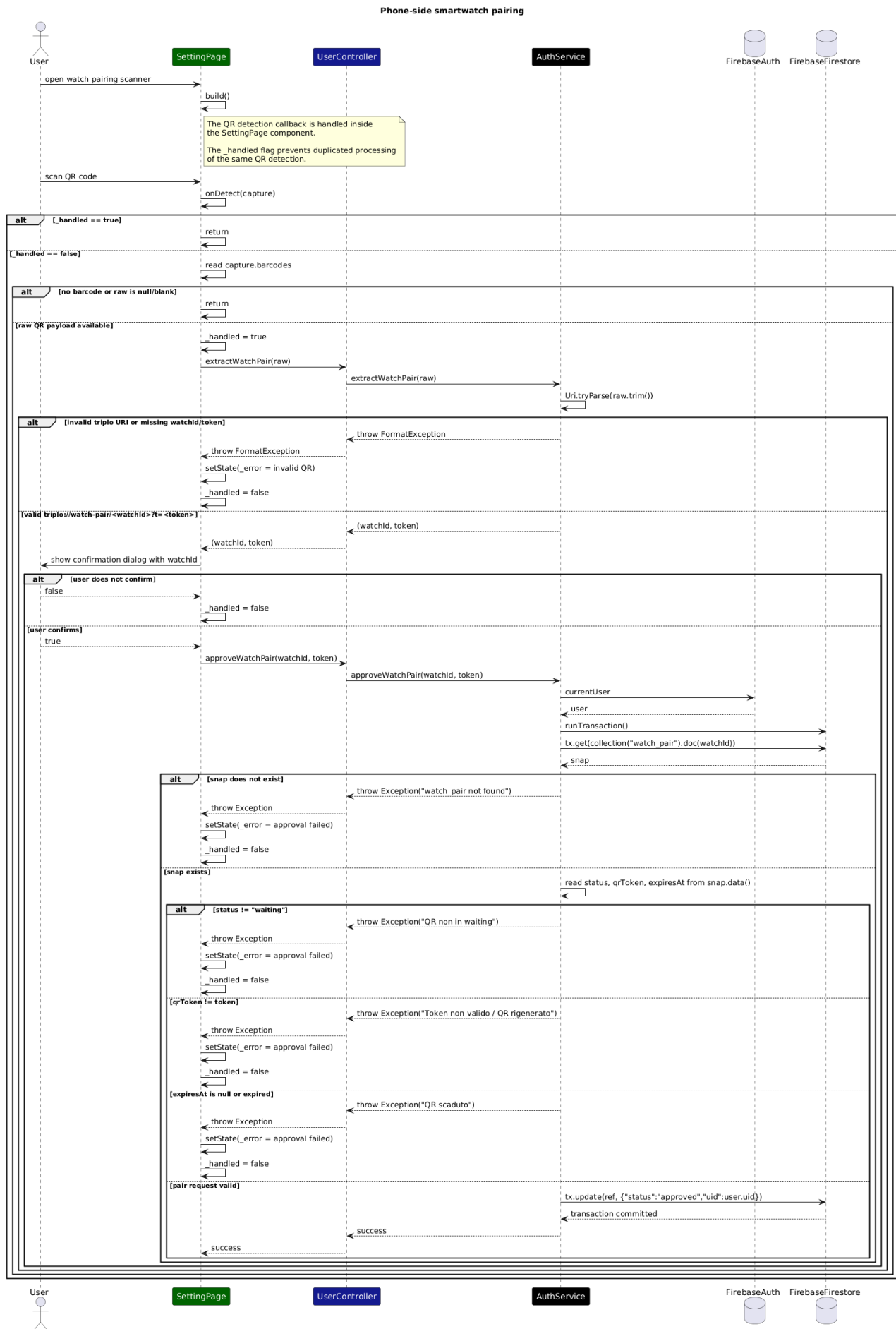


Figure 2.8: Phone-side smartwatch pairing sequence diagram

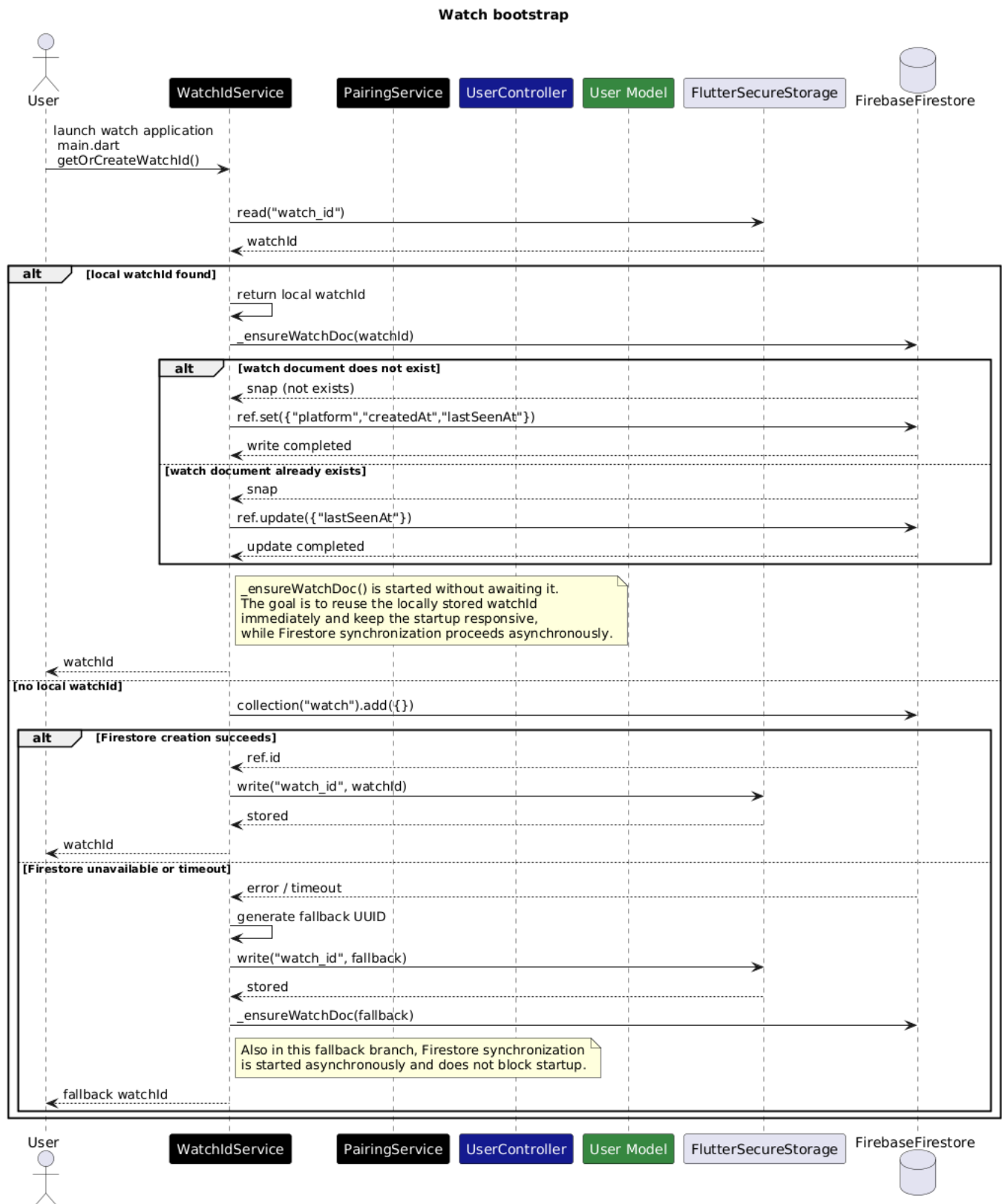


Figure 2.9: Watch bootstrap sequence diagram

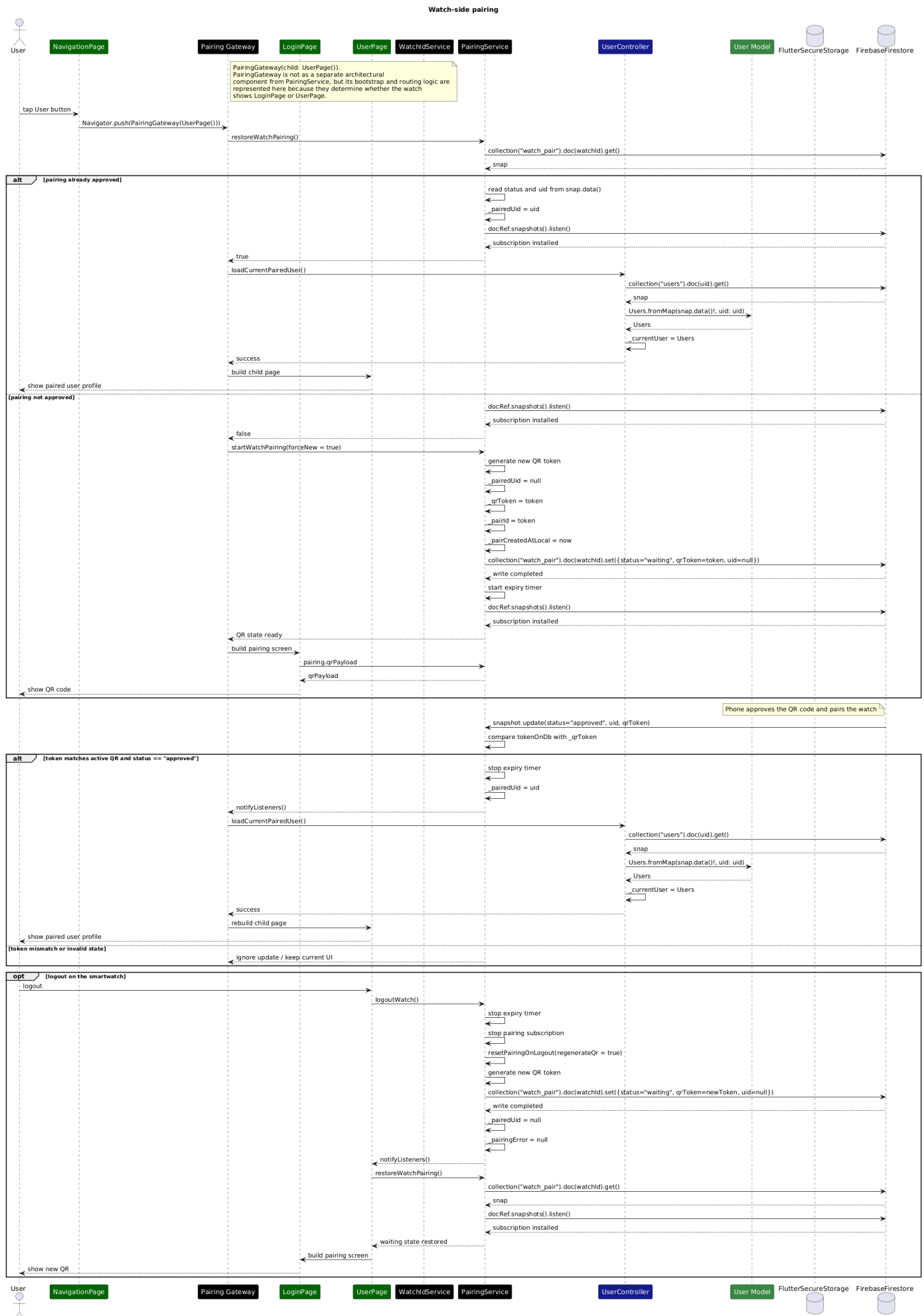


Figure 2.10: Watch-side smartwatch pairing sequence diagram

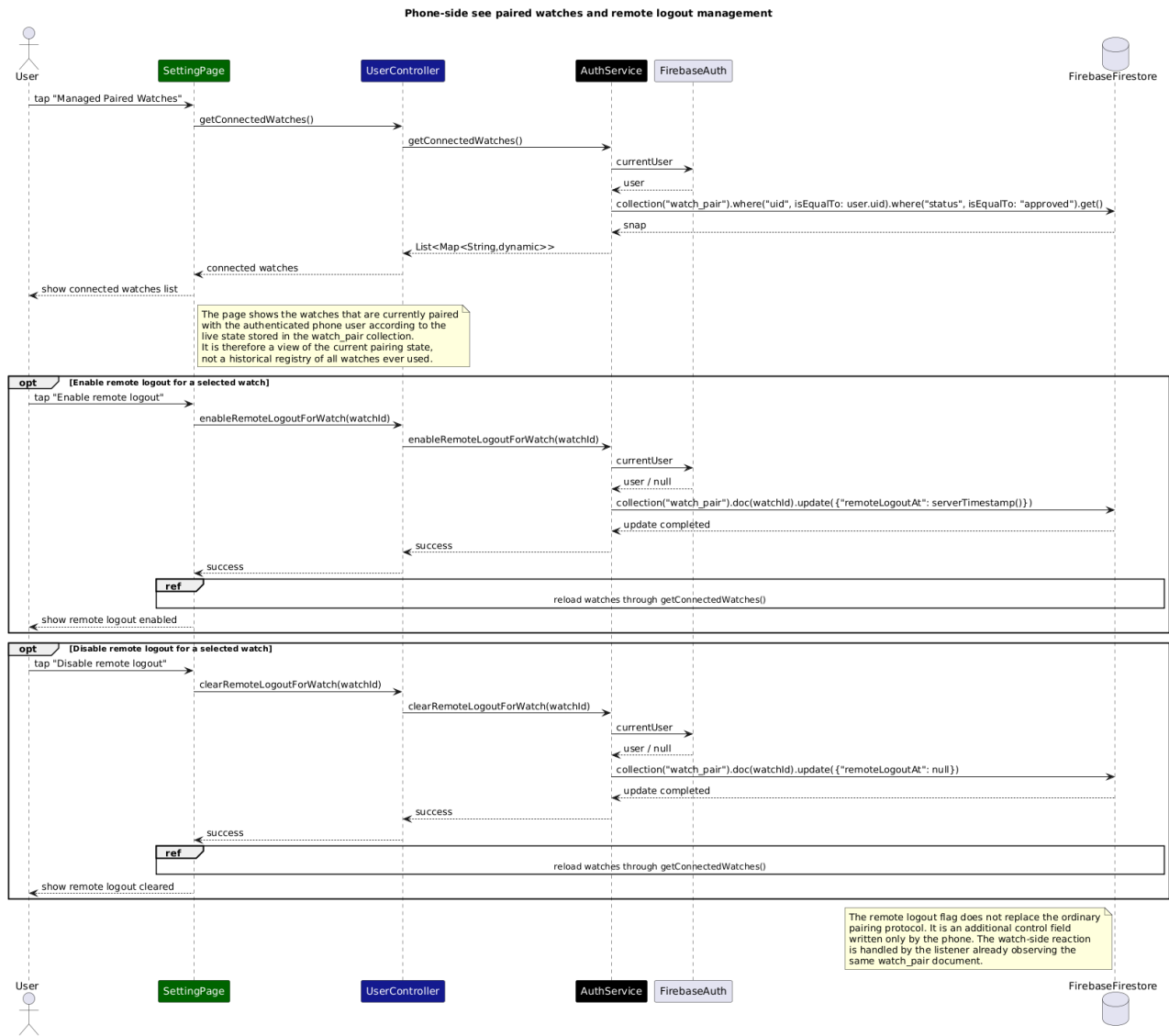


Figure 2.11: Phone-side view paired watches and remote logout management sequence diagram

2.6 Component interfaces

2.6.1 Component Interfaces Smartphone Application

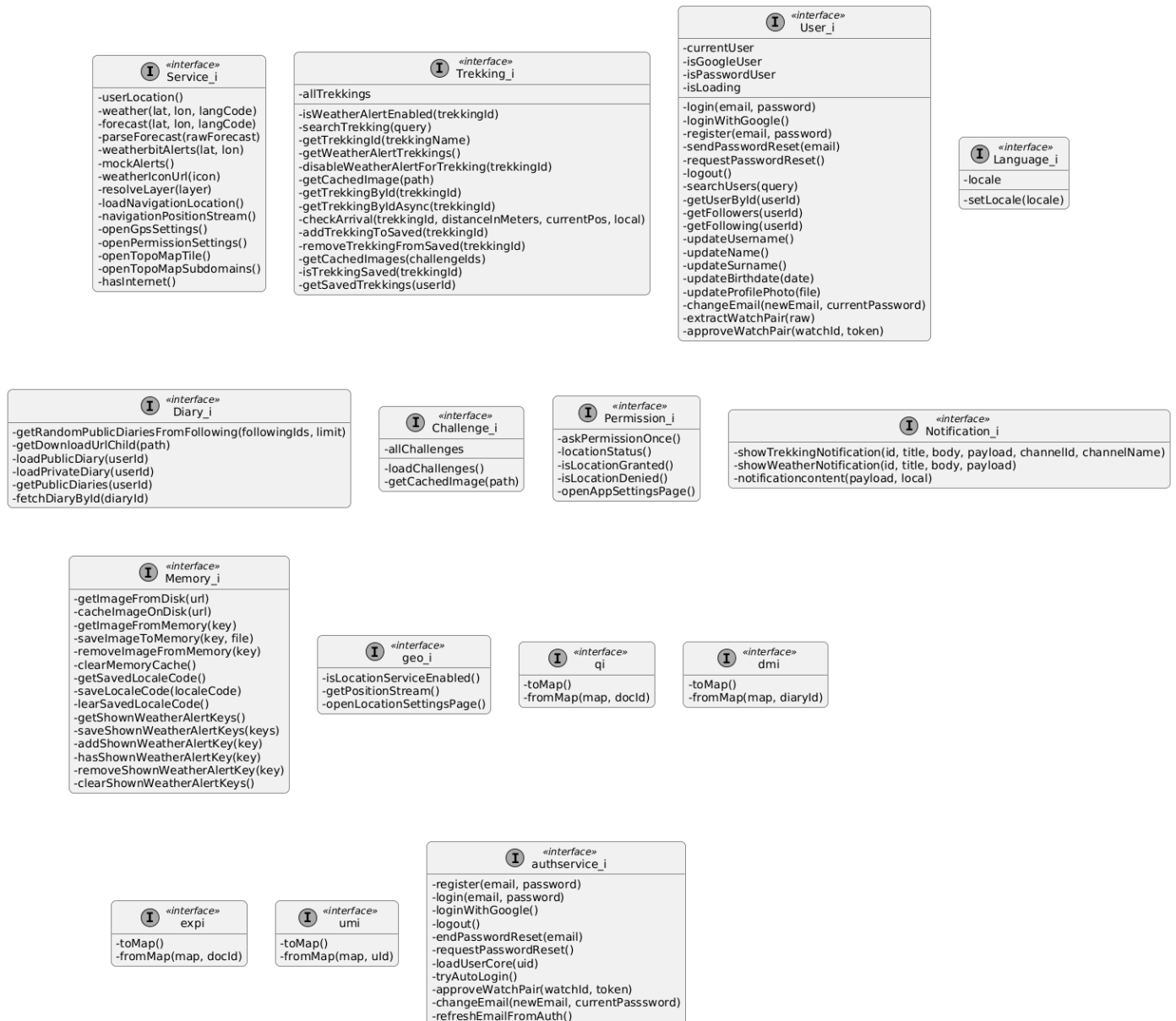


Figure 2.12: Interface diagram smartphone application

The interfaces shown in Figure 2.6.1 should not be interpreted as direct one-to-one representations of language-level constructs. Rather, they are used as an architectural abstraction to explicitly model the set of operations that each component exposes to the rest of the system. Figure 2.6.1 presents the main interfaces exposed by the application components. These interfaces define the contracts through which the different parts of the system interact, making the architecture more modular.

The diagram includes a first group of interfaces associated with the core application domains.

User_i defines the operations related to user profile management, following and followers data.

Trekking_i groups the operations required to retrieve trekking data, search routes, manage saved treks.

Diary_i exposes the operations related to diary retrieval and visualization.

Challenge_i provides access to challenge data.

Finally, **Language_i** defines the minimal contract required to manage the application locale.

A second group of interfaces regards infrastructure and platform interaction. **Service_i** acts as

the unified access point for a heterogeneous set of services that are not strictly tied to a single domain entity. As shown in the diagram, this interface includes operations for weather retrieval, forecast parsing, weather alerts, navigation-related location handling, access to map tiles, and redirection to system configuration components such as GPS and permission settings. For this reason, `Service_i` can be interpreted as a façade-oriented interface, collecting in a single access point the functionalities needed by the UI when interacting with external services and operating-system-level capabilities.

The operating system services are further represented through dedicated interfaces, namely `Permission_i`, `Notification_i`, `Memory_i`, and `geo_i`. These interfaces isolate the logic that depends on Flutter packages that communicate with device APIs or with the operating system. This design choice is particularly relevant from a maintenance point of view: if a package or platform-specific integration must be replaced, the impact can be limited to the corresponding service implementation, without requiring modifications in the rest of the application, as explained in section 2.8.4. In this sense, the interfaces shown in Figure 2.6.1 contribute to preserving a clear separation between application logic and platform-dependent concerns.

More specifically, `Permission_i` encapsulates permission requests.

`geo_i` abstracts location-service availability and position streaming

`Notification_i` defines the operations needed to issue local notifications

and `Memory_i` centralizes local storage concerns, including image caching, saving the preferred locale.

The diagram also contains a set of lightweight model-oriented interfaces, namely `qi`, `dmi`, `exp_i`, and `umi`. These define conversion contracts based on `toMap()` and `fromMap(...)` operations. Their purpose is to standardize the serialization and deserialization logic used to map application objects to and from Firestore representations.

In addition to the previously described interfaces, Figure 2.6.1 also includes the `authservice_i` interface, which encapsulates the operations related to user authentication and credential management.

This interface defines a dedicated contract for handling authentication workflows, including registration, login (both with email/password and Google Sign-In), logout, password reset, and automatic session restoration. Furthermore, it includes operations for account-related updates, such as email changes and smartwatch pairing approval.

From an architectural perspective, `authservice_i` plays a key role in separating authentication from the higher-level domain logic exposed by `User_i`. While `User_i` provides user-related functionalities at the application level, `authservice_i` is responsible for interacting with the underlying authentication provider (e.g., Firebase Authentication), thus centralizing all authentication-related operations in a single component.

This design choice improves modularity and maintainability, as it prevents other components from directly depending on authentication mechanisms and allows the authentication logic to be modified or extended with minimal impact on the rest of the system.

Overall, the interfaces represented in Figure 2.6.1 document a design in which each major responsibility is accessed through a clearly delimited contract. This improves readability of the architecture, supports separation of concerns, and makes explicit which services are expected from each component without exposing internal implementation details. For these reasons, the interface view complements the component decomposition by showing not only which elements exist in the system, but also how their responsibilities are formally made available to the rest of the application.

2.6.2 Component interfaces smartwatch application

2.7 Selected architectural styles and patterns

2.7.1 Architectural Patterns: Model–View–Controller with Mediator and Observer

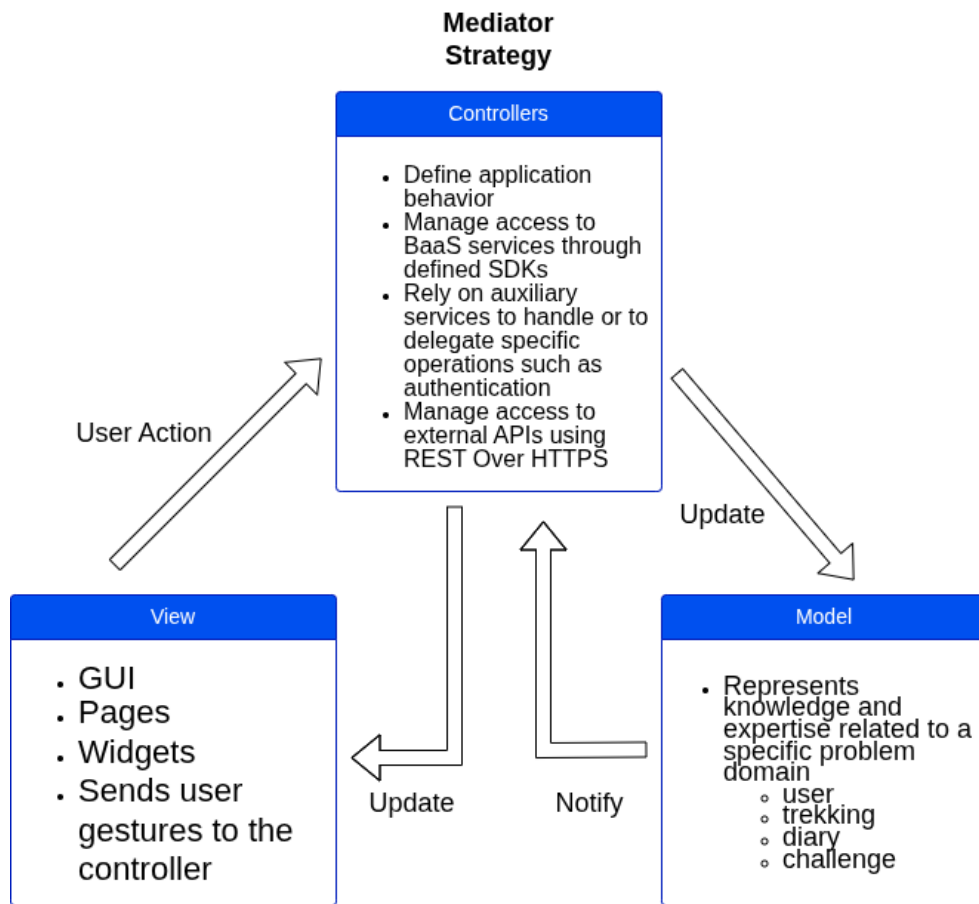


Figure 2.13: Model view controller pattern, adapted from Apple and Oracle documentation

The system adopts the Model–View–Controller (MVC) architectural pattern to separate presentation, application logic, and data management. Within this structure, each role is assigned a specific responsibility in order to improve modularity, maintainability, and clarity of interactions.

The **Model** represents the knowledge and expertise related to the application domain. It includes the core data structures of the system, such as **Diary**, **Trekking**, and **User**. The Model is independent from the user interface and does not handle user interactions directly.

The **View** is responsible for rendering the user interface and collecting user input. Its role is limited to presentation and interaction handling, without directly applying business logic or modifying domain data.

The **Controller** manages the application logic and coordinates the interaction between View and Model. In the adopted variant of MVC, consistent with the interpretation described in both Apple and Oracle documentation, the Controller also plays the role of a *Mediator*. More precisely, the View does not directly manipulate the Model; instead, user actions are forwarded to the Controller, which interprets them, executes the required logic, and updates the corresponding Model data. In this way, the Controller acts as the central coordination point of the interaction, reducing direct dependencies between presentation and domain data.

In addition, the architecture relies on the **Observer pattern** to keep the user interface consistent with state changes. When the state managed by a Controller is updated, the interested Views are notified through the listener mechanism provided by the Flutter framework, based on `ChangeNotifier`, integrated via `Provider`, so that they can refresh their representation accordingly. This notification-based interaction avoids tight coupling and supports consistency across multiple dependent Views.

Therefore, the adopted solution combines the structural organization of MVC with the use of the Mediator pattern in the Controller and the Observer pattern. This combination is particularly suitable for the Flutter-based implementation of the system, where the presentation layer, the application logic, and the domain data remain clearly separated while still cooperating through a well-defined interaction model.

2.7.2 ServiceController as a Facade

The ServiceController is designed following the Facade pattern.

It provides a unified interface that aggregates multiple underlying services, including external APIs and OS-level services. Instead of interacting directly with heterogeneous components, UI components and controllers rely on a single access point.

This design reduces coupling and simplifies the interaction model of the system. It also improves readability and consistency, as all non-domain-specific functionalities are accessed through a centralized component.

Furthermore, the facade allows internal changes to the underlying services without impacting the rest of the application, thus increasing flexibility and maintainability.

2.8 Other Design Decisions

2.8.1 Flutter as Application Framework

Flutter was selected for the implementation of the client applications because it provides an effective balance between cross-platform development, interface consistency, and maintainability. A primary requirement of the project was the possibility of deploying the smartphone application on both Android and iOS while preserving the same visual structure and interaction style across platforms. In this respect, Flutter offers a significant advantage, since it enables the development of a single shared codebase while also providing a uniform UI across platforms.

Compared with native alternatives such as Kotlin and Swift, Flutter avoids the need to maintain two separate implementations for Android and iOS, thus reducing duplication of development effort and simplifying maintenance. Compared with other cross-platform solutions, the choice of Flutter was particularly suitable for this project because it supports the construction of a coherent and controlled UI layer, which was important for preserving the same application identity and user experience on both mobile platforms.

2.8.2 Firebase as Backend-as-a-Service

The adoption of a Backend-as-a-Service solution represents a strategic architectural decision aimed at improving maintainability, scalability, and integration.

Firebase was adopted as the backend solution to support a scalable and robust system architecture while maintaining a clear separation between client-side logic and backend responsibilities. Rather than introducing a custom server-side layer, the system relies on a Backend-as-a-Service approach, where core functionalities are delegated to managed cloud services.

This choice allows the architecture to focus on the coordination of application logic on the client side, while relying on well-established and production-ready services for critical backend responsibilities such as authentication, data persistence, and media storage, integrating with the client applications through platform-specific SDKs.

From an architectural perspective, this approach improves modularity and reduces system complexity by eliminating the need to design and maintain a dedicated backend infrastructure. This approach presents notable advantages, including scalability, reliability, and security, through managed services that are designed to handle distributed workloads and concurrent access patterns.

Furthermore, the use of Firebase aligns well with the multi-platform nature of the system, as both smartphone and smartwatch applications can interact with the same backend services in a consistent and unified manner. This contributes to maintaining a coherent data model across different devices.

2.8.3 Phone Smartwatch Pairing

While the smartphone application directly relies on Firebase Authentication, the smartwatch application does not perform authentication independently. Instead, it adopts a pairing mechanism based on QR code scanning. Through this process, the smartphone securely associates an authenticated user with the smartwatch by storing the corresponding identifier in Firestore.

This design choice provides several advantages:

- avoids handling user credentials on the smartwatch or to enforce specific authentication methods, such as Google Sign-In, reducing the exposure of sensitive data on a constrained and potentially less secure device
- simplifies the user interaction model, eliminating the need for credential input on devices where text entry is inherently limited and less user-friendly

For these reasons, the interaction is intentionally defined as a pairing process rather than a traditional login, reflecting both its security properties and its role within the overall system architecture.

2.8.4 OSService Package Design

As shown in the smartphone component diagram (see Figure 2.2.2), the OSService package has been introduced to isolate all interactions with operating system functionalities and external libraries.

In particular, this package includes the following services:

- **PermissionService**, responsible for handling runtime permissions;
- **GeoService**, responsible for accessing device location;
- **MemoryService**, responsible for local storage and caching;
- **NotificationService**, responsible for managing local notifications;

Each service encapsulates the interaction with specific Flutter packages and platform APIs, such as geolocation, local storage, or notification systems. These dependencies correspond to external libraries (e.g., packages available on *pub.dev*) that internally communicate with the underlying operating system.

This design ensures that the rest of the application does not directly depend on platform-specific APIs or external libraries. Instead, all such dependencies are confined within the OSService package.

The main motivation behind this decision is maintainability. If a library or underlying implementation needs to be replaced, only the corresponding service must be modified, without requiring changes in the rest of the system. This significantly reduces the impact of technological changes.

2.8.5 Use of External Service Providers

The application relies on a set of external service providers to implement functionalities related to weather information and map visualization, which are required by several functional requirements of the system. In particular, these providers support the fulfillment of requirements related to the visualization of weather conditions associated with trekking routes, the detection of weather alerts, and the display of geographical information on maps.

From an architectural perspective, the `Service_i` interface acts as the main entry point for all operations involving third-party providers. It exposes methods such as `weather()`, `forecast()`, `weatherbitAlerts()`, and map-related operations like `openTopoMapTile()`.

This design follows a facade-oriented approach: a single interface lets other components access the required functionalities. As a result, other components interact only with abstract operations, without depending on the specific provider for the data. This approach improves modularity and ensures that changes in external services can be handled locally within the service layer, without affecting the rest of the application. Regarding weather data, two different providers have been adopted based on their respective strengths. Standard weather information and forecasts are obtained through OpenWeather, accessed via the `weather()` and `forecast()` methods. These operations are used to satisfy the functional requirements related to the visualization of current weather conditions and short-term forecasts for trekking locations.

Conversely, weather alerts are handled through Weatherbit, accessed via the `weatherbitAlerts()` operation. This choice is motivated by the availability and reliability of alert-specific data provided by Weatherbit, which are essential to fulfill the functional requirements related to user notification of critical weather conditions.

Map visualization is supported through external tile providers, such as OpenTopoMap, accessed via dedicated operations in `Service_i`. These functionalities are required to satisfy the functional requirements related to the visualization of trekking routes and geographical context. Delegating this responsibility to external providers avoids the need to manage map data internally and significantly reduces system complexity.

Alternative solutions were considered during the design phase. A first option consisted in adopting a single weather provider for both standard weather data and alerts. However, this approach was discarded due to limitations in the quality, availability, or completeness of alert data offered by individual providers. In particular, while OpenWeather provides reliable general weather information, its alert coverage is not always sufficient for the requirements of the application.

The chosen solution, based on multiple providers integrated through a unified interface, such as `Service_i`, represents a trade-off between flexibility and architectural simplicity. It allows the system to leverage the strengths of different providers while keeping responsibilities well separated and interactions with higher level components stable.

Overall, the use of external service providers, combined with the abstraction introduced by the `Service_i` interface, enables the system to satisfy the relevant functional requirements while preserving modularity, maintainability, and extensibility of the architecture.

2.8.6 Madonna di Campiglio as Application Area

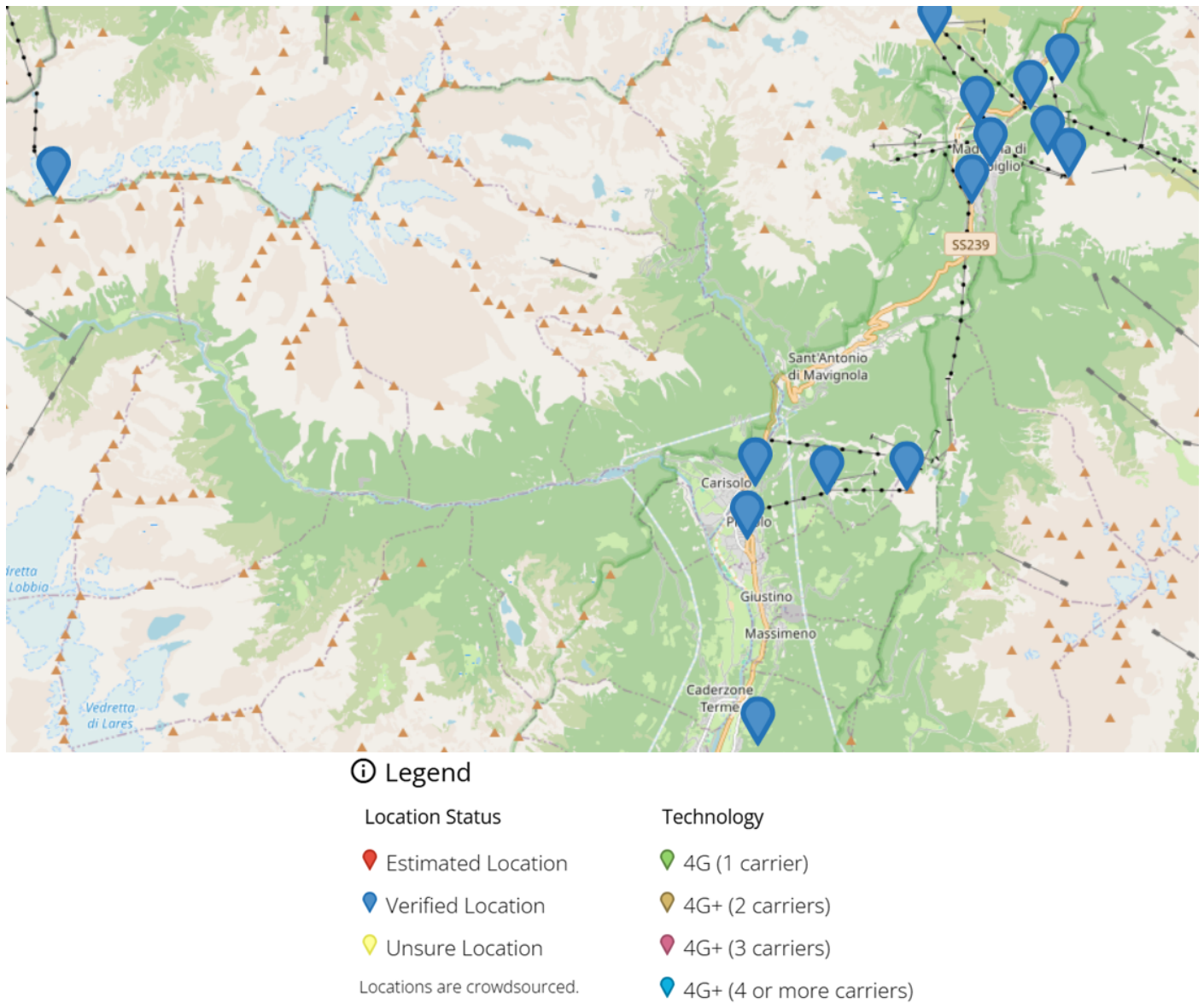


Figure 2.14: eNodeB LTE Radio Sites in the area, courtesy of IteItaly.it

The application has been evaluated for the area of *Madonna di Campiglio*. This location was selected due to its relatively good LTE network coverage compared to other mountainous regions.

Reliable connectivity is an important factor for an application that relies on external services such as weather APIs and cloud storage. The chosen area provides a realistic but favorable scenario, allowing the system to be tested under conditions that are representative of real trekking environments while still ensuring acceptable performance.

2.8.7 Network Connectivity Requirements and smartwatch Platform

The smartwatch application requires reliable access to internet connectivity in order to provide a complete and consistent user experience. In particular, network access is necessary for several core functionalities of the system.

The application relies on external services to retrieve real-time data, such as maps, which are essential for trekking activities.

Internet access enables communication with backend services and APIs, allowing the application to fetch dynamic content and maintain updated information.

Given these requirements, the choice of Wear OS as the target platform is appropriate and justified. Wear OS supports multiple network communication modes, including:

- **Bluetooth connectivity:** allows the smartwatch to access the internet indirectly through a paired smartphone.
- **Wi-Fi connectivity:** enables the device to connect directly to wireless networks, allowing standalone operation in specific scenarios.
- **Cellular connectivity (LTE with eSIM):** available on selected devices, providing full independence from the smartphone and ensuring continuous connectivity during outdoor activities.

This flexibility ensures that the application can operate under different conditions, ranging from smartphone-assisted usage to fully autonomous scenarios. In particular, the availability of LTE-enabled devices makes Wear OS especially suitable for outdoor and trekking contexts, where the user may not always have access to their smartphone.

2.8.8 Image cache strategy

A caching mechanism has been introduced to reduce latency in loading images from Firebase Storage. Since images represent one of the most bandwidth-intensive resources in the application, caching them locally significantly improves performance and responsiveness.

This decision is particularly relevant given the geographical separation between the database and storage services, as discussed in Section 2.3. By minimizing repeated downloads, the caching layer reduces both latency and network usage.

2.8.9 Firebase BaaS and GDPR compliance

From a data protection perspective, Firebase can be made compliant with GDPR regulations by offering secure data handling, encryption in transit and at rest, and configurable security rules. These features are important because they allow to enforce access control policies and protect user data, ensuring that personal information is handled according to European data protection standards. If the application is distributed commercially or made publicly available, users must be clearly informed about data processing practices through a privacy policy, and explicit consent may be required, especially when analytics services are involved or data is transferred and stored outside of the EU.

3. User Interface

3.1 Smartphone User Interface

3.1.1 Login Page

login

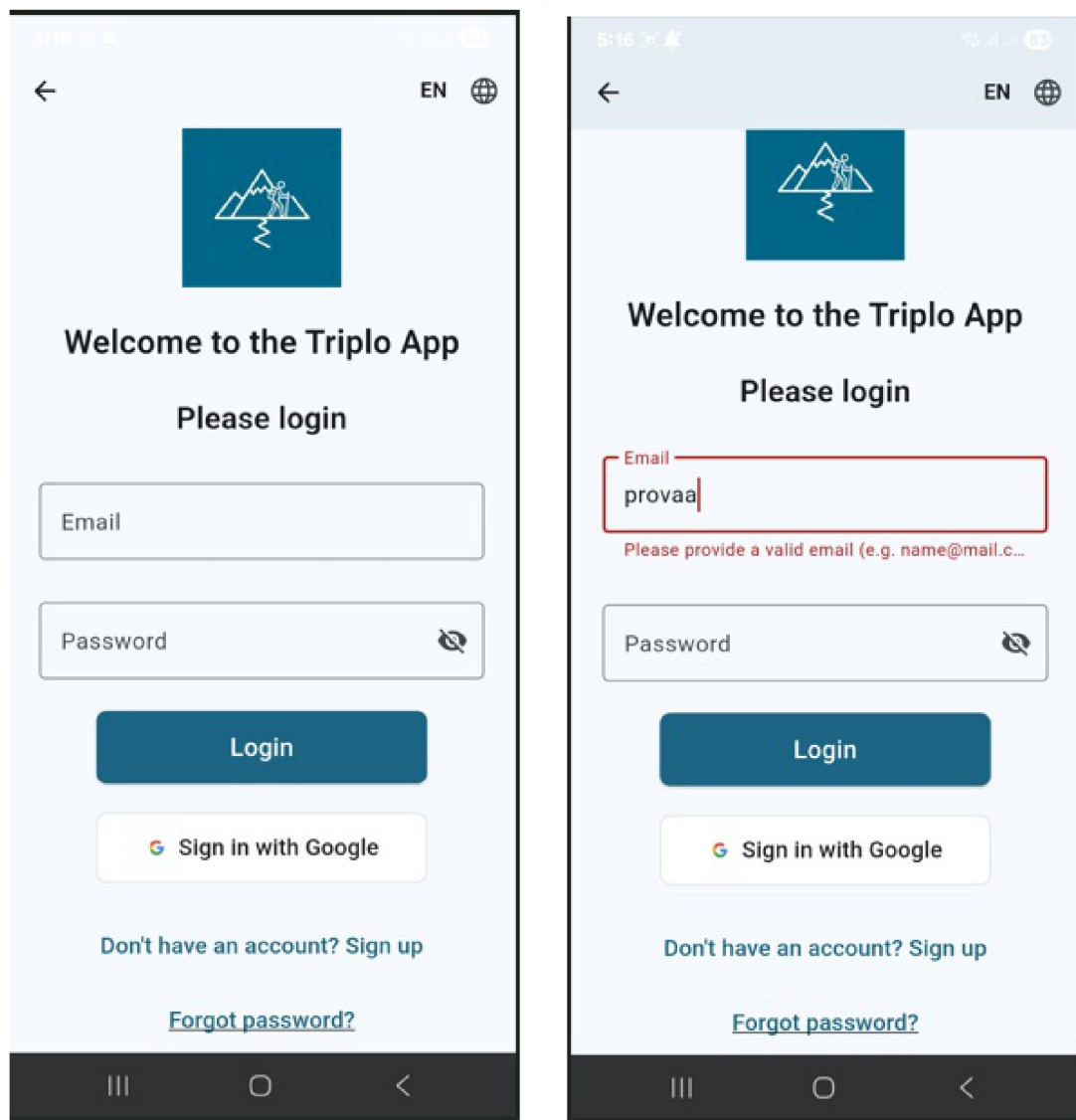


Figure 3.1: Smartphone Login Page UI

This page allows users to access the application by entering their email and password or using Google external authentication. It also provides options for password recovery and account registration.

3.1.2 Registration Page

register

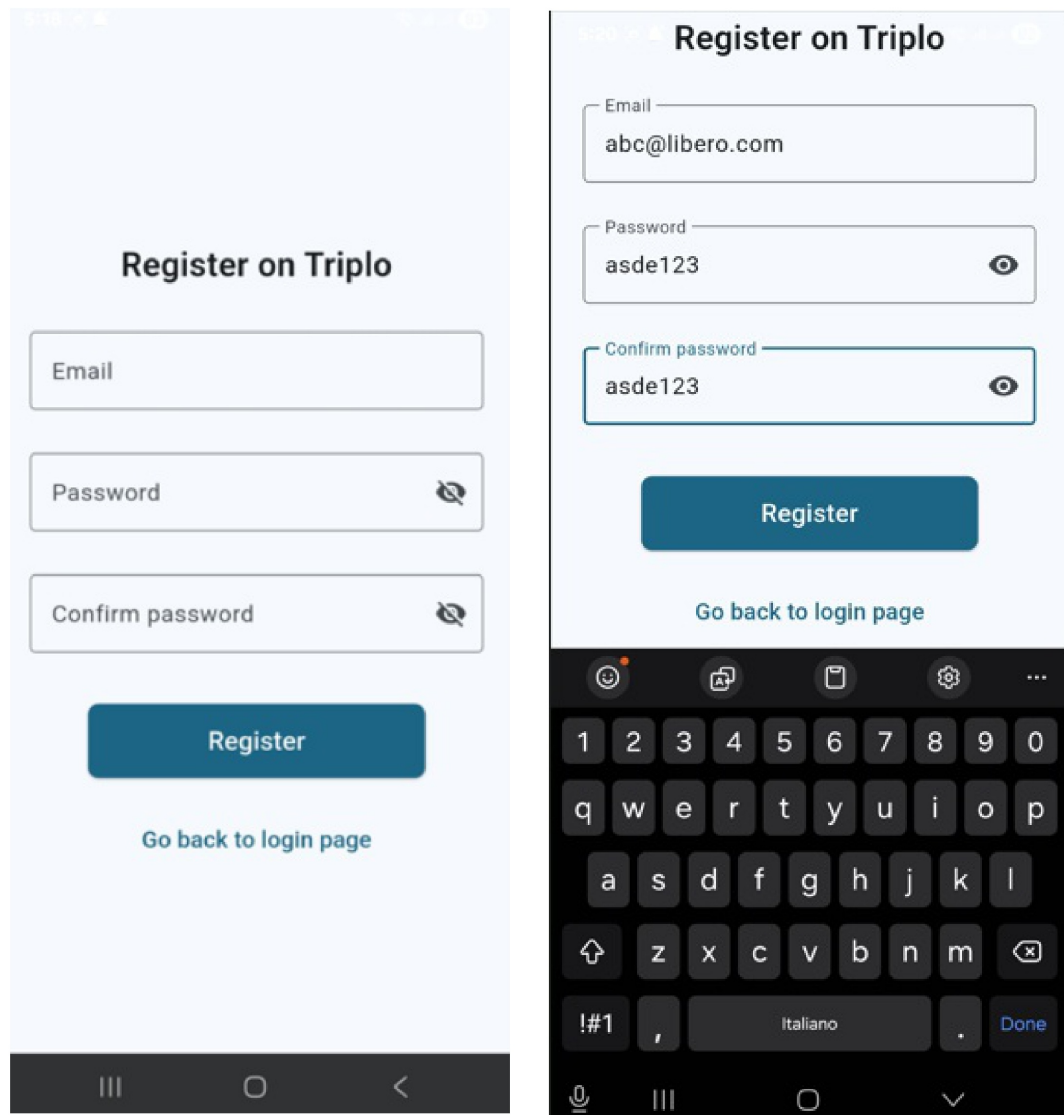


Figure 3.2: Smartphone Registration Page UI

This page enables new users to create an account by providing email, password, and password confirmation. A navigation option allows users to return to the login page.

3.1.3 Settings Page

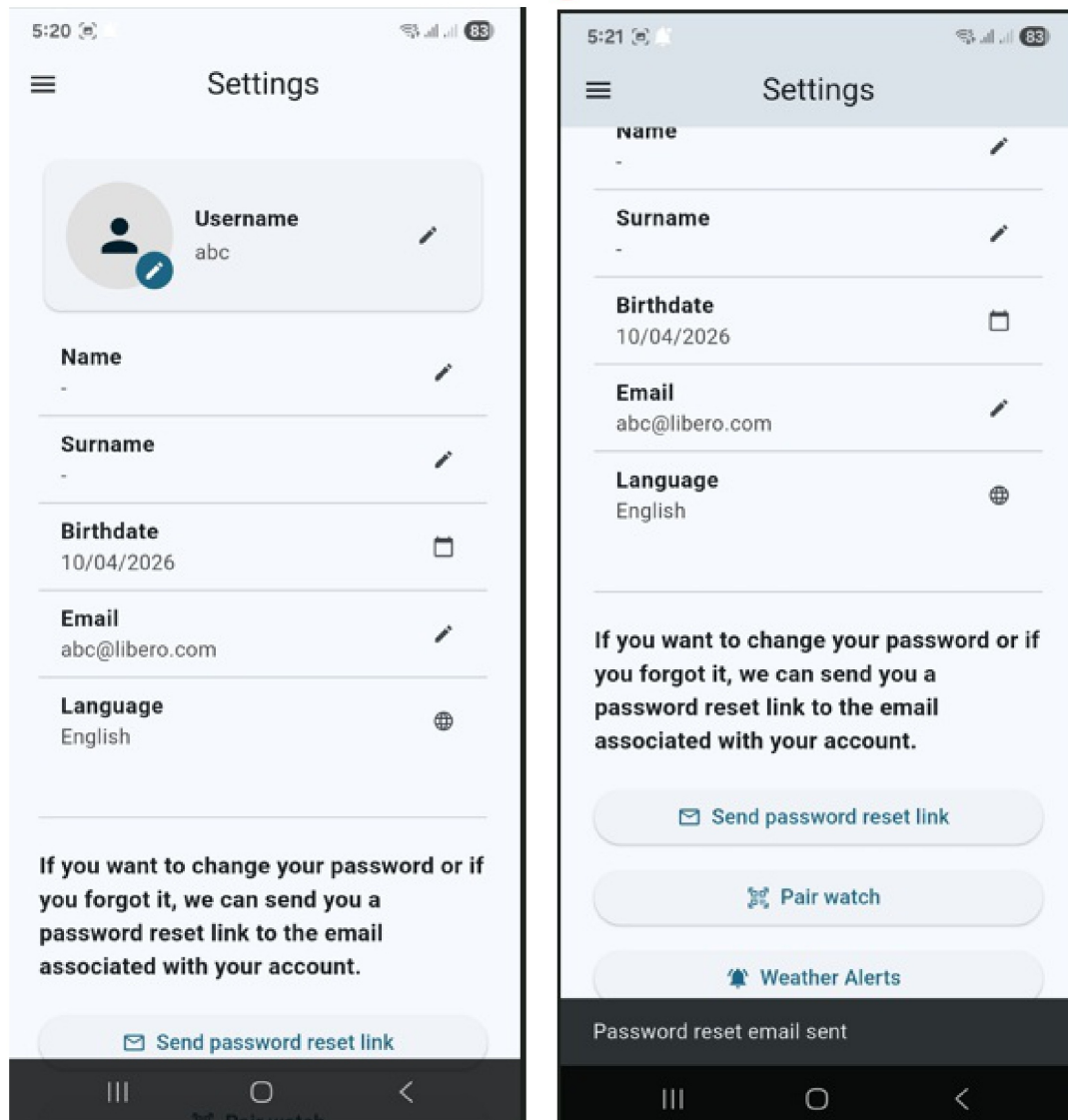
user registered

Figure 3.3: Smartphone Settings Page UI

This page displays and allows editing of user personal information such as username, name, surname, birthdate, email, and language. It also includes features like password reset, smartwatch pairing, and weather alerts configuration.

3.1.4 Offline Page

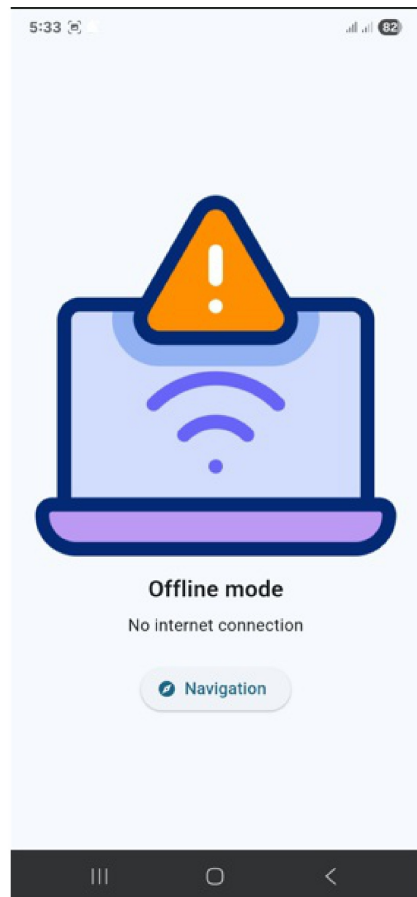


Figure 3.4: Smartphone Offline Page UI

This page is shown when no internet connection is available. It informs the user about the offline status and still allows access to essential features such as navigation.

3.1.5 Drawer Menu

drawer menu

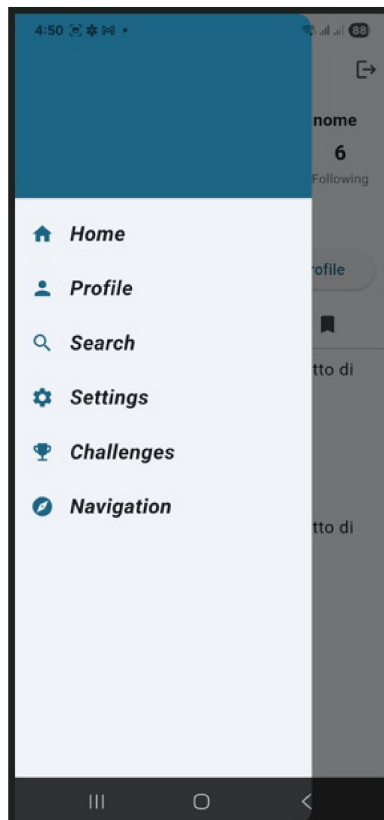


Figure 3.5: Smartphone Drawer Menu UI

The drawer menu provides access to the main sections of the app, including Home, Profile, Search, Settings, Challenges, and Navigation.

3.1.6 Home Page Path Exploration

home - path exploration

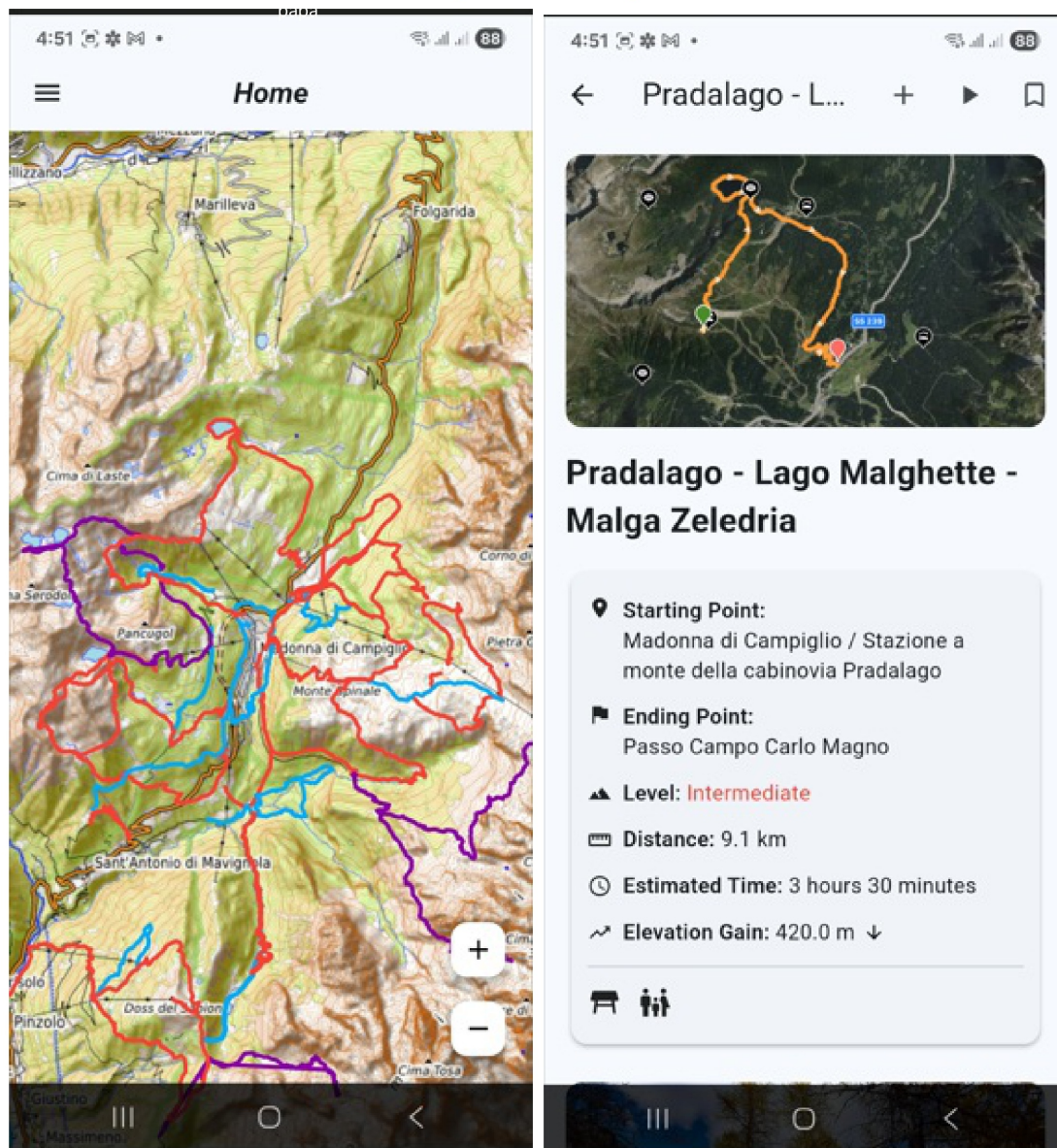


Figure 3.6: Smartphone Home Page UI

This is the main page of the application. It displays a map with available trekking routes and allows users to explore paths and select one to view detailed information.

3.1.7 Trekking Page trail details

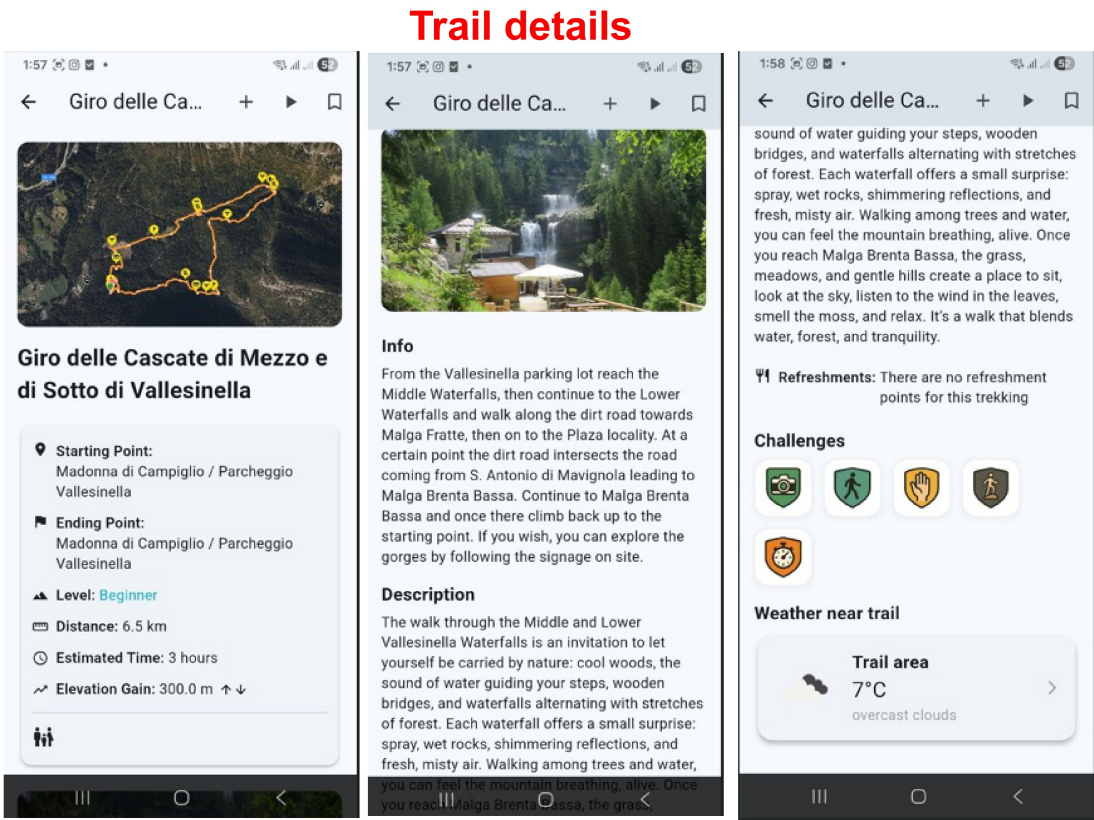


Figure 3.7: Smartphone trekking Page UI

This page shows detailed information about a selected trail, including starting and ending points, difficulty level, distance, estimated time, elevation gain, description, challenges, and weather conditions.

3.1.8 User Page

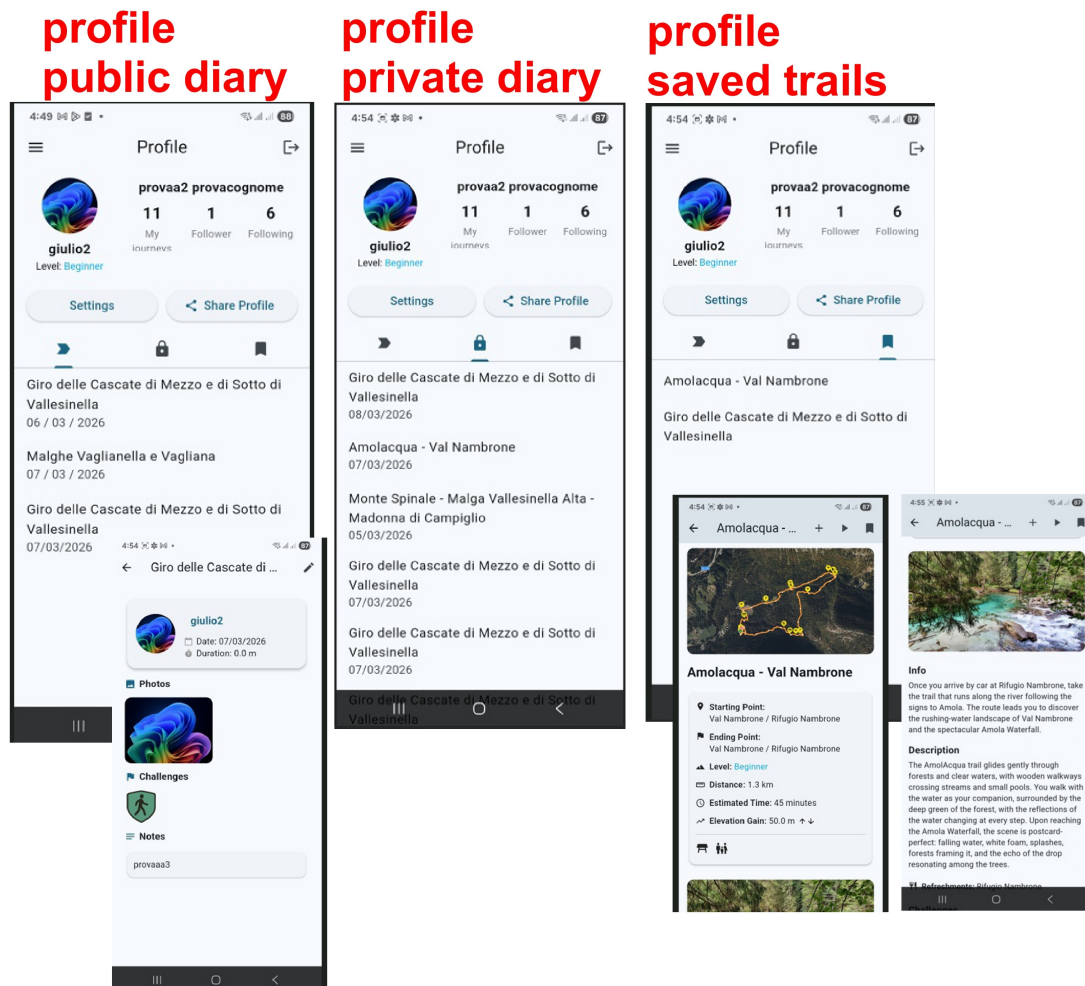


Figure 3.8: Smartphone User Page UI

This section allows users to view: Public diaries, Private diaries, Saved trails. It provides a structured overview of the user's activities and saved content.

3.1.9 Search Page to search other users or trails

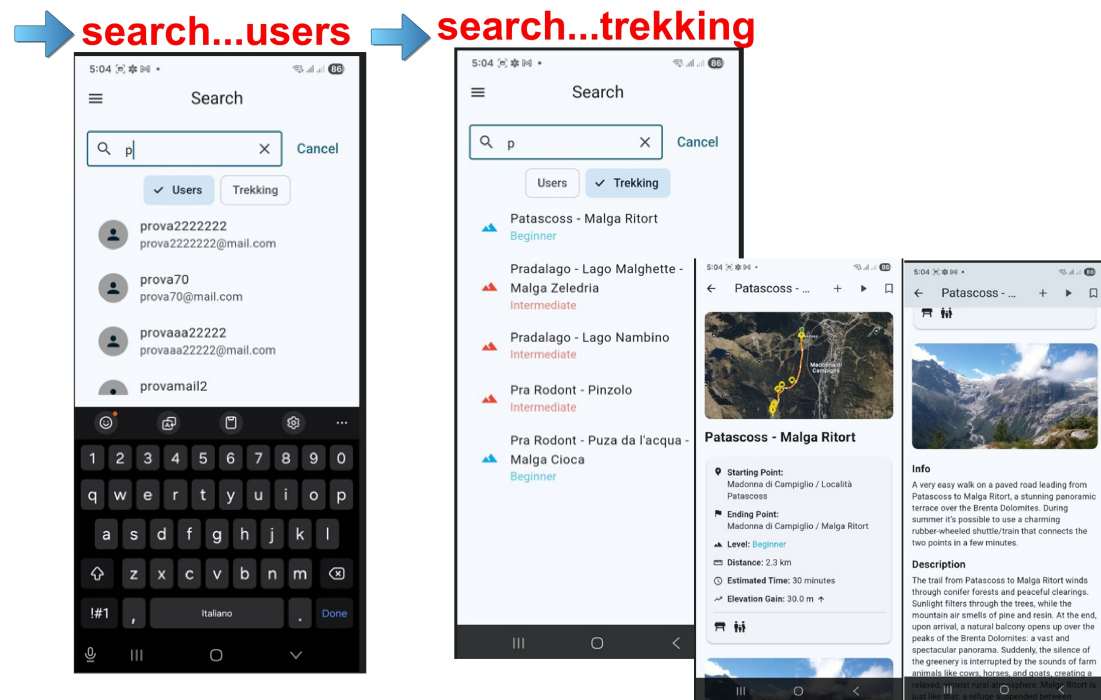


Figure 3.9: Smartphone Search Page UI

This page allows users to search for: Other users, Trekking routes. It displays results dynamically and allows navigation to detailed views.

3.1.10 Search Page to see random diaries published by followed users

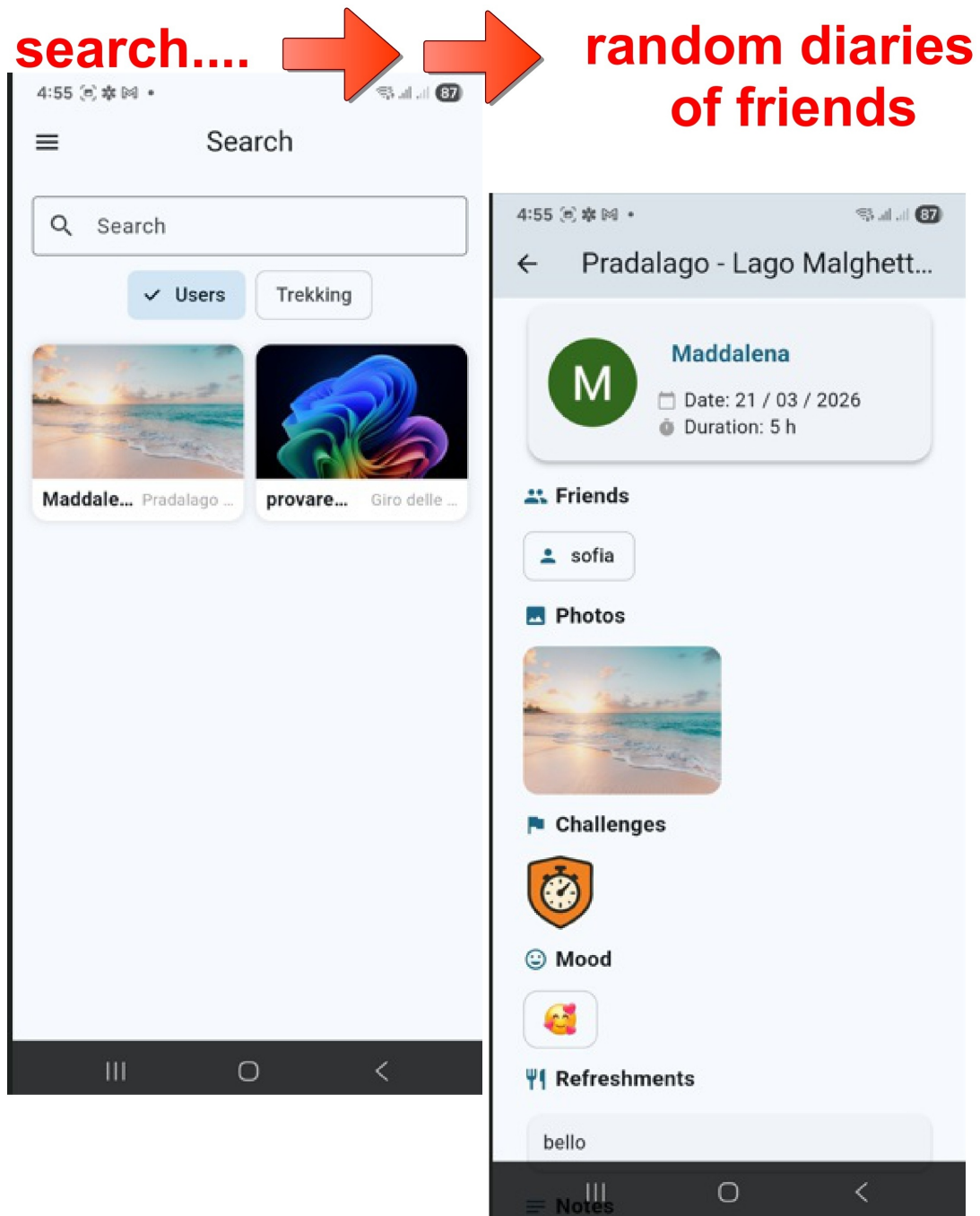


Figure 3.10: Smartphone Search Page Diaries UI

If the user does not type anything, the search page shows random diaries published by followed users.

3.1.11 Add Diary Page

add to public diary profile

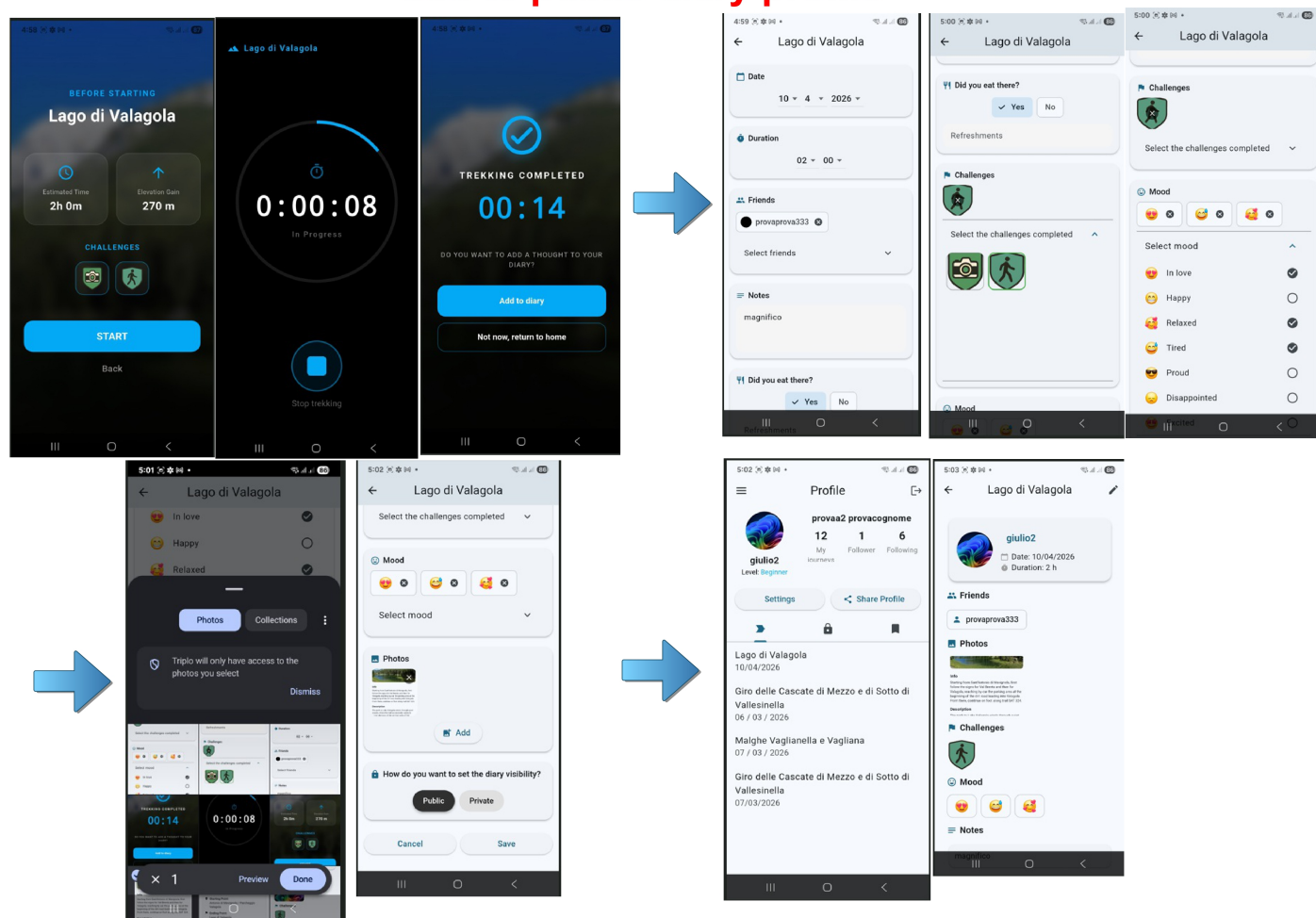


Figure 3.11: Smartphone Modify Diary Page UI

This page allows users to create a diary entry after completing a trekking activity. Users can add duration, photos, challenges, mood, and other details before saving the diary.

3.1.12 Weather Notification Preferences Page

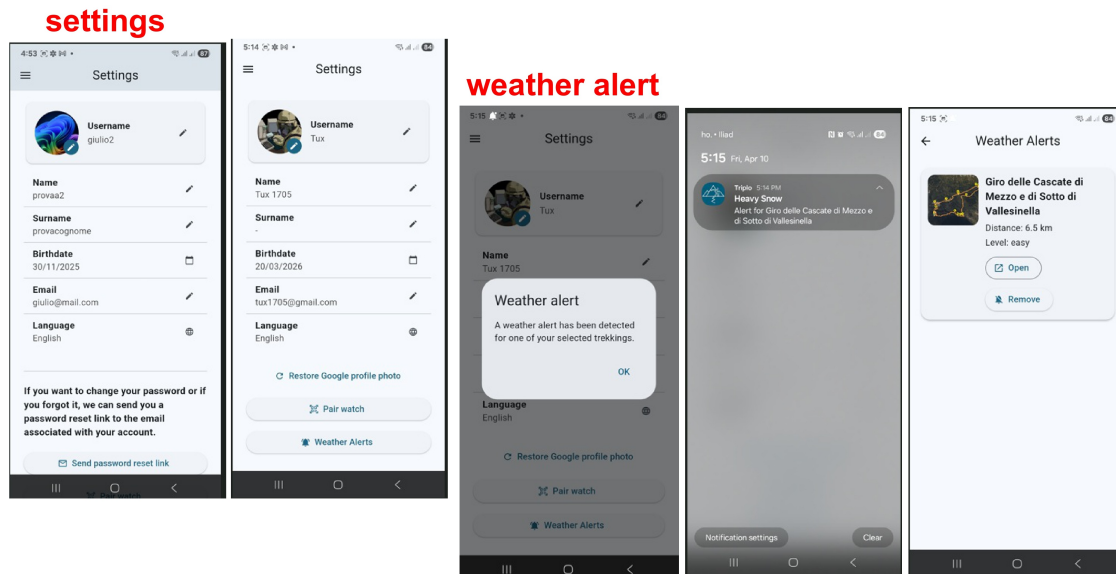


Figure 3.12: Smartphone Weather Notification Preferences Page UI

This page allows users to manage application settings and see for which trails weather alert notifications are enabled and possibly disable them for a path.

3.1.13 Application Language Dialog UI

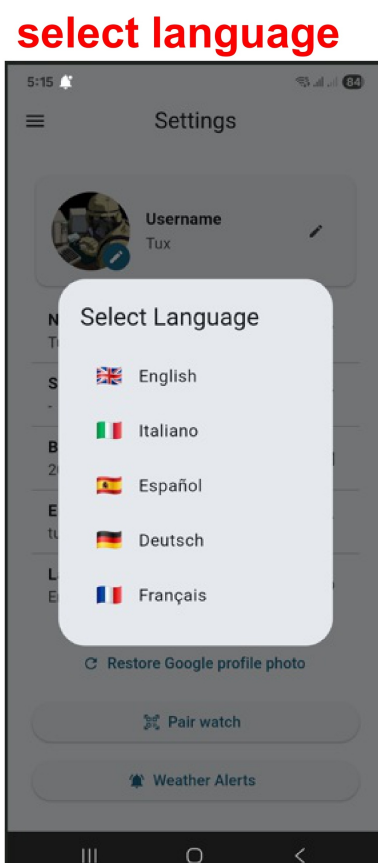


Figure 3.13: Smartphone Application Language Dialog UI

This dialog allows users to select the application language from multiple available options.

3.1.14 Overview of the challenges UI

challenges

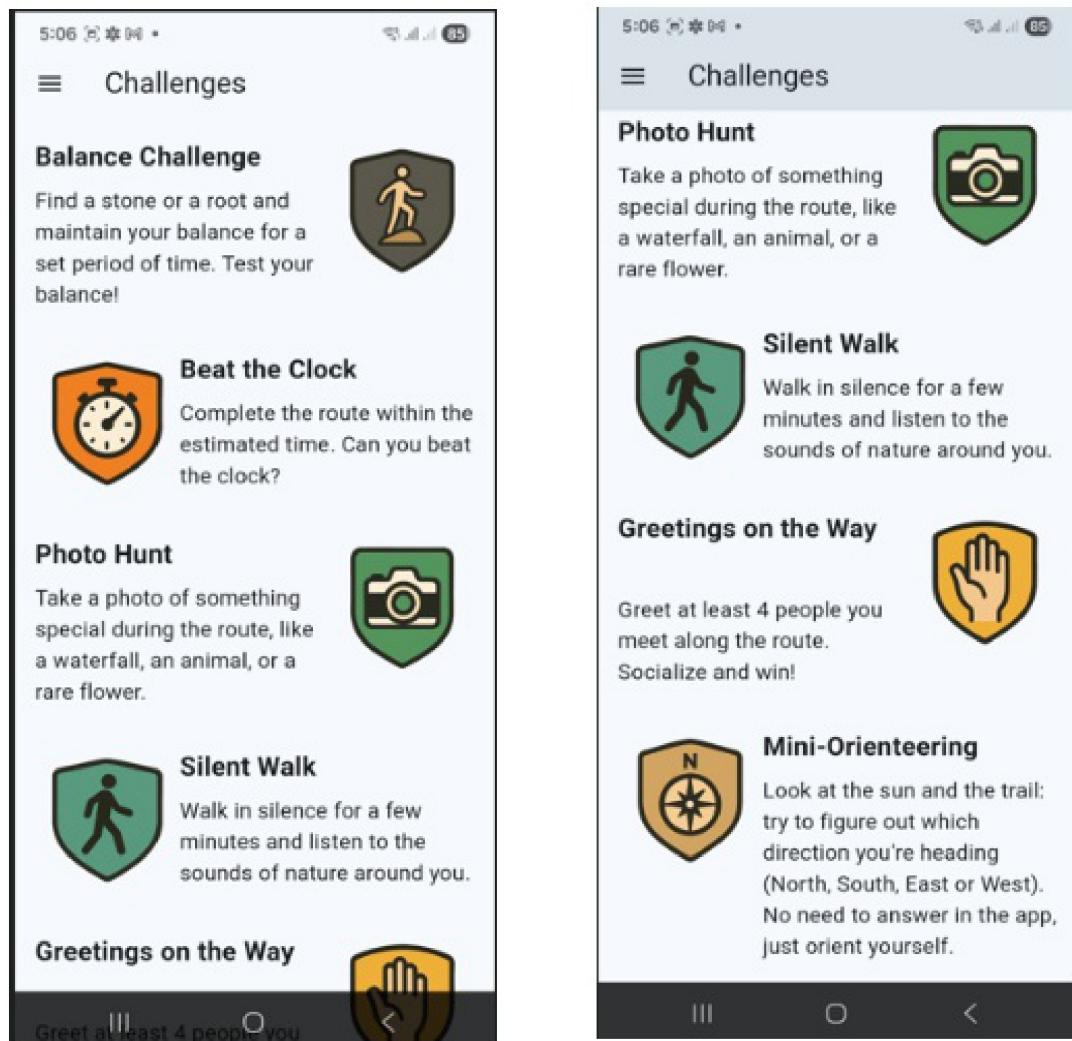


Figure 3.14: Smartphone challenge Page UI

This page displays available challenges that users can complete during trekking activities, such as time-based challenges or exploration tasks.

3.1.15 Navigation Page

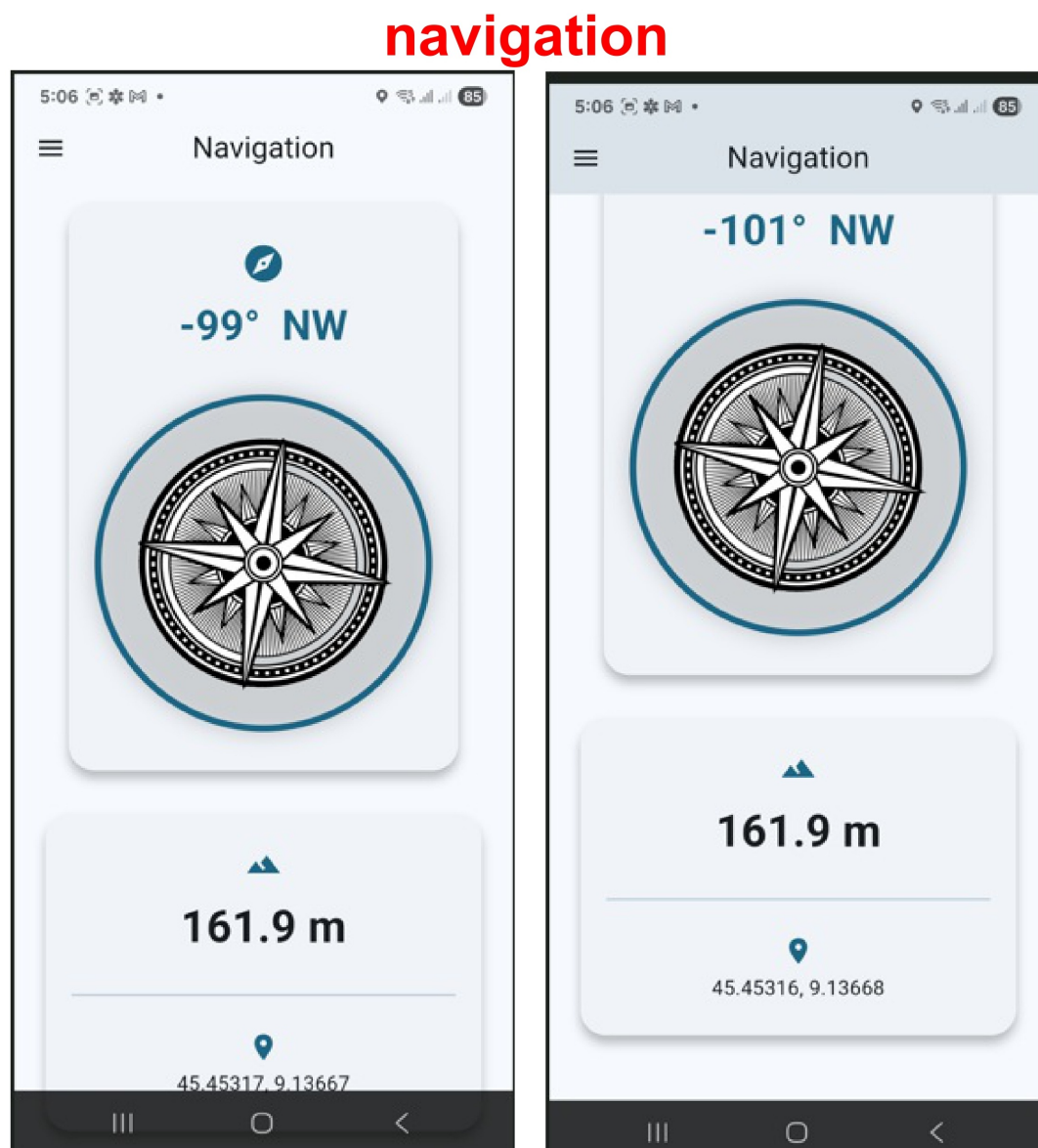


Figure 3.15: Smartphone Navigation Page UI

This page provides real-time navigation information, including compass direction, altitude, and GPS coordinates.

3.1.16 Geowatch Page

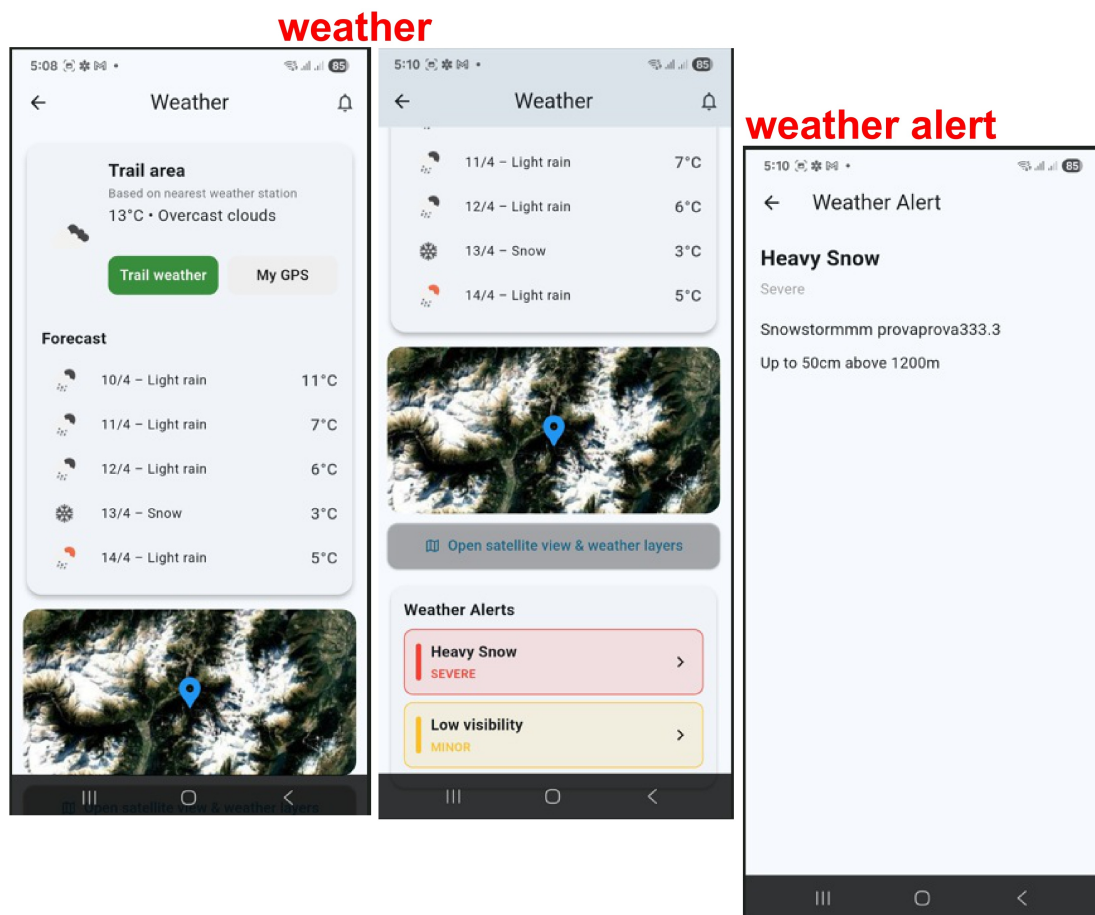


Figure 3.16: Smartphone Geowatch Page UI

This page displays weather information related to a selected trail or the user's position, including forecasts and current conditions and shows detailed information about weather alerts, such as severe conditions, including description and severity.

3.1.17 Google Satellite Page with weather layers

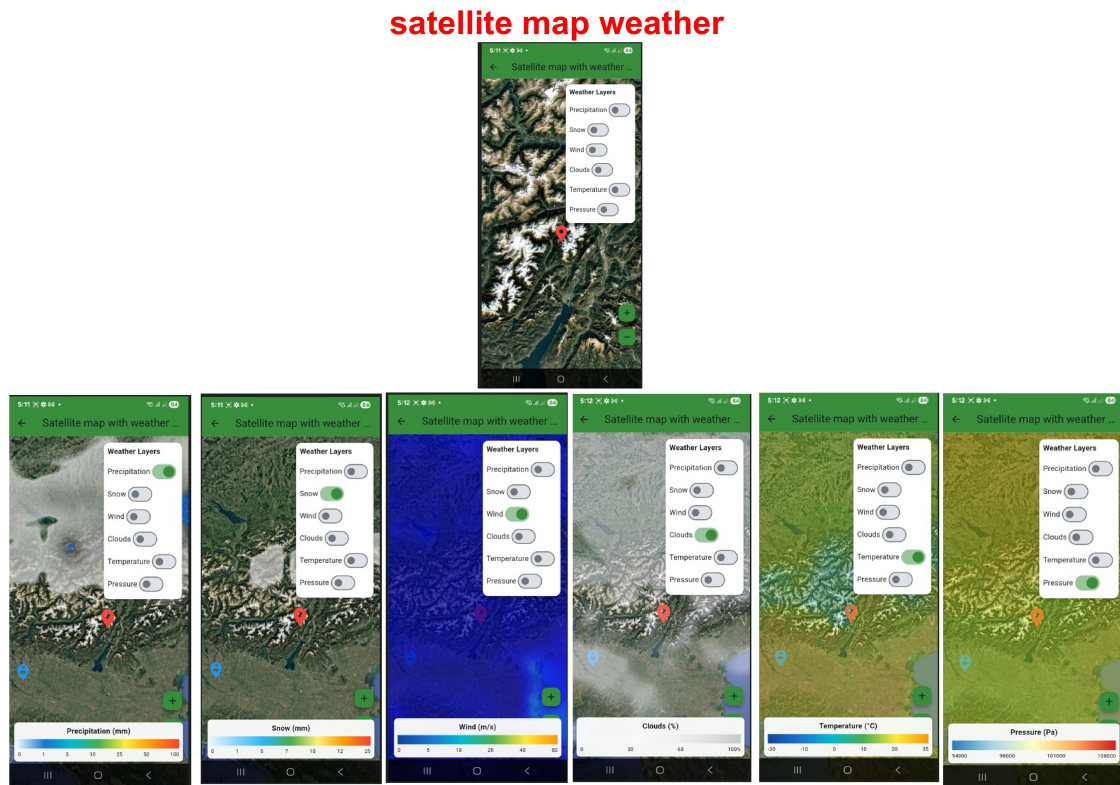


Figure 3.17: Smartphone Registration Page UI

This page displays a satellite map with optional weather overlays, for example precipitation, clouds, temperature, allowing users to visualize environmental conditions.

4. Functional Requirements

4.1 Smartphone Application Functional Requirements

4.1.1 Account and Profile Management

- FR1** The system shall allow users to register by providing email and password
- FR2** The system shall allow users to register using an external authentication provider
- FR3** The system shall allow users to log in using email and password
- FR4** The system shall allow users to log in using an external authentication provider
- FR5** The system shall allow users to reset their password
- FR6** When registration is successful, the system shall create a user profile
- FR7** When registration is successful, the system shall assign a default username based on the user's email
- FR8** The system shall allow users to update their username
- FR9** The system shall allow users to update their profile image
- FR10** The system shall allow users to update their name
- FR11** The system shall allow users to update their surname
- FR12** The system shall allow users to update their birth date
- FR13** When users are registered with email and password, the system shall allow them to update their email address
- FR14** When users are authenticated through an external provider, the system shall allow them to restore their profile image from the provider
- FR15** The system shall allow users to change the application language
- FR16** The system shall allow users to pair a smartwatch with their account
- FR17** The system shall allow users to enable weather notifications for selected routes

4.1.2 Home, Routes and Route Details

- FR18** The system shall display routes on a map
- FR19** The system shall classify routes according to their difficulty level
- FR20** The system shall display routes using different colors based on their difficulty level
- FR21** When a user selects a route, the system shall display detailed information about the selected route
- FR22** The system shall display route information including starting point, ending point, difficulty level, length, estimated time, and elevation gain
- FR23** The system shall indicate the suitability of a route for families
- FR24** The system shall display a textual description of the route
- FR25** The system shall display refreshment points associated with the route
- FR26** The system shall display challenges associated with the route
- FR27** The system shall display summary weather information for the selected route
- FR28** The system shall allow users to save a route to their profile
- FR29** The system shall allow users to remove a saved route from their profile
- FR30** The system shall allow users to start a navigation session for a selected route
- FR31** When a navigation session ends, the system shall suggest creating a diary entry for the route

4.1.3 Diary Management

- FR32** The system shall allow users to create a diary entry associated with a selected route
- FR33** The system shall allow users to specify the date of the activity in a diary entry
- FR34** The system shall allow users to specify the duration of the activity in a diary entry
- FR35** The system shall allow users to select followed users who participated in the activity
- FR36** The system shall allow users to add textual notes to a diary entry
- FR37** The system shall allow users to indicate the presence of refreshment stops during the activity
- FR38** The system shall allow users to select completed challenges in a diary entry
- FR39** The system shall allow users to select a mood associated with the activity
- FR40** The system shall allow users to upload images to a diary entry
- FR41** The system shall allow users to mark a diary entry as public or private
- FR42** The system shall allow users to save a diary entry
- FR43** The system shall allow users to discard a diary entry before saving it

4.1.4 Search and Challenges

- FR44** The system shall allow users to search for other users
- FR45** The system shall allow users to search for routes
- FR46** The system shall display search results according to the selected search type
- FR47** The system shall display public profile information of searched users
- FR48** The system shall display route information for searched routes
- FR49** The system shall display diary entries from followed users
- FR50** The system shall display the available challenges
- FR51** The system shall display the requirements for completing each challenge

4.1.5 Profile and Friends Features

- FR52** The system shall display the user's profile picture, username, and level
- FR53** The system shall display the number of diary entries created by the user
- FR54** The system shall display the number of followers of the user
- FR55** The system shall display the number of followed users
- FR56** The system shall allow users to access the public profile of a follower
- FR57** The system shall allow users to access the public profile of a followed user
- FR58** The system shall allow users to share a public link to their profile

4.1.6 Navigation and Sensor Information

- FR59** The system shall display compass information
- FR60** The system shall display the current heading in degrees
- FR61** The system shall display the user's current altitude
- FR62** The system shall display the user's current geographic coordinates

4.1.7 Weather Information

- FR63** The system shall allow users to access detailed weather information for a selected route
- FR64** The system shall display a five-day weather forecast related to the selected route
- FR65** The system shall allow users to request weather information using their current GPS position
- FR66** When the user selects route-based weather information, the system shall display weather data for the route area
- FR67** When the user selects GPS-based weather information, the system shall display weather data for the user's current position
- FR68** The system shall display a map preview associated with the currently selected weather location
- FR69** The system shall display weather alerts related to the currently selected weather location
- FR70** The system shall allow users to enable weather alert notifications for a selected route

- FR71** The system shall allow users to access a satellite map view displaying both the selected route position and the user's current position, using distinct visual indicators
- FR72** The system shall allow users to display weather layers on the map
- FR73** The system shall allow users to display precipitation information on the map
- FR74** The system shall allow users to display snow information on the map
- FR75** The system shall allow users to display wind information on the map
- FR76** The system shall allow users to display cloud information on the map
- FR77** The system shall allow users to display temperature information on the map
- FR78** The system shall allow users to display pressure information on the map
- FR79** The system shall display a legend associated with the selected weather layer

4.1.8 Offline mode

- FR80** The system shall detect when the internet connection is not available
- FR81** When the internet connection is not available, the system shall display a dedicated offline page
- FR82** In offline mode, the system shall allow users to access compass functionalities
- FR83** In offline mode, the system shall allow users to access their current GPS position
- FR84** In offline mode, the system shall allow users to access altitude information
- FR85** When the internet connection is restored, the system shall allow users to resume online functionalities

4.2 Smartwatch Application Functional Requirements

4.2.1 Access & Synchronization

- FR1** The system shall allow users to access smartwatch functionalities after pairing the device with their account.
- FR2** The system shall synchronize route and user data with the associated smartphone application.
- FR3** When synchronization fails, the system shall notify the user.
- FR4** The system shall allow users to enable weather notifications for selected routes.

4.2.2 Routes & Map Visualization

- FR5** The system shall display routes on a smartwatch map.
- FR6** The system shall display refuge points as spots on the route.
- FR7** The system shall allow users to select a refuge point on the map.
- FR8** When a refuge point is selected, the system shall display associated information.
- FR9** The system shall allow users to scroll through route-related information.
- FR10** The system shall classify routes according to their difficulty level.
- FR11** The system shall display routes using different colors based on their difficulty level.

4.2.3 Route Details

- FR12** The system shall display key information about a selected route.
- FR13** When a user selects a route, the system shall display detailed information about the selected route.
- FR14** The system shall display route information including difficulty level, distance, estimated time, and elevation gain.
- FR15** The system shall indicate the suitability of a route for families.
- FR16** The system shall display a concise textual description of the route.
- FR17** The system shall display refreshment points associated with the route.
- FR18** The system shall display challenges associated with the route.
- FR19** The system shall display summary weather information for the selected route.
- FR20** The system shall allow users to start a navigation session for a selected route.

4.2.4 Weather Information

- FR21** The system shall display weather information for a selected route.
- FR22** The system shall display current weather conditions.
- FR23** The system shall display a short-term forecast.
- FR24** When weather information is unavailable, the system shall notify the user.
- FR25** The system shall allow users to receive weather alert notifications for selected routes.
- FR26** The system shall display weather alerts related to the currently selected weather location.
- FR27** The system shall allow users to enable weather alert notifications for a selected route.

4.2.5 Friends' Routes

- FR28** The system shall display routes published by followed users.
- FR29** The system shall allow users to browse routes shared by friends.
- FR30** The system shall allow users to select a route shared by a followed user.
- FR31** The system shall display basic information about routes shared by friends.

4.2.6 Interaction & Smartwatch Constraints

- FR32** The system shall present information in a format optimized for a small screen.
- FR33** The system shall allow users to navigate content through touch-based scrolling.
- FR34** The system shall minimize user interaction steps to access route information.
- FR35** The system shall maintain the selected route context during navigation between views.

4.2.7 Offline Mode

- FR36** The system shall detect when the internet connection is not available
- FR37** When the internet connection is not available, the system shall display a dedicated offline page

5. Implementation and Integration

5.1 Implementation Strategy

The development of the Triplo application followed a structured and incremental approach, aligned with the architectural design of the system. The implementation was carried out using a bottom-up strategy, where foundational components were developed first and progressively extended towards higher-level functionalities.

In particular, the implementation process evolved through a set of stages that were not strictly sequential but partially overlapped during development.

First, the **model components** were defined. These components represent the core domain entities of the application and include `challengeModel`, `trekkingModel`, `UserModel`, and `DiaryModel`. They provide a structured representation of the data managed by the system and exchanged with the backend.

Second, the **controller components** were introduced. Controllers were developed early in the process and progressively refined as new functionalities were added. Rather than being completed in a single phase, they evolved incrementally together with the introduction of application features. This includes `languagecontroller`, `challengecontroller`, `trekkingcontroller`, `UserController`, and `DiaryController`. Over time, controllers were improved to better coordinate logic and interactions among components.

As development progressed, the implementation of the **view components** and controllers proceeded together, following the introduction of new functionalities. The user interface components, such as `GeowatchPage`, `HomePage`, `trekkingPage`, `SettingsPage`, `LoginRegistrationPage`, `SearchPage`, `UserProfilePage`, and `DiaryPage`, were developed incrementally to support the growing feature set of the application.

Finally, the **service components** were introduced. Services such as `Background_service`, `InternetService`, `permissionservice`, `geoservice`, `memoryservice`, `notificationsservice`, and `authservice` were implemented to reduce the responsibilities of controllers and to delineate and isolate individual technical duties. In particular, services encapsulate interactions with the operating system (e.g., permissions, geolocation, notifications) and provide reusable functionality. This design improves modularity and maintainability by decoupling application logic from platform-specific operations.

5.1.1 Implementation Process Overview

The incremental construction of the system components is summarized in Figure 5.1.1

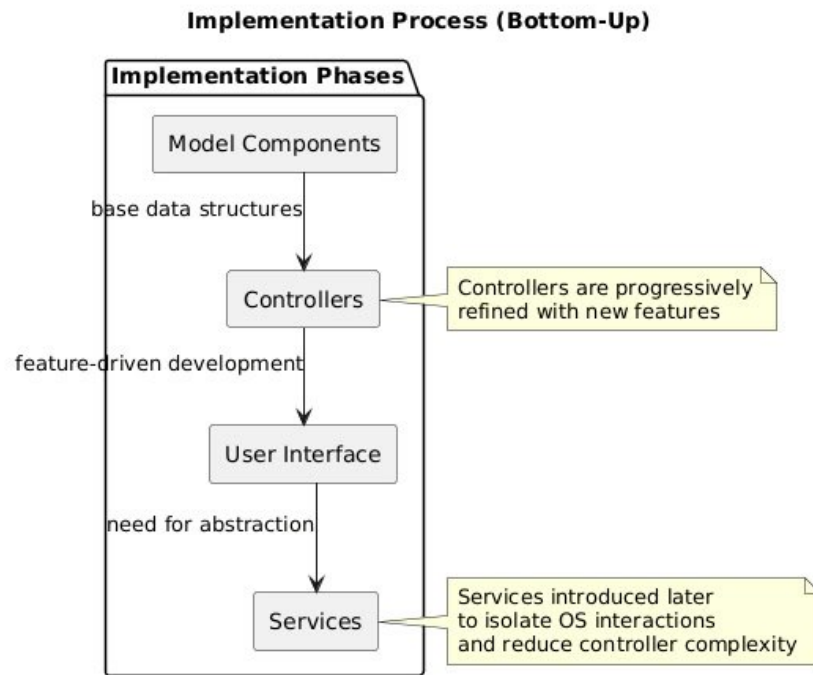


Figure 5.1: Implementation Process

5.2 Integration of Software Components

After the implementation of the individual components, the system was progressively assembled through an integration phase. This phase followed a clear incremental strategy.

Initially, **Firestore** was integrated together with the controller components. Controllers directly interact with Firestore services (such as authentication, database, and storage) through platform-specific SDKs, enabling early validation of backend-related functionalities.

Subsequently, the integration of **view components** and controllers continued in parallel with the implementation of functionalities. This allowed the application to evolve incrementally, ensuring that each feature was fully integrated before introducing new ones.

External services, such as weather and map providers, were integrated at a later stage of development, once the corresponding user interface components were available. As the number of external dependencies increased, a dedicated **ServiceController** was introduced following the Facade design pattern. This controller centralizes access to external APIs and OS-level services, simplifying interactions and reducing coupling across the system.

Finally, **device-level services** were fully integrated through the service components, which manage interactions with the operating system in a controlled and modular way.

Overall, the integration process was incremental: core backend functionality was integrated first, followed by user interface components, and finally external services and OS interactions.

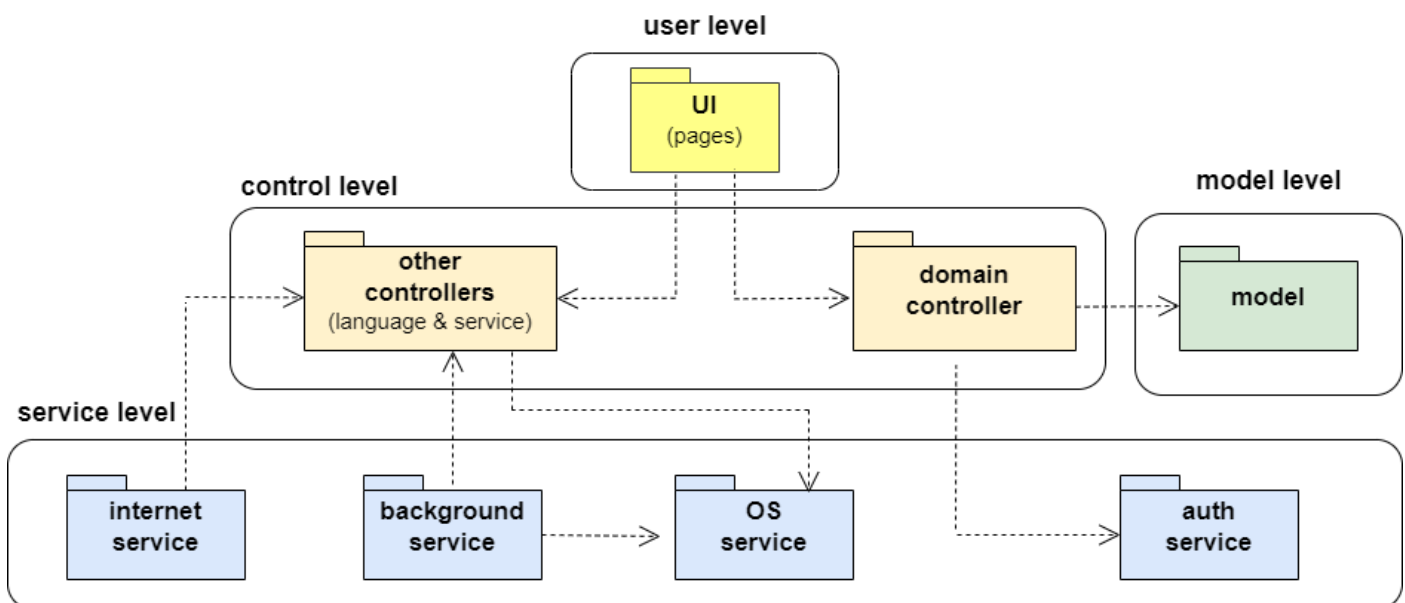


Figure 5.2: Package diagram representing the dependencies in the smartphone application

5.3 Testing

The testing strategy is defined according to the testing levels recommended in the Flutter documentation.

5.3.1 Unit Tests

Unit tests (also referred to as module tests) focus on verifying the correctness of individual components in isolation. In the context of the Triplo application, unit tests target models, controllers, and services, ensuring that each component behaves as expected when provided with controlled inputs.

5.3.2 Widget Tests

Widget tests are specific to Flutter and are used to verify the behavior of individual UI components. These tests simulate user interactions and validate the rendering and state changes of widgets without

requiring a full application runtime. Widget tests are particularly useful to ensure that UI components correctly interact with controllers and display the expected data.

5.3.3 Integration Tests

Integration tests validate the interaction between multiple components of the application. In Flutter, these tests run the application in a simulated or real environment and verify complete user flows, including navigation, data retrieval, and interaction with services. Integration tests are essential to ensure that controllers, services, and UI components work together correctly.

5.3.4 User Testing

User testing represents the final stage of validation, where the application is tested by end users in realistic scenarios. This type of testing focuses on usability, user experience, and overall system behavior in real-world conditions.

5.3.5 Test Environment

The testing and validation activities are performed using a combination of physical devices and emulators, as summarized in Table 5.1.

Device	Type	OS	Architecture
Android Emulator (Medium Phone API 36)	Emulator	Android	x86_64
Wear OS Emulator (Small Round)	Emulator	Android Wear OS	x86_64
Samsung Galaxy A34 5G	Real Device	Android	ARM64
iPhone 15 Plus	Real Device	iOS	ARM64

Table 5.1: Testing environment devices and architectures